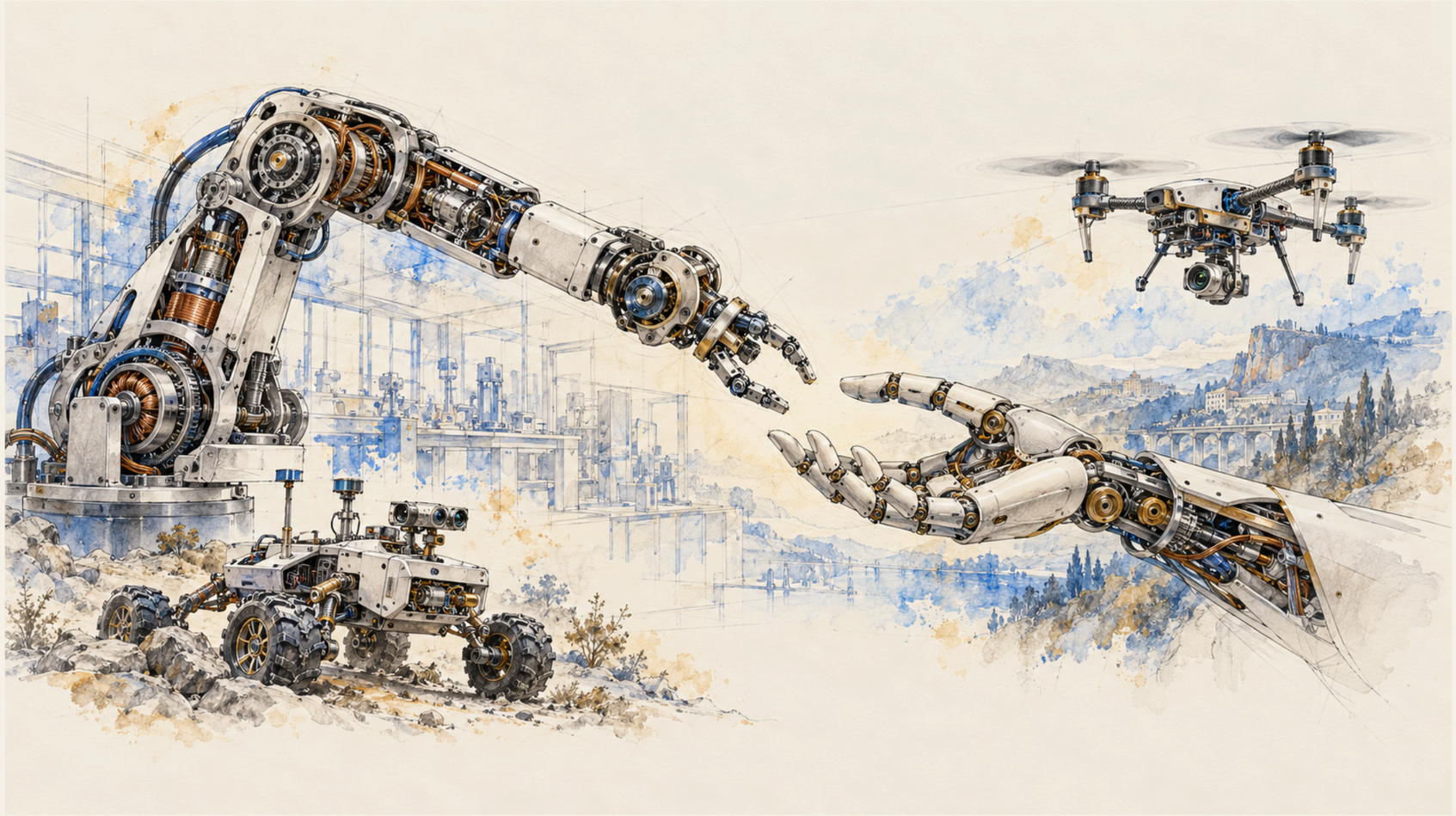
Designing the Physical Robot

WEEK 02



Mechanics · Modeling · Actuation · Power · Embedded Systems

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From mathematical model to physical robot

A robot couples continuous physics to discrete computation through feedback.

1 Two time scales of a cyber-physical system

The mechanical plant moves continuously, whereas a digital controller measures and computes at sampling instants. State x contains the model variables needed to predict the future from an initial condition; input u represents an applicable command. Measurement y is the sensor-visible part of the state. Measurement noise ν and disturbance d enter at different points. Position alone cannot predict the next position of a moving joint; present velocity is also needed. A dot denotes a time derivative: $\dot{x}$ describes how the state changes. Function f represents physical evolution, whereas h specifies what the sensor observes.

$$\dot{\mathbf{x}} = f(\mathbf{x}, \mathbf{u}, \mathbf{d}), \qquad \mathbf{y} = h(\mathbf{x}) + \boldsymbol{\nu}$$

2 Command, measurement and delay

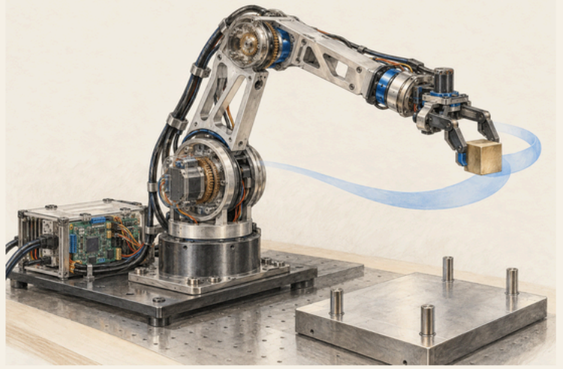
The controller computes a command from the error between reference r and measurement y. The drive converts command to current, and the motor converts current to torque. Holding the command between samples, communication delay and saturation modify an ideal continuous model. A delayed proportional command reacts to past error; sampling period alone does not specify total latency.

$$e_k = r_k - y_k, \qquad u_k = \mathrm{sat}(K_p e_k)$$

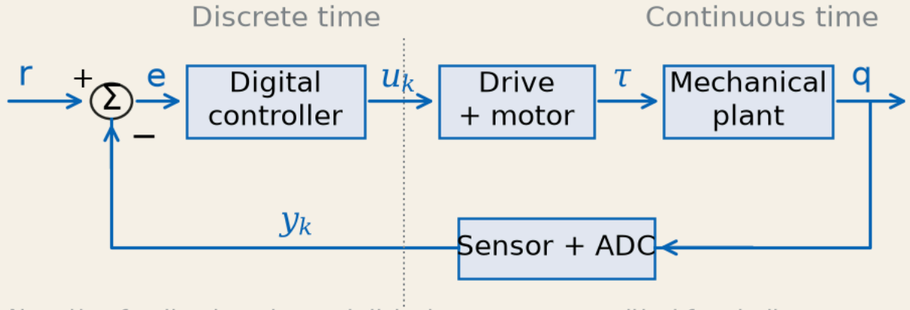
$$u(t) = u_k, \quad kT_s \le t < (k+1)T_s$$

3 Verify physical implementation

A joint model may use q and $\dot{q}$ as states, while its motor drive actually receives a PWM duty ratio. Define the current loop, gearbox and encoder scaling between these interfaces. Record measured position, current, timestamps and saturation status alongside commanded position. These observations help separate model mismatch from implementation errors.



The mechanical plant and electronic controller operate together. The diagram identifies signal directions in the physical feedback loop.



Negative feedback; noise and disturbance arrows omitted for clarity.

Model scope

State, measurement and command are different quantities. Choose model detail for the task: planning may use a velocity model, while current-loop design needs winding dynamics. This week connects models to implementation, from mechanics to embedded software.

Anatomy of a robot

System performance depends on interfaces as well as individual component properties.

1 Mechanical structure and motion freedom

The base, links and joints define load paths and admissible motion. An actuator converts energy and applies force or torque to some of that motion. Degrees of freedom need not equal motor count: constraints can reduce freedom, and a floating base adds coordinates without direct actuation. The end effector is the physical interface to the task.

2 Sensing, computation and power layers

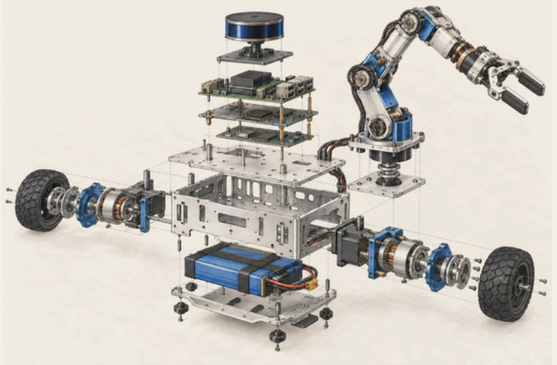
Encoders measure joint motion, current sensors electrical loading, and cameras or lidar the environment. A real-time microcontroller runs fast loops; a higher-level computer may handle perception and planning. Supply, protection and converters meet subsystem voltage and current needs. Communication must convey units, timestamps, validity and fault status, not merely data bits.

3 Interface contract: a wheel module

The motor shaft, gearbox output and tire contact are different system boundaries. For an ideal fixed-radius wheel, circumferential speed and tractive force follow the relations below. Gear losses, tire slip and bearing loads require separate treatment. Module interfaces must specify mounting stiffness, peak supply current and measurement latency alongside the torque–speed envelope. These equations describe the same contact: a larger wheel travels farther per revolution but produces less traction for the same shaft torque. Changing wheel diameter therefore changes the required motor operating point as well as the geometry.

$$v = r_w \omega_w, \qquad F_t = \frac{\tau_w}{r_w}$$

$$F_t v = \tau_w \omega_w \quad \text{(ideal)}$$



Subsystems are separated to expose assembly relationships; this is not a manufacturing drawing.

Interface	Contract quantity	Example unit
Mechanical	Torque / speed	N·m / rad·s⁻¹
Electrical	Voltage / current	V / A
Sensing	Resolution / latency	count / ms
Data	Scale / timestamp	SI / s

Consequence of architecture

Mechanical, electrical and data interfaces must be mutually consistent. Adequate motor torque cannot compensate for poor structural stiffness or delayed feedback. The exploded view explains assembly; the technical table identifies quantities to check at each interface.

Energy flow and information flow

Feedback regulates behavior; the energy required for motion comes through the physical power path.

1 Power is the rate of energy transfer

Voltage and current in an electrical port, torque and angular speed in rotation, and force and speed in translation are power-conjugate variables. A drive command is not energy itself; it regulates transfer from source to load. Input power splits into output power, change of stored energy and losses. Omitting storage invalidates transient energy accounting. Storage includes the increasing kinetic energy of an accelerating rotor; losses do not remain available as recoverable mechanical energy. During a transient, the difference between input and output power therefore cannot all be interpreted as heat.

$$P_e = Vi, \quad P_r = \tau\omega, \quad P_t = \mathbf{F}^{\mathrm{T}}\mathbf{v}$$

$$P_{\mathrm{in}} - P_{\mathrm{out}} = \dot{E}_{\mathrm{stored}} + P_{\mathrm{loss}}$$

2 Efficiencies multiply; losses become heat

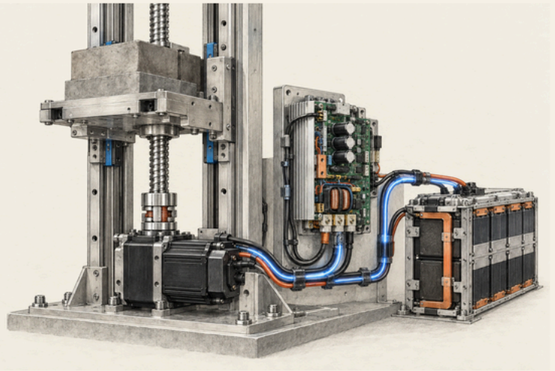
For steady motoring through a drive, motor and gearbox in series, overall efficiency is the product of stage efficiencies. Values of 0.95, 0.85 and 0.90 give 0.727 overall. A 100 W electrical input produces about 72.7 W mechanical output and 27.3 W heat loss. Data-sheet efficiencies are comparable only with consistent system boundaries.

$$\eta_{\mathrm{tot}} = \eta_d \eta_m \eta_g = 0.72675$$

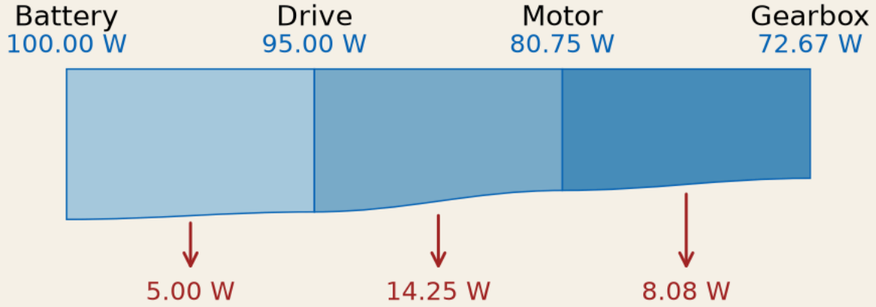
3 Where does braking energy go?

A descending load or decelerating robot can drive the motor as a generator. If the battery cannot accept returned energy, DC-link capacitor voltage rises. A braking resistor, regenerative supply or adequate charge acceptance therefore belongs to system design. A stop command on the information link does not remove energy; stopping needs a physical energy path.

$$\Delta E_C = \tfrac{1}{2} C (V_2^2 - V_1^2)$$



While a vertical axis lowers a load, mechanical energy can return through the drive to the DC bus.



Downward arrows: heat loss. Steady motoring example.

Signs and boundaries

Motoring and regeneration signs depend on the chosen power direction. Do not apply an efficiency ratio blindly at zero power or during sign-changing transients. The diagram quantifies the power balance; command and measurement connections form the information path in the preceding feedback diagram.

Energy and power equations

From the definition of work to a robot energy balance

1 Differentiate work in time

For a small displacement, force does work dW = F·dx. Dividing by time gives F·v because dx/dt is velocity. At a rotary joint, work is torque times angle increment; the same operation gives torque times angular velocity. A perpendicular force may transfer no instantaneous power. The dot product selects only the force component along velocity. Aligned force and velocity give positive power; opposite directions give negative power. Even a large force produces zero instantaneous mechanical power at zero velocity, separating static load support from mechanical work.

$$P = \frac{dW}{dt} = \mathbf{F} \cdot \mathbf{v}, \quad P_{rot} = \tau \dot{q}$$

2 From electric charge to power

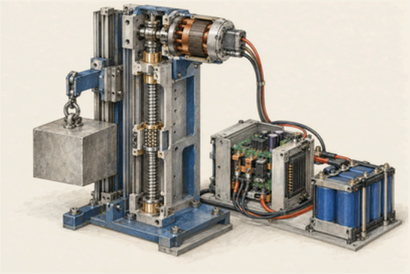
Voltage is transferred energy per unit charge: dE = V dQ. Current is dQ/dt, giving electric power VI. With positive power entering the system, the rate of stored energy equals input minus output and losses.

$$P_{elec} = V\frac{dQ}{dt} = VI, \quad \dot{E} = P_{in} - P_{out} - P_{loss}$$

3 Sign and numerical balance

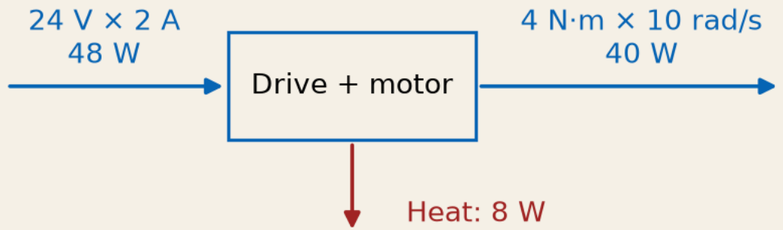
An input of 24 V and 2 A is 48 W. A joint with 4 N·m and 10 rad/s in the same direction delivers 40 W. With no storage change, loss is 8 W and efficiency 83.3%. Opposite torque and velocity make mechanical power negative, indicating braking or reverse energy flow.

$$P_{loss} = 48 - 40 = 8 \text{ W}, \quad \eta = 40/48 = 0.833$$



Load, shaft and wiring form one physical energy-transfer chain.

$$\dot{E} = 0, \quad \eta = 0.833$$



Assumptions and physical limits

Watt is a transfer rate, not energy; joule = watt·second. Define efficiency and allowed current direction separately during regeneration.

Robot design starts from the task

A requirement needs a measurable quantity, operating condition and acceptance criterion.

1 Translate the task into physical requirements

Payload, speed, acceleration, position error, terrain and runtime are coupled choices. More payload changes traction and joint torque; greater speed affects power and stopping distance; longer operation increases battery energy demand. Replace “precise robot” with permitted loaded tip error, measurement method and trial count. Accuracy and repeatability are separate criteria.

2 Establish the example platform load case

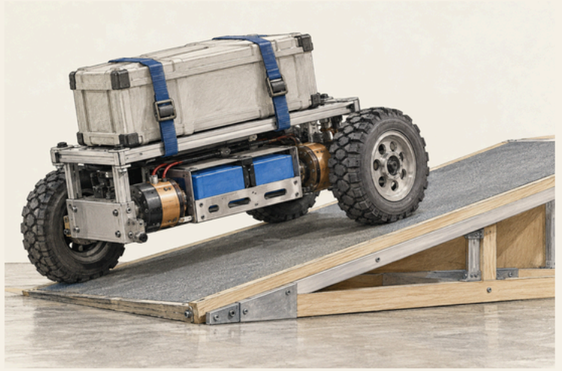
Choose 40 kg including payload, 0.10 m wheel radius, 1 m/s speed, 0.5 m/s² acceleration and a 5° slope. Longitudinal force includes acceleration, grade and rolling resistance. The simplified model below omits aerodynamic drag and assumes a rolling coefficient of 0.02. Requirements determine load-side force before choosing a motor. The terms add here because the demanding case is acceleration uphill. At constant speed, ma vanishes; downhill, gravity changes sign relative to travel. A single required-torque number therefore cannot replace the operating profile and its load cases.

$$F = ma + mg\sin\theta + c_{rr}mg\cos\theta$$

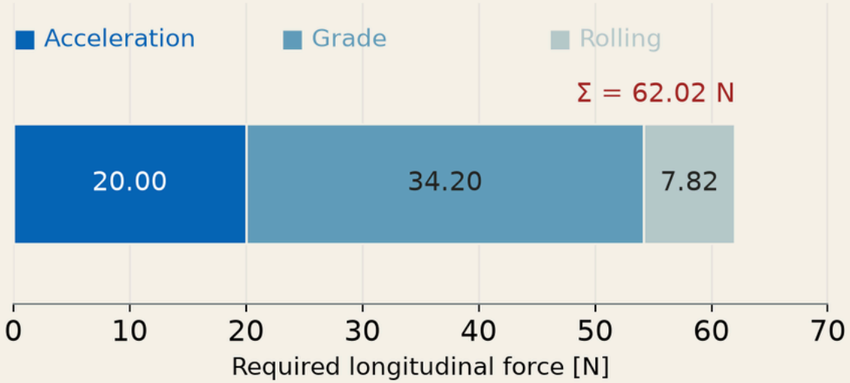
$$F = 20 + 34.20 + 7.82 \simeq 62.02\ \text{N}$$

3 Close each requirement with a test

Test slope speed at full payload on a specified surface. Verify a two-hour runtime claim using measured battery power over the mission, not label capacity alone. Tie peak current, motor temperature and stopping distance to that same duty profile. Add design margin for stated uncertainties; one unspecified safety factor does not cover every failure mechanism.



A ramp test evaluates payload, surface, slope and speed requirements together.



Design trade-off

A larger battery may increase runtime while also increasing mass and traction demand. Sizing is iterative: mission → load → actuator/power → mass → revised load calculation. These values define a shared teaching example, not validated product specifications.

Mechanical loads in robots

A free-body diagram connects a physical component to force and moment equations.

1 Isolate the system boundary

Isolate the link of interest and replace removed connections by reaction forces and moments. Weight acts at the center of mass; contact forces act at contact points. Distinguish an ideal revolute joint’s allowed motion from the reactions it supports. Joint internal forces cancel when the whole arm is selected, but must appear when an individual link is isolated. Action and reaction forces from one interaction act on different bodies; do not place both on the same free-body diagram. Choosing the isolated body before marking external loads prevents counting the same interaction twice.

$$\sum \mathbf{F}_{\text{ext}} = m\mathbf{a}_G$$

2 Force direction and moment arm

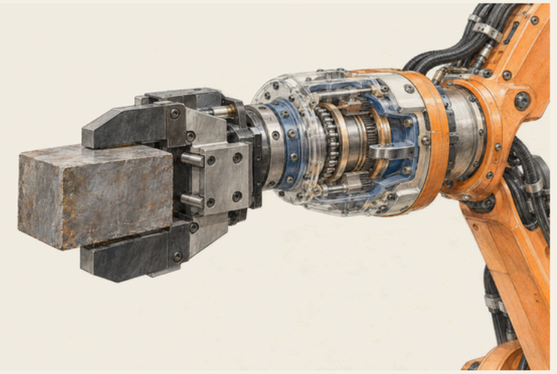
The moment about O is the cross product of position vector r and force. Its magnitude equals force times perpendicular distance to its line of action. A force passing through the axis has zero moment. For planar rigid-body motion, net moment about the center of mass equals polar inertia times angular acceleration; general three-dimensional rotation also includes gyroscopic terms.

$$\mathbf{M}_O = \mathbf{r} \times \mathbf{F}, \quad |M_O| = Fd_\perp$$

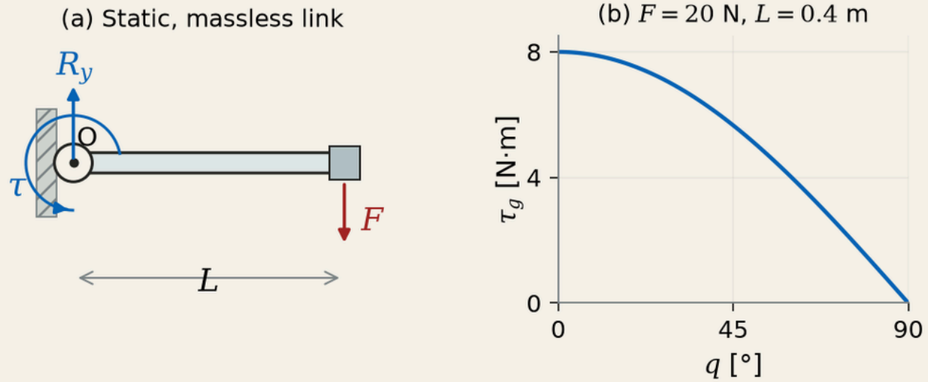
$$\sum M_{G,z} = I_{G,z}\alpha_z \quad \text{(planar)}$$

3 Same force, different joint loading

A 20 N vertical load produces 2 N·m at a 0.10 m horizontal offset and 8 N·m at 0.40 m. Bearing force loading remains in both cases. q denotes link angle from horizontal. Tilting the link changes the gravitational moment arm; maximum static joint torque often occurs at horizontal extension. Dynamic sizing adds acceleration, impact and duty cycle to the static load case.



The wrist and gripper reveal the load path; the free-body diagram identifies force application points.



Boundary and unit checks

Mass is measured in kg, force in N and torque in N·m. State moment origin and positive direction in the diagram. Support reaction is not actuator torque. The figure is a planar free-body model; real bearing pairs also carry three-dimensional loads.

Payload and joint torque

Compute payload weight, link self-weight and inertia contributions separately.

1 Model and sign convention

Consider a uniform link of length L = 0.40 m rotating about a fixed axis. Link mass is m_l = 1 kg and tip point mass m_p = 2 kg. Let q be positive upward from horizontal. The link center of mass is at L/2. Applied torque is positive in this direction; gravity acts oppositely at q = 0.

$$\tau_g(q) = g\left(m_p L + \frac{m_l L}{2}\right)\cos q$$

2 Sum inertia about the same axis

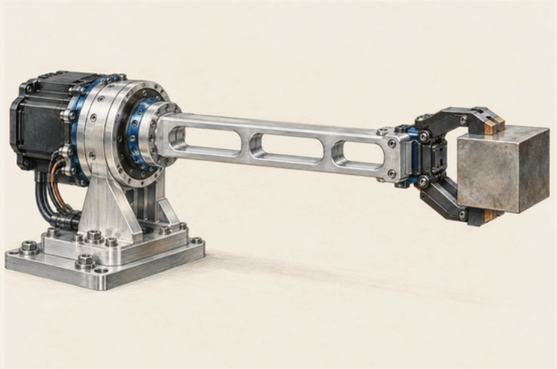
The point payload contributes $m_p L^2$. A uniform slender link contributes $m_l L^2/3$ about its end; the centroidal $m_l L^2/12$ cannot be used directly. Inertia therefore depends on distance from the axis as well as mass. Doubling payload distance quadruples its inertia but only doubles its gravity moment.

$$J = m_p L^2 + \frac{m_l L^2}{3} = 0.3733 \text{ kg m}^2$$

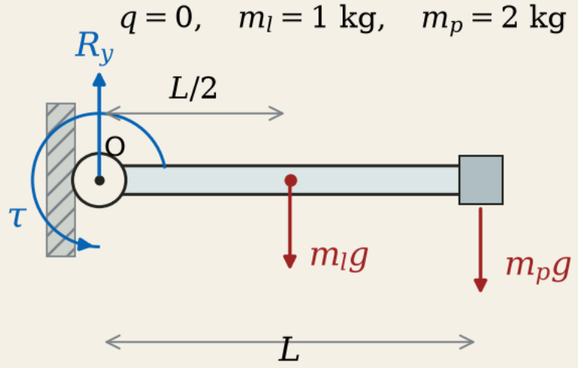
3 Complete calculation at horizontal pose

At q = 0, $\dot{q}$ = 0 and $\ddot{q}$ = 2 rad/s², gravity contributes 2·9.81·0.4 + 1·9.81·0.2 = 9.81 N·m. Inertia contributes 0.3733·2 = 0.7467 N·m. The frictionless model requires 10.5567 N·m output torque. This is not motor-shaft torque; reduction ratio, efficiency and rotor inertia enter the subsequent sizing calculation. Positive acceleration here acts against gravity, so the two torque contributions add. Requesting negative acceleration at the same position reverses the inertia term. A torque requirement based only on angle misses the distinction between accelerating and decelerating motion.

$$\tau = J\ddot{q} + \tau_g(q) = 0.7467 + 9.81 = 10.5567 \text{ N m}$$



The physical link provides context; loads and centers of mass are defined by the ideal model below.



Interpretation and assumptions

At vertically upward pose, cos q = 0 eliminates gravity torque, but acceleration torque remains. Assume one rigid link, a point payload and no friction. R_y is the vertical support reaction. Include gripper mass in the real payload; the illustration is not a geometrical replica of the uniform-link idealization.

Stress, strain, strength and stiffness

A component can survive its load yet violate robot accuracy requirements through deformation.

1 Separate material and component behavior

Under axial normal force, average engineering stress σ is force divided by original area. Engineering strain ε is length change divided by original length and is dimensionless. In the linear elastic range, Young's modulus E is the slope of Hooke's law. E is a material property; component stiffness also depends on section, length and supports.

$$\sigma = \frac{F}{A}, \qquad \varepsilon = \frac{\Delta L}{L}, \qquad \sigma = E\varepsilon$$

2 Strength and stiffness impose different limits

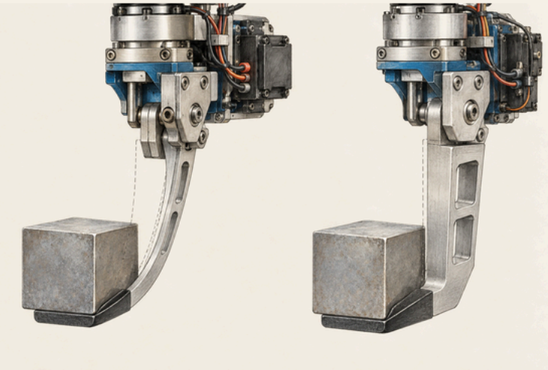
A strength check prevents a selected failure criterion from being exceeded; in a ductile material, yield marks the onset of permanent deformation. A stiffness check limits displacement under load. Combining the three axial relations gives extension FL/(EA). Increasing area increases axial stiffness; increasing length decreases it even with unchanged material. The extension equation follows by substituting ΔL/L for strain and F/A for stress, then isolating ΔL. This shows why the material modulus E and component stiffness EA/L have different units and different physical meanings.

$$\Delta L = \frac{FL}{EA}, \qquad k_{\text{axial}} = \frac{F}{\Delta L} = \frac{EA}{L}$$

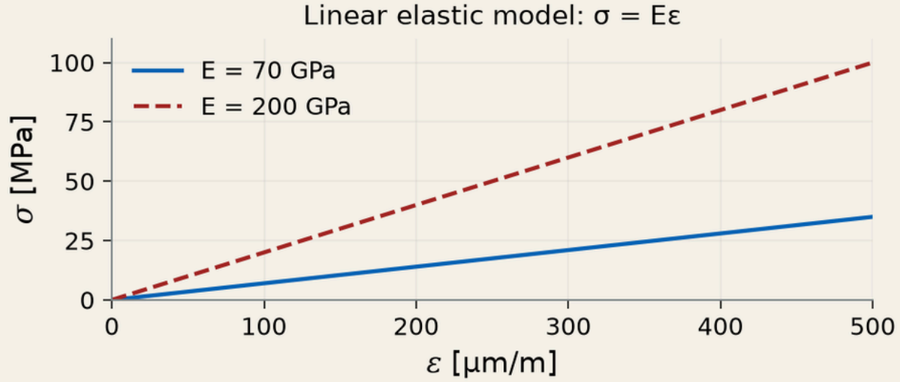
3 Connect deformation to robot tip error

A component deflecting 0.20 mm under 100 N has translational stiffness 500 kN/m. If allowed tip error is 0.10 mm, it fails the requirement even when stress remains below yield. Compliances add for elastic components in series; the softest connection may dominate total error. Joint and bearing stiffness matter as much as link material.

$$\frac{1}{k_{\text{eq}}} = \sum_i \frac{1}{k_i} \quad \text{(series)}$$



The qualitative deformation comparison is exaggerated. The graph is a dimensioned example of linear elasticity.



Validity limit

These relations assume small deformation, linear elasticity and ideal loading. Bending, torsion, buckling, contact and fatigue require other models. Stress and E are in Pa = N/m²; converting mm and m correctly is essential.

Robot bracket deflection

Section geometry and boundary conditions change how the same material behaves on a robot.

1 Cantilever beam model

Idealize the sensor bracket as a beam of length L with constant bending rigidity EI. Displacement and slope vanish at the fixed left end; a downward force F acts at the right end. Euler–Bernoulli theory assumes plane sections, negligible shear deformation and small slopes. E is Young’s modulus and I is the second moment of area about the bending axis, not mass inertia.

$$I = \frac{bh^3}{12}, \qquad [I] = \mathrm{m}^4$$

2 From curvature to deflection

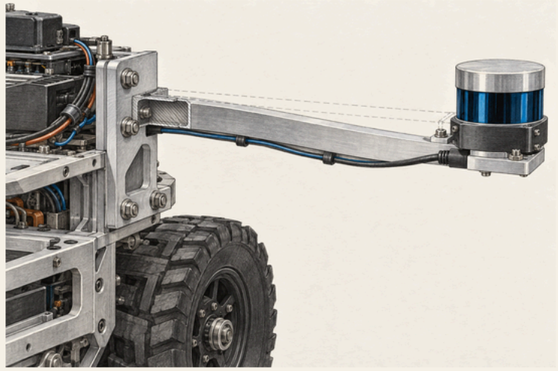
With downward deflection taken positive, the magnitude form gives curvature F(L−x)/(EI). Integrate twice and apply zero slope and zero displacement at the fixed end. The result describes the entire deflection curve, not just its tip. Different supports or distributed loading change both boundary conditions and the solution. Integration constants represent how the bracket is supported. A hinge cannot impose the zero-slope condition of a clamp. This explains why identical material and dimensions can produce different deflection when the mounting arrangement changes.

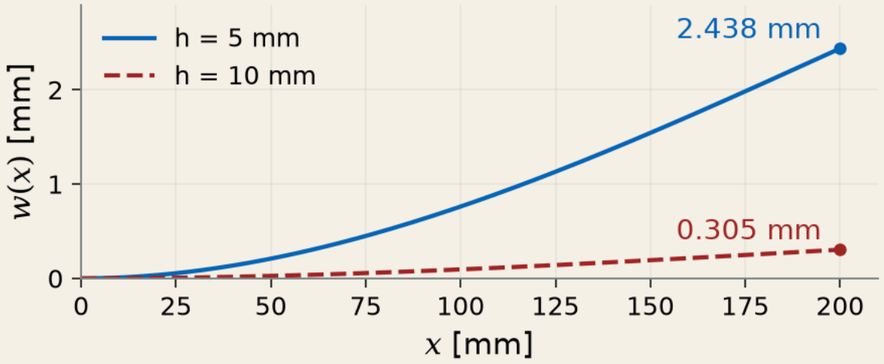
$$w(x) = \frac{Fx^2(3L - x)}{6EI}, \qquad \delta = w(L) = \frac{FL^3}{3EI}$$

3 Change geometry and check the result

For F = 20 N, L = 0.20 m, E = 70 GPa, b = 30 mm and h = 5 mm, I = 3.125·10⁻¹⁰ m⁴ and tip deflection is 2.438 mm. Doubling thickness in the bending direction to 10 mm increases I eightfold and reduces deflection to 0.305 mm. Mass only approximately doubles: section design can bring major gains before changing material.

$$\delta \propto \frac{L^3}{h^3}, \qquad \sigma_{\max} = \frac{FL(h/2)}{I} = 32\ \mathrm{MPa}$$





Bracket application and calculated deflection curve. Vertical axis is in mm; deformation and geometry use different scales.

Transfer to the physical component

Compare root stress with the material’s allowable stress separately. Bolt compliance, holes, stress concentrations and sensor off-axis loading are absent from this ideal beam. Compare FEA and experiment using matched loads and supports.

Deriving cantilever deflection

Moment → curvature → slope → displacement

1 Section moment and curvature

Choose x from clamp to tip and y positive downward. The downward tip force gives bending magnitude F(L−x). For small slopes, curvature is approximately y″. With consistent moment/displacement signs, EI y″ = F(L−x), where E is Young's modulus and I is the second moment of area.

$$EIy'' = F(L - x)$$

2 Two integrations, two boundary conditions

Integrate once to obtain slope; y′(0)=0 at the clamp removes the first constant. Integrate again to obtain displacement; y(0)=0 removes the second. Substituting x=L yields the tip-deflection formula. The first integral gives dimensionless small-angle slope; the second gives displacement in metres. Checking both conditions at x=0 and the tip value at x=L separately catches missing constants or an incorrect power of length.

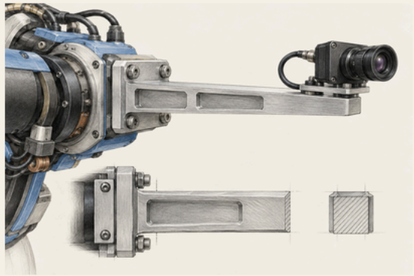
$$EIy' = F(Lx - x^2/2)$$

$$y(x) = \frac{F}{EI}\left(\frac{Lx^2}{2} - \frac{x^3}{6}\right), \quad y(L) = \frac{FL^3}{3EI}$$

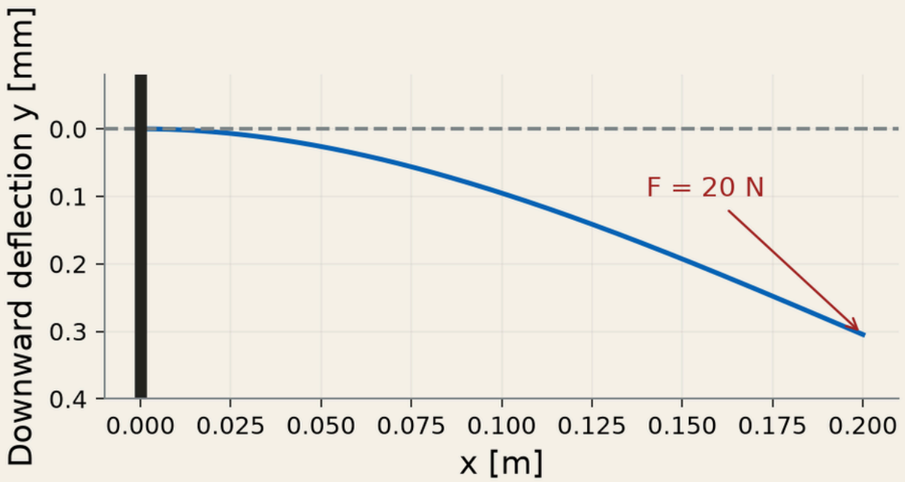
3 The cubic effect of thickness

For a rectangle, b is width and h thickness in the bending direction. Since I=bh³/12, doubling h reduces deflection to one eighth. For F=20 N, L=0.2 m, E=70 GPa, b=0.03 m and h=0.01 m, I=2.5×10⁻⁹ m⁴ and tip deflection is 0.305 mm.

$$I = \frac{bh^3}{12}, \quad \delta = 3.0476 \times 10^{-4}\ \text{m}$$



The curve is the analytical beam solution; displayed deformation is magnified for visibility.



Assumptions and physical limits

Assume Euler–Bernoulli behavior, small deflection, constant section and rigid clamping. Shear deformation and joint compliance are omitted.

Material selection for robotics

Select material, cross-section and manufacturing together; low density alone does not make a good robot link.

1 Separate material properties from part behavior

Density ρ [kg/m³] determines mass per volume; Young's modulus E [Pa] characterizes resistance to small elastic strain. Yield strength concerns the onset of permanent deformation. Component stiffness also depends on section and length. Steel has a high modulus, but its required mass for a task cannot be compared without considering geometry.

$$m = \rho A L, \qquad k_{\text{axial}} = \frac{EA}{L}$$

2 Compare mass at equal stiffness

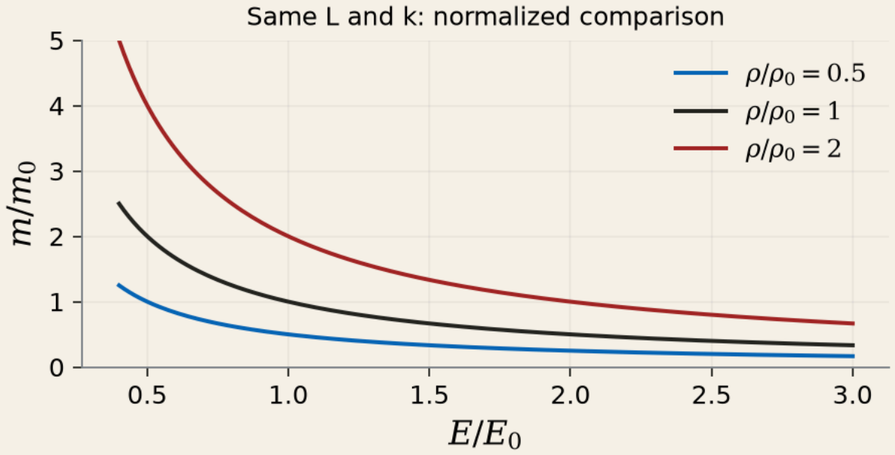
For an axial member of fixed length L and target stiffness k, first obtain area A = kL/E. Substitution into the mass equation shows dependence on ρ/E. This result applies to axial loading at equal length. Bending instead involves second moment of area I [m⁴]; a hollow section that places material away from the neutral axis may be highly effective. The comparison holds function constant rather than cross-section: both candidates must give the same extension under the same load. Lower density alone is insufficient; a lower modulus may require a larger section and reduce the expected mass saving.

$$m = kL^2 \frac{\rho}{E}, \qquad k_{\text{bend}} = \frac{3EI}{L^3}$$

3 Translate the decision to a robot

Aluminum is a candidate for machinable lightweight links, steel for wear and high contact loads, polymers for covers and lightly loaded parts, and composites for directionally stiff lightweight beams. Fiber direction and metal inserts govern composite joints; temperature and creep matter for polymers. Equal material indices do not imply equal tool error when joint stiffness differs.

Different link sections show physical choices; the plot compares mass ratios at equal axial stiffness.



Assumptions and physical limits

Any numerical property needs a grade, temperature, fiber direction and source. The plot is a scaling model under fixed requirements, not a universal material ranking. Joints, fatigue and manufacturing require separate checks.

Safety factor and fatigue

A single maximum load cannot characterize damage from repeated loading.

1 Split the stress cycle into two components

Consider a robot joint cycling from −20 MPa to +80 MPa. Mean stress is half the sum of extrema; amplitude is half their difference. The mean is 30 MPa and amplitude 50 MPa. Equal peak stresses can produce different fatigue behavior when mean and amplitude differ. Notches increase local stress.

$$\sigma_m = \frac{\sigma_{\max} + \sigma_{\min}}{2}, \qquad \sigma_a = \frac{\sigma_{\max} - \sigma_{\min}}{2}$$

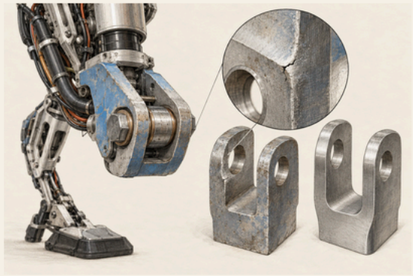
2 Interpret the Goodman relation

Corrected endurance strength S_e and ultimate tensile strength S_{ut} are distinct from applied stress. The Goodman line models reduced allowable alternating stress as tensile mean stress increases. Surface, size, temperature and reliability effects enter S_e. For proportional scaling of a specified stress state, the safety factor n follows the relation below. Each ratio is dimensionless. A sum of one identifies the model boundary; a smaller sum lies inside it. A factor n=3.33 does not mean 3.33 times the fatigue life; it describes the stated proportional stress margin.

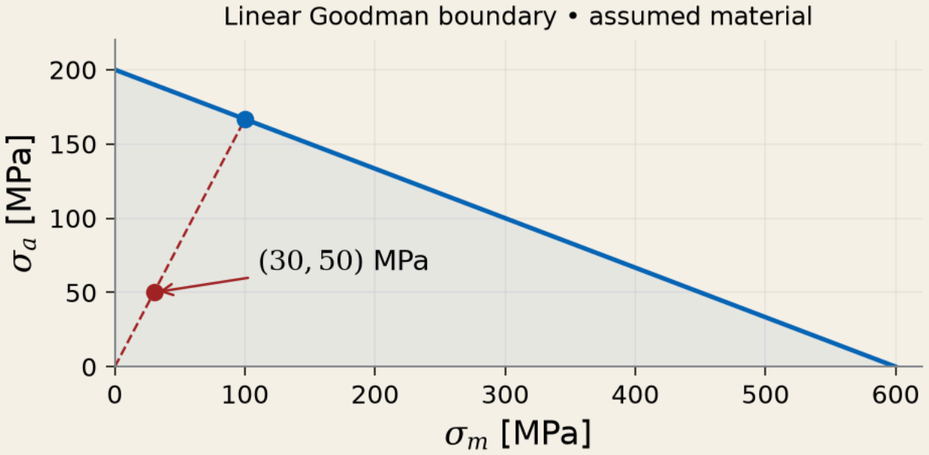
$$\frac{\sigma_a}{S_e} + \frac{\sigma_m}{S_{ut}} = \frac{1}{n}$$

3 What calculation and testing establish

For an instructional example choose S_e = 200 MPa and S_{ut} = 600 MPa: 50/200 + 30/600 = 0.30, giving n = 3.33. These are not catalog values for a specified alloy. Static yielding, contact damage and impact need separate checks. The same impact energy stopped over a shorter distance produces a larger average force; a static-load check alone is insufficient.



The source Goodman diagram’s axis meanings are retained; lines and operating point are calculated for this example.



Assumptions and physical limits

Goodman is a simplified fatigue model; not every material has an endurance limit. Duty cycle, notches and manufacturing details require validation. A safety factor is meaningful only with its failure criterion.

Structural modes and vibration

Elastic deformation can accompany rigid joint motion, and controller commands can excite these modes.

1 Represent one dominant mode

Let x [m] denote a flexible link's tip displacement. Use equivalent moving mass m [kg], stiffness k [N/m] and damping c [N·s/m]. Force balance equals m ẍ. The spring opposes displacement and the damper opposes velocity. This small-deformation model describes a selected elastic mode, not the complete joint configuration. Here x is a small elastic displacement superimposed on the planned motion, not the joint coordinate producing that motion. Separating these variables explains why a camera can vibrate even when the robot follows its commanded joint trajectory.

$$m\ddot{x} + c\dot{x} + kx = F(t)$$

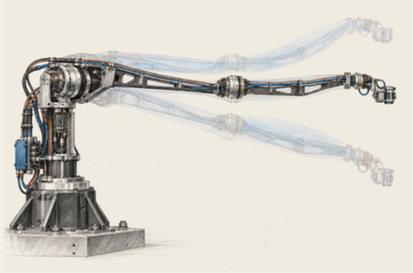
2 Natural frequency and damping

Substituting a sinusoid into undamped free motion balances inertia against spring force, giving natural angular frequency √(k/m). Divide by 2π to obtain hertz. Damping ratio ζ scales actual damping against critical damping. For m = 2 kg and k = 8000 N/m, ω_n = 63.25 rad/s and f_n = 10.07 Hz.

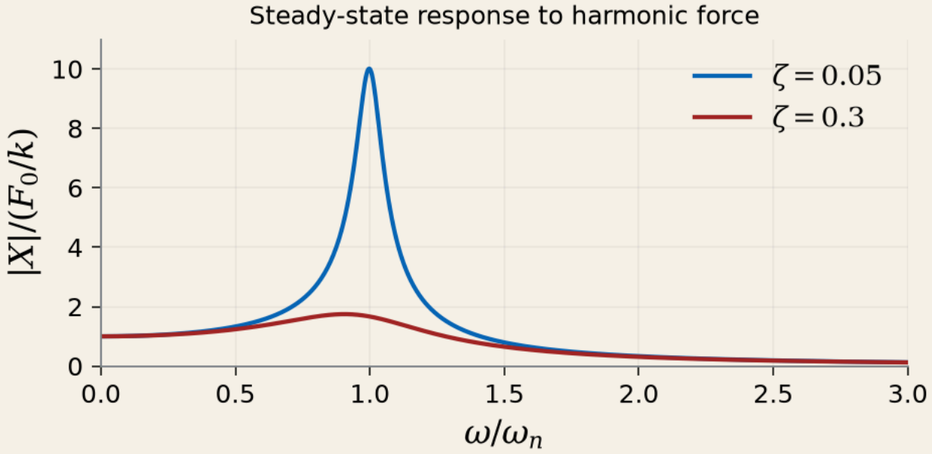
$$\omega_n = \sqrt{\frac{k}{m}}, \quad f_n = \frac{\omega_n}{2\pi}, \quad \zeta = \frac{c}{2\sqrt{km}}$$

3 What resonance means for a robot

Periodic commands or rotating components may produce large displacement near a natural frequency. The plot compares ζ = 0.05 and 0.30 at equal mass and stiffness; more damping reduces the peak. Increasing k raises natural frequency, while increasing m lowers it. Raising servo gain alone does not make a flexible mechanism more accurate; control bandwidth interacts with structural modes.



Read the link's elastic mode together with its calculated normalized force-to-displacement response.



Assumptions and physical limits

The plot is calculated from a linear single-mode model, not measured. Mode-shape deformation may be exaggerated. Real joint stiffness and payload change the modes; modal tests and multiple load poses are needed.

Vibration equation and resonance

Describe one structural mode with mass, stiffness and damping

1 Write Newton’s equation

The spring opposes displacement with −kx, and viscous damping opposes velocity with −c ẋ. Newton’s law gives m ẍ+c ẋ+kx=F(t). Here m and k may be equivalent modal values, not necessarily the robot’s total mass and stiffness.

$$m\ddot{x} + c\dot{x} + kx = F_0 \cos \Omega t$$

2 Natural frequency and damping ratio

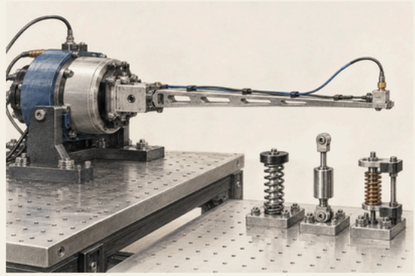
For undamped free motion, ẍ+(k/m)x=0, so natural angular frequency is √(k/m). Normalize damping by 2√(km) to obtain ζ. With m=2 kg and k=8000 N/m, ωn=63.25 rad/s and fn=10.07 Hz; rad/s and Hz are not numerically identical.

$$\omega_n = \sqrt{k/m}, \quad f_n = \omega_n/(2\pi), \quad \zeta = \frac{c}{2\sqrt{km}}$$

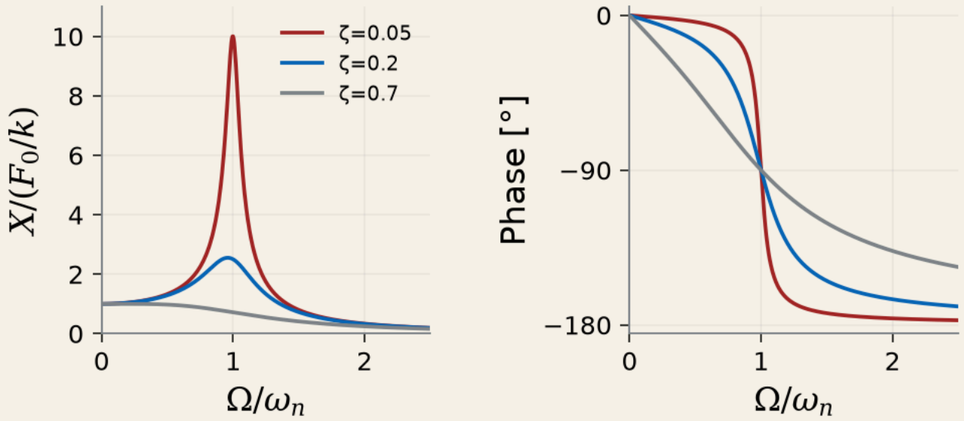
3 Amplitude under harmonic forcing

In sinusoidal steady state, derivatives scale with frequency; dynamic stiffness k−mΩ² and damping cΩ determine amplitude. Using r=Ω/ωn gives amplification relative to static deflection F0/k. For ζ=0.05 at r=1, amplification is 10. At low frequency, inertia is small and amplification approaches one, the static response. At high frequency, inertia dominates. The phase curve shows that force and displacement do not necessarily peak together; amplitude alone does not describe their timing relationship.

$$\frac{X}{F_0/k} = \frac{1}{\sqrt{(1-r^2)^2 + (2\zeta r)^2}}$$



Phase lag relative to force matters to control design as well as steady amplitude.



Assumptions and physical limits

With damping, the forced-response peak need not occur exactly at ωn. Select the frequency band in which a single-mode approximation is valid.

Center of mass and static stability

Where a payload is carried can matter as much as its mass.

1 From components to total center of mass

Let each component have mass m_i and position r_i in the same world frame. The total center of mass is their mass-weighted position average. A 30 kg body at z = 0.20 m and a 10 kg load at z = 0.80 m give zC = (30·0.20 + 10·0.80)/40 = 0.35 m. Lowering the load to 0.40 m lowers the combined center to 0.25 m.

$$\mathbf{r}_C = \frac{\sum_i m_i \mathbf{r}_i}{\sum_i m_i}$$

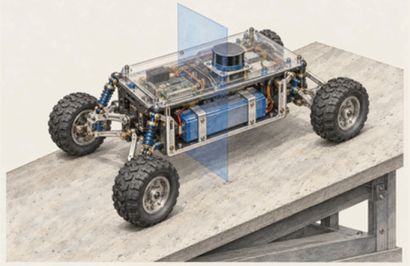
2 Contacts define the support region

For planar support, draw the convex hull of contact points. In quasi-static balance, if the line of gravity lies outside it, nonnegative normal reactions cannot maintain equilibrium. Tipping begins about an edge. Projection follows gravity; on a slope, projection normal to the ground would be incorrect. The condition Ni≥0 arises because nonadhesive ground contact can push a wheel but cannot pull it downward. If algebraic equilibrium requires a negative normal reaction, that contact assumption is physically invalid and the lifted-contact model must be used.

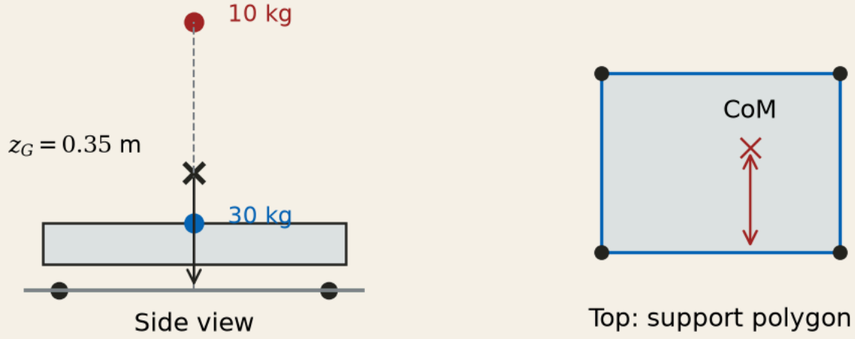
$$\sum \mathbf{F} = \mathbf{0}, \qquad \sum \mathbf{M} = \mathbf{0}, \qquad N_i \geq 0$$

3 Quantify the margin

Distance from the gravity line to the support boundary is a static geometric margin. Lowering a battery helps on slopes and during acceleration, but does not change its projection on level ground if horizontal position is unchanged. Sliding requires a separate friction check. Motion introduces inertia and changing momentum, so this static test alone is insufficient.



The side view shows CoM height; the plan view shows support region and gravity projection.



Assumptions and physical limits

Assume maintained contacts, rigid ground and sufficient friction. Humanoid dynamic balance later uses centroidal momentum and ZMP; the static CoM test does not replace them.

Payload placement and tipping limit

Horizontal acceleration creates an overturning moment through center-of-mass height.

1 Moments about the outer support edge

For a rigid robot on level ground, let h be CoM height and b the horizontal distance from center to the outer support edge. b is half the track, not full track. In a quasi-static load-transfer calculation, inertial force m a_y acts opposite acceleration at the CoM. At impending tip, inner contact reaction vanishes and moments m a_y h and m g b balance. Taking moments about the outer edge eliminates its unknown normal reaction because its moment arm is zero. Weight retains lever arm b, and the inertial term retains lever arm h. The ratio b/h therefore follows directly from the geometry.

$$ma_y h = mgb \quad \Rightarrow \quad a_{y,\,\text{tip}} = g\frac{b}{h}$$

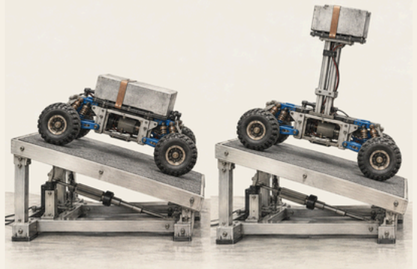
2 Same robot, two payload heights

For b = 0.25 m and h = 0.50 m, the threshold is 9.81·0.25/0.50 = 4.905 m/s². Raising h to 0.80 m lowers it to 3.066 m/s². Although mass cancels in the equation, payload placement changes h and therefore the result. The plot shows this inverse relationship: a higher load creates a larger moment arm at the same acceleration.

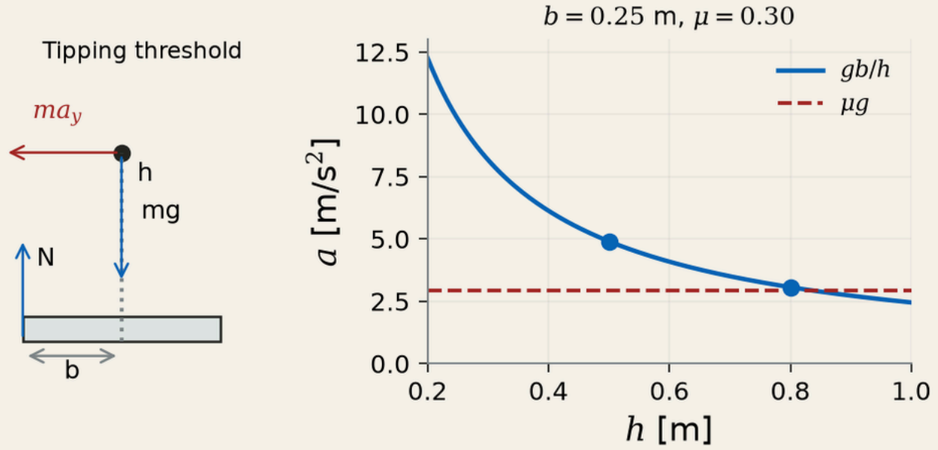
3 Sliding may occur first

If total available friction on level ground is μmg, the sliding threshold is a_y = μg. With μ = 0.30 this is 2.943 m/s², below both tipping thresholds. Tipping analysis alone therefore cannot select a safe command. First identify the active physical constraint, then allow margin for uncertainty and operating conditions.

$$a_{y,\,\text{slide}} = \mu g, \qquad a_y < \min\left(gb/h, \mu g\right)$$



Free-body geometry and sliding–tipping bounds are calculated for two payload heights.



Assumptions and physical limits

Assume a rigid body, level ground and slowly varying load transfer. Suspension motion, impact, tire deformation and slope modify these simple bounds. These thresholds are not validated operating limits for a robot.

Mechanical elements and power transmission

Power-transmitting components and load-supporting components serve different roles.

1 Complete the load path

A shaft transmits torque; bearings transfer radial and, depending on type, axial forces to the housing while allowing rotation. Shoulders and retaining rings locate parts axially. Couplings accommodate limited misalignment; bolt preload affects slip and separation. Making a motor’s small internal bearings carry all wheel and ground loads is unsuitable for many architectures.

2 Belt tension difference creates torque

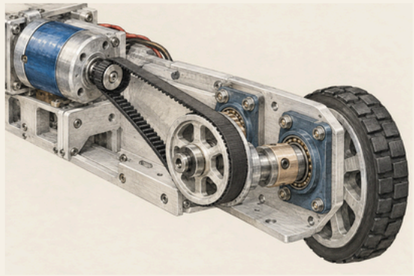
Tight- and slack-side belt tensions F_1 and F_2 differ. Pulley torque equals radius times their difference. With nearly parallel spans, radial bearing load is approximately their sum. At r = 0.04 m, F_1 = 150 N and F_2 = 50 N, torque is 4 N·m and radial load about 200 N. More pretension can increase bearing load without changing transmitted torque. The force difference produces torque because the two belt branches tend to rotate the pulley in opposite directions. Their lateral shaft loads add because they pull approximately in the same direction. Different combinations follow from different equilibrium equations.

$$\tau = (F_1 - F_2)r, \qquad F_{\text{bearing}} \approx F_1 + F_2$$

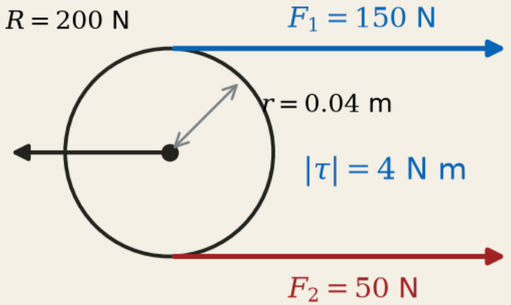
3 Relate motion conversion to work

Gears and belts change speed–torque ratio; chains provide positive engagement; screws convert rotation to translation. If a screw advances p [m] per revolution, input work is 2πτ and output work Fp. Efficiency η gives F = 2πητ/p. With 1 N·m, 5 mm lead and η = 0.90, force is about 1131 N; buckling and bearing loads still need checking.

$$Fp = \eta\, 2\pi\tau$$



The two belt tensions and bearing reaction are separate forces on the same free body.



Assumptions and physical limits

p denotes screw lead, not necessarily thread pitch for a multi-start screw. Belt shafts are separate and parallel. Actual efficiency, preload, lubrication and backdriving require manufacturer data.

Operating principles of robot gearboxes

Planetary, strain-wave and cycloidal reducers obtain high ratios through different geometries.

1 The fixed member matters in a planetary train

The sun, planets, internal ring and carrier are four functional members. With fixed ring, sun input and carrier output, $N = 1 + Z_r/Z_s$, where Z is tooth count. For $Z_s = 20$ and $Z_r = 60$, the ratio is 4. Changing the fixed member changes the speed relation even with identical teeth; the gearbox family name is insufficient. In a frame rotating with the carrier, subtract carrier speed from both sun and ring speeds. Meshing imposes opposite relative motion in that frame. Setting ring speed to zero then produces the extra one in the reduction-ratio formula.

$$N = \frac{\omega_s}{\omega_c} = 1 + \frac{Z_r}{Z_s} \quad (\omega_r = 0)$$

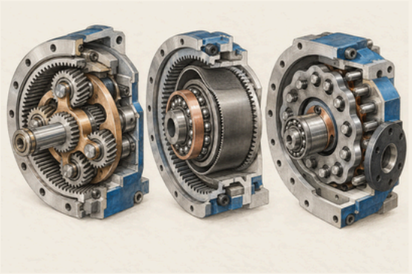
2 Strain wave: small tooth difference, large ratio

An elliptical wave generator elastically deforms a thin flexspline, engaging a rigid circular spline at two opposing regions. With 202 teeth on a fixed circular spline and 200 on the flexspline, one input revolution advances the output two teeth in the opposite direction. Output rotation is $-2/200 = -0.01$ revolution: ratio magnitude 100. Elastic deformation enables operation; it does not imply infinite torsional stiffness.

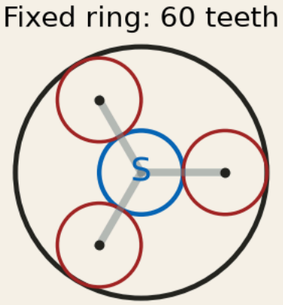
$$\frac{\omega_f}{\omega_w} = -\frac{Z_c - Z_f}{Z_f}$$

3 Cycloidal reducers and selection criteria

In a cycloidal reducer, eccentric input rolls a lobed disc along housing pins; output pins extract the disc’s slow rotational component. Multiple contacts share load. Selection considers impact, output support, hysteresis, stiffness, efficiency and life as well as ratio. A product’s low backlash does not establish zero error or lossless operation for an entire mechanism family.



Three distinct contact and output arrangements are compared; fixed members are identified in the diagram.



Assumptions and physical limits

Ratios apply to the stated fixed/input/output members. Cutaways explain mechanism relationships, not manufacturing tooth profiles. Catalog performance applies to a specified product, load and lubrication.

Selecting a gearbox for a joint

Even with adequate rated torque, compliance, output support or duty cycle can limit the design.

1 State requirements at the output shaft

First define joint torque–speed history, maximum speed, impact cases and service life. Compare them with the catalog's continuous, acceleration and momentary ratings. Rated torque and emergency peak torque are different limits. Bending moment and axial/radial forces at the output flange impose bearing requirements separate from transmission torque.

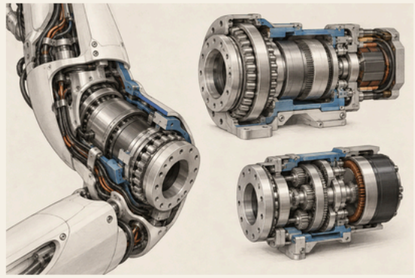
2 Torsion becomes tool-position error

Torsional stiffness kθ [N·m/rad] is the local torque–angle slope. For small elastic displacement, Δθ = τ/kθ. A point at distance L from the axis moves tangentially by approximately LΔθ. With τ = 10 N·m, kθ = 10 000 N·m/rad and L = 0.50 m, Δθ = 0.001 rad and Δx = 0.0005 m = 0.5 mm. The tip-error relation is the small-angle arc length: with angle in radians, displacement is LΔθ. The same angular compliance produces a larger position error on a longer arm. Evaluate gearbox stiffness at the task point, not only at the joint.

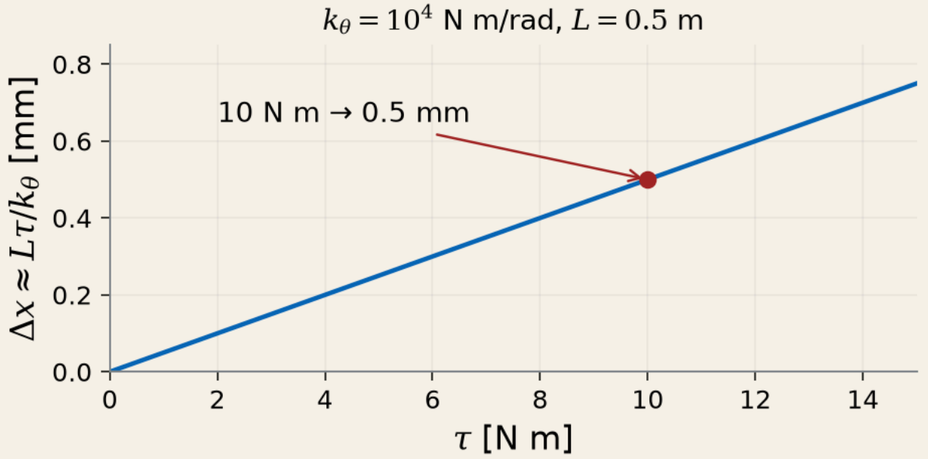
$$\Delta\theta \approx \frac{\tau}{k_\theta}, \qquad \Delta x \approx L\Delta\theta$$

3 Distinguish the error mechanisms

Backlash is clearance on reversal, elastic torsion is load-dependent deflection, and transmission error is departure from ideal gear motion. A motor encoder does not directly measure all effects downstream of the gearbox. Output sensing and loaded calibration may be needed. Excessive reduction brings other costs in speed, backdriving and reflected inertia.



The joint section and torque–twist curve distinguish load capacity from positioning error.



Assumptions and physical limits

The example kθ is assumed; catalog stiffness may be piecewise nonlinear. Repeatability differs from absolute accuracy. Validate selection under load, speed, temperature, lubrication and mounting conditions.

Real transmission effects

Friction, backlash and compliance are different mechanisms; one efficiency number cannot describe them all.

1 Friction depends on speed and motion state

A simple Coulomb term changes sign with motion direction, while viscous friction grows with speed. Parameters are τc [N·m] and b [N·m·s/rad]. With τc = 0.20 and b = 0.03, opposing torque at ω = 2 rad/s is 0.26 N·m. At rest, static friction can balance a range of loads; a sign function alone cannot represent that behavior. Here τf is the signed resisting term used in the motion equation; physical friction opposes motion. At positive speed the motor must overcome both load and loss. Without a consistent sign convention, an incorrect friction term can appear to generate energy.

$$\tau_f = \tau_c \operatorname{sgn}(\omega) + b\omega \quad (\omega \neq 0)$$

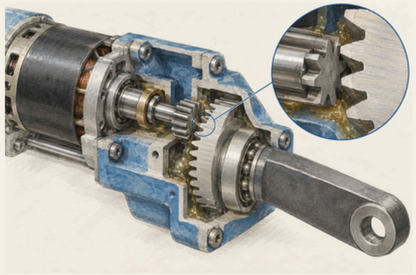
2 Separate clearance from elastic displacement

On tooth-flank reversal, the motor may rotate before the output moves: backlash. Once contact is established, increasing torque elastically twists gears and shafts. With motor angle θm, output angle q and ratio N, spring deflection is θm/N − q. Enforcing ideal q = θm/N in a compliant system hides load-dependent displacement.

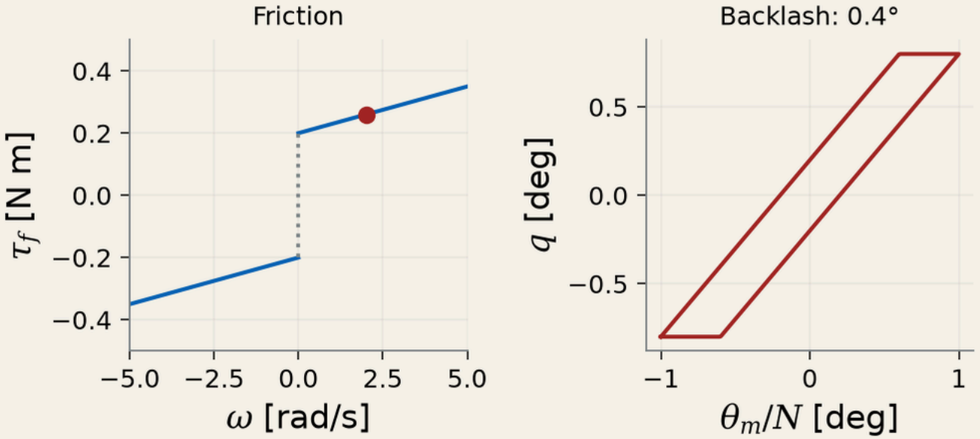
$$\tau_{\text{spring}} = k_\theta\left(\frac{\theta_m}{N} - q\right)$$

3 Follow the system-level consequences

Backlash creates a dead zone on reversal, friction may cause low-speed stick–slip, and compliance adds dynamic modes. Lubrication and temperature alter losses; wear may increase clearance over time. Efficiency measured in forward drive does not determine regenerative or backdriven losses. Evaluate reversal tests using both position and torque measurements.



Tooth-flank clearance, friction versus speed and input–output angle behavior occupy distinct panels.



Assumptions and physical limits

The plots isolate idealized mechanisms; they are not measured aggregate gearbox characteristics. Rest friction and backlash contact require piecewise models. Parameters can vary with load and temperature.

Gear ratio is not free torque

Reduction transforms torque and speed; it does not generate energy.

1 Define the ratio and reference shaft

Let N = ωm/ωout > 1 denote reduction. If the motor turns 20 revolutions per output revolution, N = 20. Angular displacements over equal times have this ratio, so output speed is motor speed divided by N. Revolutions per minute and rad/s express the same quantity in different units; convert using n·2π/60 before calculating power.

$$\omega_{\text{out}} = \frac{\omega_m}{N}, \qquad \omega = \frac{2\pi n}{60}$$

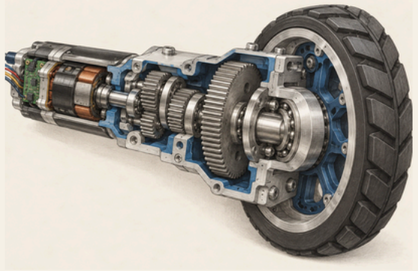
2 Power balance determines torque

Input mechanical power is τmωm and output power is τoutωout. For motor-to-load flow, Pout = ηPin. Substituting the speed ratio yields τout = ηNτm. At ideal η = 1, torque increase exactly balances speed reduction. For η < 1, the remaining power becomes heat and other losses. Torque measures rotational loading; power measures how rapidly that torque performs work. A gearbox transfers energy at a lower output speed. Its torque multiplication therefore does not remove the motor's electrical or thermal limits.

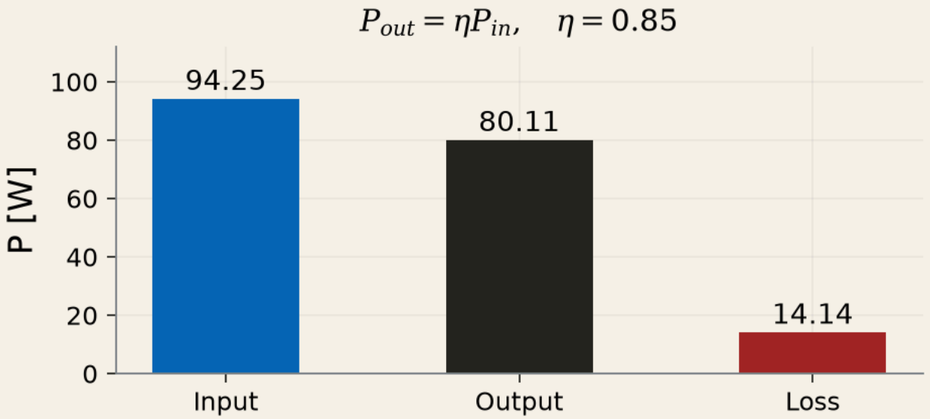
$$\tau_{\text{out}}\omega_{\text{out}} = \eta\tau_m\omega_m$$

3 Check torque, speed and power together

Let N = 20, η = 0.85, τm = 0.30 N·m and nm = 3000 rpm. Output torque is 0.85·20·0.30 = 5.10 N·m and speed 150 rpm. Input power is 94.25 W, output power 80.11 W and loss 14.14 W. Torque increased, but power did not. If required output speed exceeds the motor's allowed speed after scaling, the ratio is infeasible.



The physical drivetrain is matched to motor/output operating points and the loss balance.



Assumptions and physical limits

This example assumes constant efficiency and forward power flow. Startup also requires rotor/gear inertia and static friction. Zero mechanical power at zero speed does not imply zero motor current or copper loss.

Reduction: torque, speed and efficiency

Combine gear geometry with power balance

1 Tangential speed at the mesh

For ideal external gears rolling at the mesh, radius times angular-speed magnitudes match. Tooth count is proportional to radius. External gears rotate oppositely; defining positive N as a speed-magnitude ratio means direction must be stated separately. Tooth-count ratio equals pitch-radius ratio for meshing gears of the same module. Absolute values deliberately separate magnitude from direction. In a multistage train, speed ratios multiply, while the number of external meshes determines the overall direction.

$$r_m|\omega_m| = r_o|\omega_o|, \quad N = \frac{|\omega_m|}{|\omega_o|} = \frac{z_o}{z_m}$$

2 Obtain output torque from power

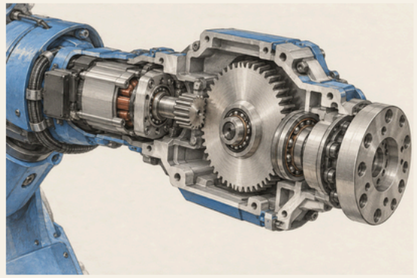
For power flowing from motor to output, let Pout=ηPin. Substituting the speed relation into torque times speed shows that output torque grows by ηN. If efficiency is below one, the remaining input power is dissipated through friction and other losses.

$$\tau_o\omega_o = \eta\tau_m\omega_m, \quad |\tau_o| = \eta N|\tau_m|$$

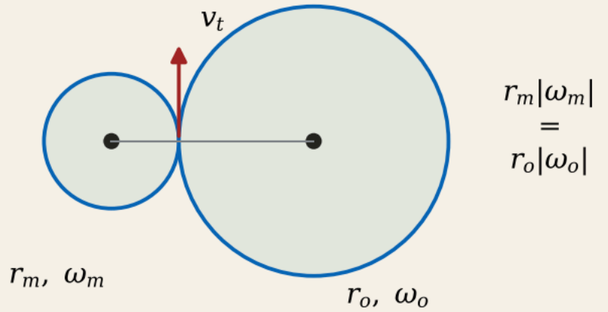
3 A 20:1 reduction example

A 0.3 N·m motor at 3000 rpm gives 5.1 N·m at 150 rpm with N=20 and η=0.85. Convert rpm using 2π/60. Input is 94.25 W, output 80.11 W and loss 14.14 W; gearing does not create power.

$$\omega_m = 3000\frac{2\pi}{60} = 314.16 \text{ rad/s}$$



Mesh velocity and shaft power relations are used in separate derivation steps.



Assumptions and physical limits

This efficiency applies when the motor drives the load. For backdriving, define power direction and backdrive efficiency rather than blindly inserting 1/η.

Gear ratio and reflected inertia

Referring inertia to another shaft introduces the square of the ratio; choosing the wrong direction corrupts sizing.

1 Express the same energy at another speed

Load kinetic energy is Jlωout²/2. Represent the same energy at the motor as Jref,mωm²/2. Because ωout = ωm/N, the motor-referred load inertia is Jl/N². The physical load mass has not decreased; the speed coordinate used to represent its energy has changed.

$$\frac{1}{2}J_L\omega_{\text{out}}^2 = \frac{1}{2}J_{\text{ref},m}\omega_m^2, \quad J_{\text{ref},m} = \frac{J_L}{N^2}$$

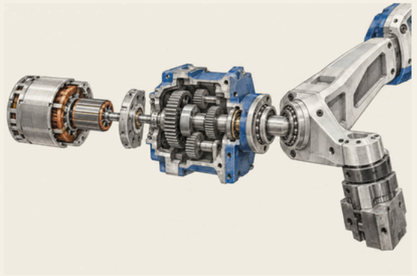
2 Rotor influence grows in the opposite direction

Referring motor rotor inertia Jm to the output gives N²Jm. With Jm = 10^{-4} kg·m² and N = 50, its output-referred value is 0.25 kg·m². A Jl = 0.40 kg·m² load appears as 0.00016 kg·m² at the motor. Both expressions describe the same energy transformation from different shaft coordinates.

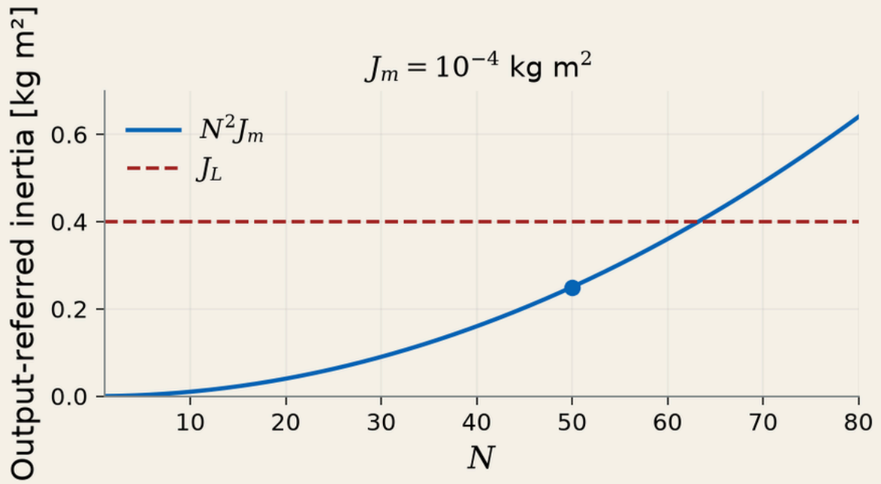
$$J_{\text{eq, out}} = J_L + N^2 J_m$$

3 Why more reduction is not always better

Load torque referred to the motor decreases with ratio, but for a prescribed output acceleration the motor accelerates N times faster. Rotor acceleration torque JmNq̈ grows. In this ideal example output-equivalent inertia is 0.65 kg·m², so q̈ = 2 rad/s² requires 1.30 N·m from inertia alone. Gravity and losses must be added separately. Backdriving the output also rotates the rotor N times faster. A physically small rotor can therefore resist rapid output acceleration substantially. This is why reflected inertia matters for force control and physical interaction, not just for motor sizing.



Different rotor/load speed scales representing equal energy explain the squared ratio.



Assumptions and physical limits

The energy equivalence assumes an ideal rigid transmission. Do not insert efficiency arbitrarily into inertia definitions; model loss torque/power separately. Gear-stage inertias are also referred using their own speed ratios.

Deriving reflected inertia

The squared ratio comes from squared speed in kinetic energy

1 Express the same energy in motor coordinates

For rigid ideal gearing let ωL=ωm/N. Load energy is JLωL²/2. Expressing speed on the motor side gives equivalent load inertia JL/N². Physical mass does not change; the same energy is represented in another coordinate. The squared ratio comes from squared speed in kinetic energy, not an arbitrary gearbox correction. A derivation using acceleration introduces the second ratio through torque transformation. Both methods must agree when referred to the same shaft.

$$E_L = \frac{1}{2} J_L \left(\frac{\omega_m}{N}\right)^2 = \frac{1}{2} J_{L\to m} \omega_m^2$$

$$J_{L\to m} = J_L / N^2$$

2 Refer the rotor to the load side

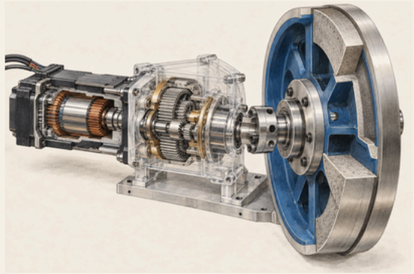
In the reverse coordinate change, ωm=NωL. Rotor energy then gives an output-side equivalent inertia JmN². Total inertia includes rotor and intermediate gears as well as the load. A larger N increases the rotor's effect on output acceleration.

$$J_{m\to L} = J_m N^2, \quad J_{eq,L} = J_L + J_m N^2$$

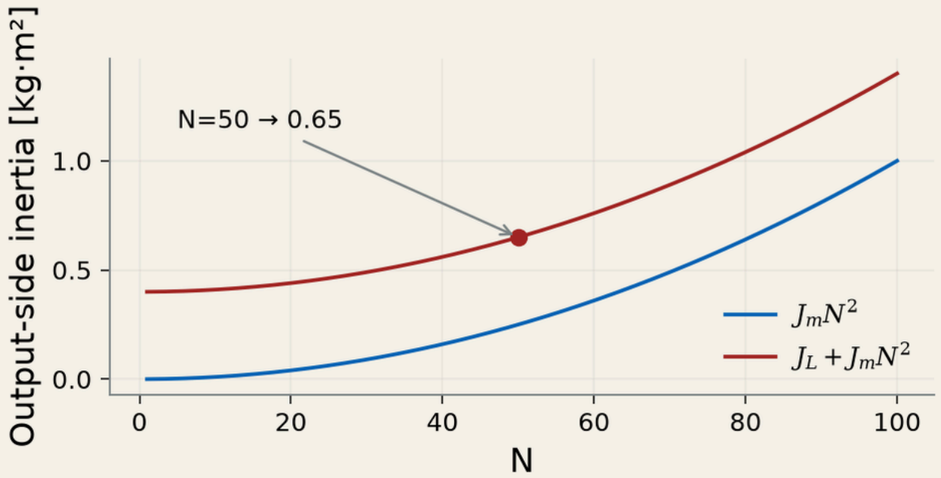
3 Numerical interpretation

With Jm=10⁻⁴ kg·m², JL=0.4 kg·m² and N=50, the rotor contributes 0.25 kg·m² at the output. Total is 0.65 kg·m². An output acceleration of 2 rad/s² needs 1.3 N·m for inertia alone, excluding gravity, friction and transmission loss.

$$\tau_{inertia} = J_{eq,L} \alpha_L = 0.65(2) = 1.3 \text{ N m}$$



Load contribution decreases on the motor side while rotor contribution grows on the load side.



Assumptions and physical limits

Inertia transformation follows ideal rigid kinematics. Efficiency describes loss; do not arbitrarily multiply physical inertia by η.

Mobile robot mechanisms

Wheel placement determines instantaneous motion constraints as well as physical architecture.

1 Differential drive and skid-steer

An ideal differential-drive robot turns by varying the speeds of two driven wheels on one axle; a passive caster supports the load. With body x forward and y left, lateral body velocity is zero. A four-fixed-wheel skid-steer platform requires lateral contact slip while turning. The same ideal rolling equations therefore cannot be applied directly to both architectures. Zero lateral velocity applies to the instantaneous body orientation. Turning first and then driving forward can still produce lateral displacement in world coordinates. A nonholonomic constraint limits instantaneous motion directions rather than simply forbidding a position target.

$$v_y^b = 0, \qquad -\sin\psi\,\dot{X} + \cos\psi\,\dot{Y} = 0$$

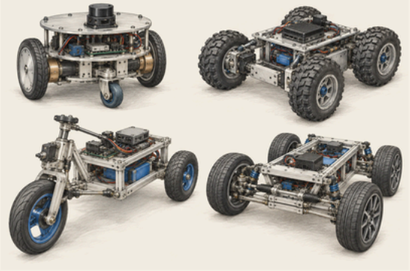
2 Steered wheels and Ackermann geometry

A tricycle layout can use one steered wheel. In an Ackermann layout the inner and outer front wheels steer through different angles; wheel-plane normals intersect at one instantaneous center. L is wheelbase [m], t is track width [m], and δi and δo are inner and outer steering angles. The relation describes ideal low-speed rolling geometry.

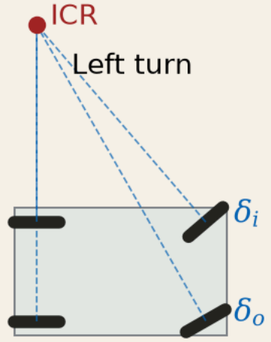
$$\cot\delta_o - \cot\delta_i = \frac{t}{L}$$

3 Choose the mechanism from the task

Differential drive supports on-the-spot turns indoors; skid-steer may offer mechanically simple rugged mobility if slip losses are acceptable. Ackermann steering reduces tire scrub for road vehicles. Minimum turning radius, support area, tire forces and motor loading still require separate checks. No architecture is best for every task.



Four physically distinct mechanisms. The diagram shows the common instantaneous center of inner and outer Ackermann wheels in a left turn.



Assumptions and physical limits

An instantaneous lateral-velocity constraint does not prevent reaching a lateral position through maneuvers. Slip, tire compliance and terrain interaction are outside these kinematic models.

Omnidirectional mobile robots

Passive rollers release one wheel constraint; wheel geometry determines the full chassis-velocity map.

1 From one wheel to the chassis

Body axes are x forward and y left; yaw rate r is positive counterclockwise. Wheel i lies at body position (xi, yi). Rigid-body kinematics first gives its center velocity. For a wheel whose forward direction aligns with body x, γ determines the passive sliding direction; R is wheel radius [m] and si is wheel angular speed [rad/s].

$$v_{ix} = v_x - r y_i, \quad v_{iy} = v_y + r x_i$$

$$R s_i = v_{ix} + v_{iy} \tan \gamma_i$$

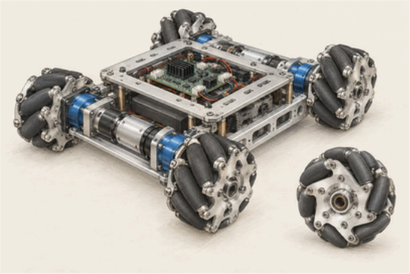
2 Derive wheel signs from geometry

Order the wheels front-left, front-right, rear-left, rear-right at (a,b), (a,−b), (−a,b), (−a,−b). Choose roller angles −45°, +45°, +45°, −45°, respectively. Substitution yields the first-wheel equation below. The remaining rows follow identically; three independent columns allow all three planar chassis velocities. Four motors do not create four independent planar body motions; there are only three body velocities. Arbitrary wheel-speed commands may therefore be incompatible. Generating all four through H from one body command preserves the ideal kinematic consistency.

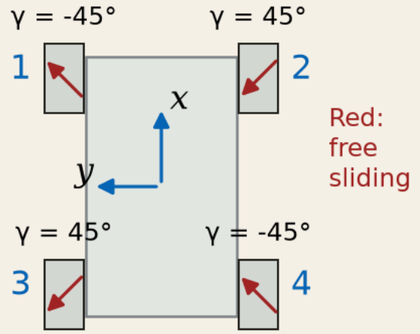
$$R s_1 = v_x - v_y - (a+b) r, \qquad \mathbf{s} = H \boldsymbol{\xi}$$

3 Interpret the commands

Pure forward motion drives all wheels with the same sign. Pure leftward motion uses (−,+,+,−); positive yaw uses (−,+,−,+). For R = 0.10 m and a desired leftward speed of 0.20 m/s, wheel speeds are (−2,+2,+2,−2) rad/s. This ideal mapping requires maintained roller contact; thresholds, soft ground and uneven loads can invalidate it.



Free sliding direction d = (−sin γ, cos γ); the constrained/traction direction is perpendicular. Wheel numbers match the equations.



Assumptions and physical limits

The sign table applies only to the stated roller angles, wheel order and axes. Re-derive the mapping for a different X/O mounting; visual resemblance is insufficient.

Mecanum wheel velocity map

From the free-roller direction to four wheel equations

1 Velocity at each wheel center

Use body x forward, y left, and positive yaw rate ω for a left turn. A wheel at (xi,yi) has center velocity (vx−ωyi, vy+ωxi) from rigid-body kinematics. Wheel radius is r and angular speed is u̇i.

$$\mathbf{v}_i = (v_x - \omega y_i,\ v_y + \omega x_i)^T$$

2 Eliminate free roller motion

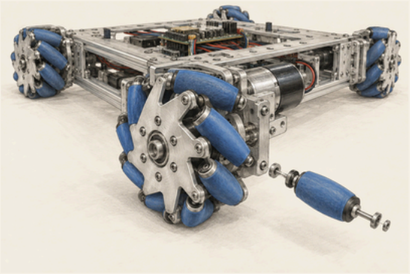
Choose the free roller direction as (−sinγ,cosγ). Projecting onto its normal (cosγ,sinγ) eliminates unknown free roller speed. After subtracting main-wheel rolling speed, residual velocity must lie in the free direction. For nonzero cosγ the resulting row is: Projection does not assume that the roller is stationary; it eliminates its unmeasured free motion. Resolving velocity perpendicular to that free direction leaves the constraint controlled by the wheel motor. Roller geometry thereby determines the signs in the wheel-speed map.

$$r\dot{u}_i = v_x + \tan\gamma_i\, v_y + (-y_i + x_i \tan\gamma_i)\omega$$

3 Check all four wheels

Use front-left, front-right, rear-left, rear-right positions (a,b),(a,−b),(−a,b),(−a,−b), with γ = −45°, +45°, +45°, −45°. Pure left motion gives command signs −,+,+,−. For r=0.1 m and vy=0.2 m/s, magnitude is 2 rad/s.

$$r\dot{\mathbf{u}} = (v_x - v_y - (a+b)\omega,\ v_x + v_y + (a+b)\omega,$$
$$v_x + v_y - (a+b)\omega,\ v_x - v_y + (a+b)\omega)^T$$



Each wheel row follows from position and roller normal, rather than copying a memorized matrix.

$$v_x = 0,\ v_y = 0.2 \text{ m/s},\ \omega = 0,\ r = 0.1 \text{ m}$$

Wheel	v_y	ω	$\dot{u}_i$ [rad/s]
FL	−1	−(a+b)	−2
FR	+1	+(a+b)	+2
RL	+1	−(a+b)	+2
RR	−1	+(a+b)	−2

Assumptions and physical limits

Roller-angle definition determines signs. This arrangement supports three independent body velocities. Re-derive angles and mapping for a different X/O mounting.

Manipulator mechanical architectures

Serial and parallel chains realize the same tool task with different load paths and motion constraints.

1 Motion and load in a serial chain

A serial arm has one open chain from base to tool. Each joint affects downstream links; proximal actuators carry the loads of downstream links and motors. q denotes joint coordinates and T the tool pose. Reachable positions, reachable orientations and collision-free workspace are different sets.

$$T_{EE} = T(q), \qquad q \in \mathcal{Q}$$

2 Loop closure in a parallel chain

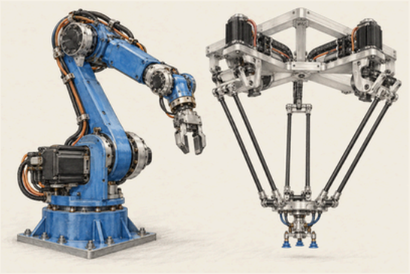
Multiple chains connect to a parallel robot’s moving platform. Their coordinates cannot be chosen independently: they must close at the same platform pose. Φ(q) = 0 expresses geometric loop closure. Differentiation places admissible velocities in the constraint Jacobian’s null space. A rank change can create unexpected mobility or loss of load support. Φ is the position-level loop-closure error and must vanish in a compatible assembly. JΦ measures its sensitivity to joint changes. An admissible small motion must not increase closure error; the velocity null space expresses that physical condition mathematically.

$$\Phi(q) = 0 \quad \Rightarrow \quad J_\Phi(q)\dot{q} = 0$$

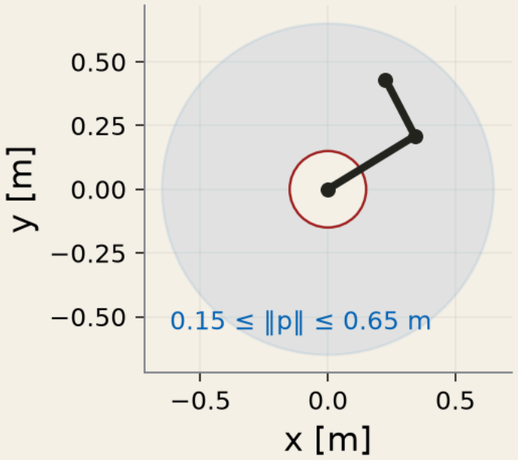
3 Compute workspace before comparing

For a planar serial arm with L1 = 0.40 m and L2 = 0.25 m, tool distance lies between 0.15 and 0.65 m if joint limits and collisions are ignored. Parallel layouts such as delta robots can provide low moving mass for fast handling, but may restrict workspace and orientation. Any stiffness claim must specify loading direction and configuration.

$$|L_1 - L_2| \le \|p\| \le L_1 + L_2$$



Physical serial/parallel comparison; below, an analytical 2R reach annulus and one link configuration.



Assumptions and physical limits

The annulus is only the position workspace of an ideal unrestricted 2R arm. Real robots also impose cable, joint, orientation, collision and singularity restrictions.

End-effectors and physical interaction

Successful grasping depends on contact-force direction and object surface as well as force magnitude.

1 Force balance in a two-jaw gripper

Consider two jaws squeezing a vertical object from its sides. Each jaw applies horizontal normal force Fn; upward friction at each contact is at most μFn. Weight mg acts downward. In a symmetric static grasp with balanced moments, total friction capacity must at least equal the weight. Friction is not always equal to μFn; before slip it takes the value needed up to that bound. The inequality is therefore a capacity condition. Equating jaw normal force directly to weight overlooks that vertical support is transmitted through friction.

$$2\mu F_n \geq mg, \qquad F_{n,\min} = \frac{mg}{2\mu}$$

2 Numerical example and dynamic loading

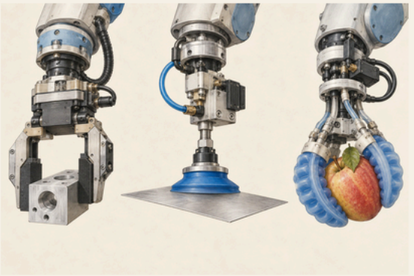
For m = 1 kg and μ = 0.40, each jaw requires at least 12.26 N; total normal force is 24.53 N. These ideal limits contain no safety margin. Upward acceleration a = 2 m/s² increases effective loading to m(g+a), raising the per-jaw threshold to 14.76 N. Excessive squeezing may crush a compliant object, so an upper force limit also matters.

$$F_n \geq \frac{m(g+a)}{2\mu}$$

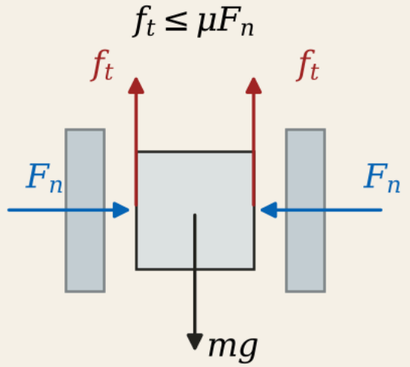
3 Vacuum and compliant grasping

A suction gripper produces force from pressure difference Δp [Pa] over sealed effective area A [m²]. For example, 40 kPa over 4 cm² ideally gives 16 N. Porosity, leaks, peeling and curvature reduce this value. Compliant fingers can improve contact distribution, but force closure requires suitable contact geometry against disturbing forces and moments, not merely strong squeezing.

$$F_{\text{vac}} \approx \Delta p A$$



Forces shown act on the object: inward normals, upward gripping friction and downward weight.



Assumptions and physical limits

The model assumes Coulomb friction, symmetric contacts and a centered load. Verify the constant-μ assumption on slippery surfaces; analyze unbalanced moments and impacts separately.

Suspension and mechanical compliance

Compliance can be an intentional design variable for contact retention and force sensing.

1 Spring and damping in suspension

A spring generates force from relative displacement and a damper from relative velocity. With compression δ [m], stiffness k [N/m] and damping c [N·s/m], the constitutive magnitude follows below. The force sign on each body depends on the chosen axis. A spring stores energy; an ideal viscous damper always dissipates energy. A spring can return stored energy during release; a damper dissipates power in either direction because velocity is squared. Softness and damping are therefore different properties: one changes force versus displacement, the other force versus velocity.

$$F_s = k\delta + c\dot{\delta}, \qquad E_s = \tfrac{1}{2}k\delta^2, \quad P_d = c\dot{\delta}^2$$

2 Series-elastic robot joint

A measurable spring between motor and load allows output torque estimation from deflection. Let θm be motor angle [rad], N reduction ratio, and q load angle [rad]. Both angles must refer to the output shaft. With kθ = 100 N·m/rad and deflection 0.05 rad, torque is 5 N·m. Subtracting load angle directly from motor angle without dividing by N is physically wrong.

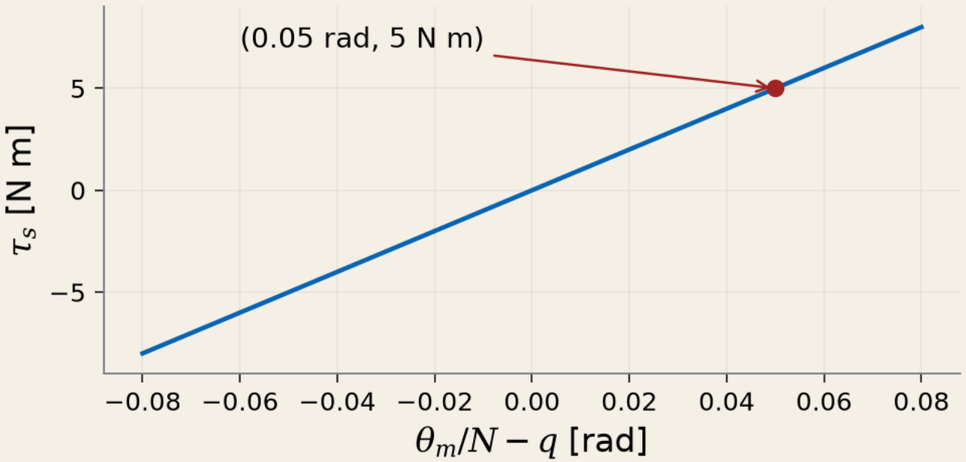
$$\delta_\theta = \frac{\theta_m}{N} - q, \qquad \tau_s = k_\theta \delta_\theta$$

3 Balance impact tolerance and precision

A softer spring may reduce impact force and increase the torque-sensing deflection signal, but creates greater load deflection and a lower mechanical resonance. A differential-angle resolution of 0.001 rad gives an ideal torque increment of 0.10 N·m. This is not total accuracy: spring calibration, hysteresis and both encoder errors also contribute.

$$\Delta\tau \approx k_\theta \Delta\delta_\theta$$

Suspension supports ground contact; a series spring supports torque sensing. The graph shows the selected joint spring's torque–deflection relation.



Assumptions and physical limits

This is a small-deflection linear model. Maintaining wheel contact requires adequate suspension travel and damping; excessive compliance does not automatically improve precision.

CAD and manufacturability for robots

CAD geometry becomes engineering design through load paths, tool access and measurable assembly interfaces.

1 Start dimensions from function

For a robot sensor bracket, first specify where the sensor axis must lie relative to the robot base. Mounting face A, side reference B and end reference C may define the physical measurement references. Locate bearing or sensor bores from functional references rather than arbitrary chained edges. Every dimension must be accessible for manufacturing and inspection. Selecting datum surfaces also determines how the part is located during inspection. If sensor orientation matters, checking hole diameter alone is insufficient; the hole-axis relationship to the mounting surface must be maintained. Functional dimensioning connects the drawing to assembly behaviour.

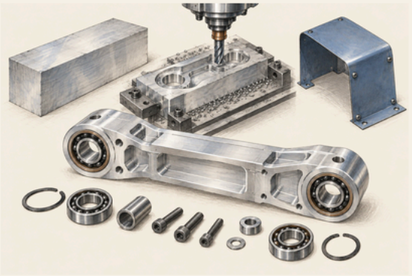
2 Manufacturing constrains geometry

In CNC milling, an internal corner cannot be smaller than the cutter radius; deep pockets increase tool deflection. Sheet metal requires attention to bend radius, bend order and fastener access. Additive manufacturing requires layer orientation, support and finishing decisions. Even a seemingly drillable hole can be inaccessible because of fixtures or neighboring walls.

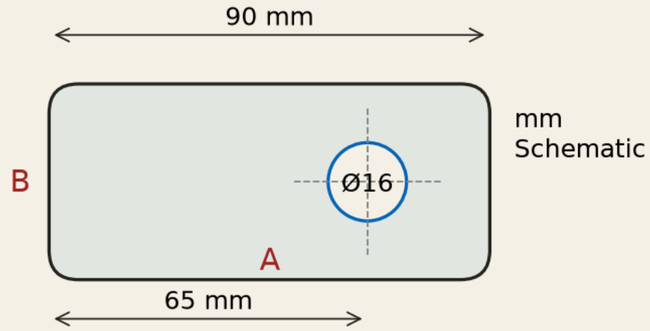
$R_{\text{inside}} \geq R_{\text{tool}}$ (ideal milling geometry)

3 Design assembly and inspection

Fewer machining setups can reduce reference-transfer error. Reserve wrench clearance for fasteners, an insertion direction for bearings and a removal path for maintenance. A 6 mm cutter cannot directly machine a 2 mm internal radius; change the tool or geometry. Unnecessarily tight tolerances raise cost; derive tolerances from permissible sensor-pose error.



An original bracket drawing locates the bore from a common reference and shows the cutter-limited internal radius.



Assumptions and physical limits

A/B/C identify functional references here; this is not a production-ready GD&T specification. Formal geometric tolerancing requires a chosen standard edition and inspection method.

Tolerance accumulation in assembly

Individually accepted parts may fail to deliver the required assembled gap or alignment.

1 Write the signed dimension chain first

Let encoder gap G equal housing internal length A minus shaft shoulder length B and sensor height C. Nominal values 20, 12 and 7.5 mm give G = 0.5 mm. Errors follow the same signed relation. Confirm from the drawing that increasing B decreases the gap rather than increasing it.

$$G = A - B - C, \qquad \Delta G = \Delta A - \Delta B - \Delta C$$

2 Worst case: combine all bounds

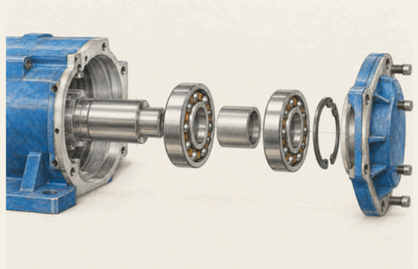
With ±0.05 mm per dimension, the smallest gap is 19.95−12.05−7.55 = 0.35 mm; the largest is 20.05−11.95−7.45 = 0.65 mm. Add absolute tolerance bounds rather than relying on error cancellation. This is a deterministic assembly bound provided every part remains within its own tolerance.

$$T_{WC} = \sum_i |a_i| t_i = 0.15 \text{ mm}$$

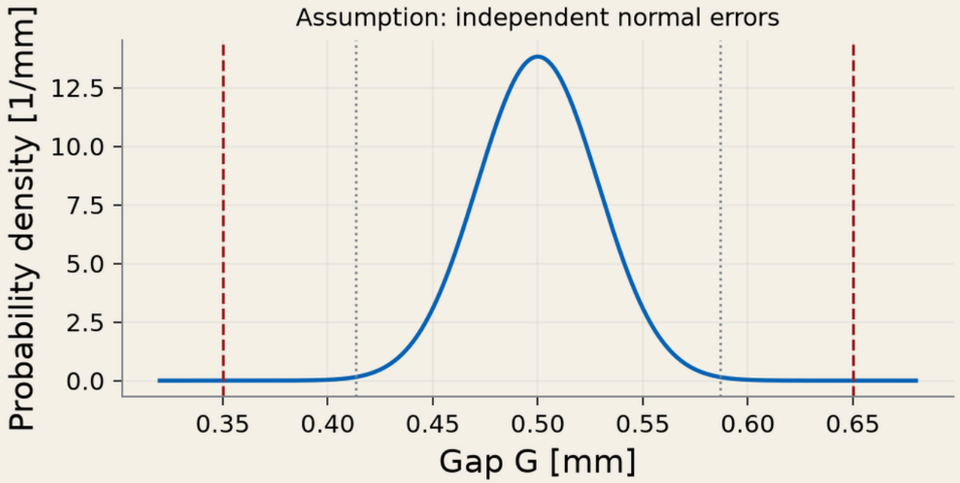
3 RSS requires statistical assumptions

For independent centered normal manufacturing errors, variances add. A ±0.05 mm drawing tolerance is not automatically one standard deviation. If these bounds are separately verified to represent ±3σ processes, each σ is 0.01667 mm and gap σG is 0.02887 mm; the ±3σG interval is ±0.0866 mm. This probabilistic interval does not replace the worst-case guarantee. Under independence, cross-covariances vanish and the minus signs become positive when squared in the variance calculation. Correlated errors, such as shared thermal expansion, do not permit that simplification. RSS is not an arbitrary way to obtain more optimistic tolerances.

$$\sigma_G^2 = \mathbf{a}^T \Sigma \mathbf{a}, \qquad \mathbf{a} = (1, -1, -1)^T$$



Deterministic bounds and an assumed ±3σ interval are shown separately for the same nominal gap; neither guarantees the other.



Assumptions and physical limits

A shared tool, temperature or datum error creates correlation, so cross-covariance terms cannot be discarded. Mean shifts also change the nominal gap.

Mechanical integration and verification

Once assembled, cables, seals, thermal paths and maintenance access become parts of the moving system.

1 Include cables in the motion envelope

Collision-free metal links alone are insufficient. A cable harness must avoid tension at both travel limits, pinching between them and bending below its permitted minimum radius. Obtain that radius from the cable manufacturer’s dynamic-use specification rather than an arbitrary diameter multiple. Strain relief should carry tensile loading at connectors.

$$R_{\text{route}}(q) \geq R_{\min}, \qquad d_{\text{clear}}(q) \geq d_{\min}$$

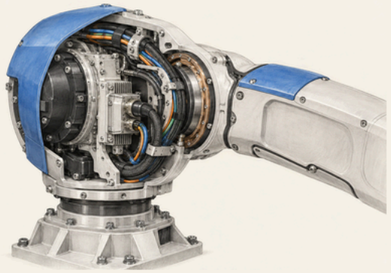
2 Clearance and length budgets

dclear is clearance [mm] to the nearest moving surface and q the joint configuration. Find its minimum over the motion, not just at end poses. A simple circular service loop has arc length s = Rφ. For R = 40 mm and 90° travel, the arc allowance is 62.8 mm; actual routing also includes straight segments, connector approach and assembly tolerance. The arc angle φ must be in radians; multiplying R directly by 90 is incorrect. This calculation describes one circular segment, not a complete cable cut length. Check changing slack and attachment points throughout the motion separately.

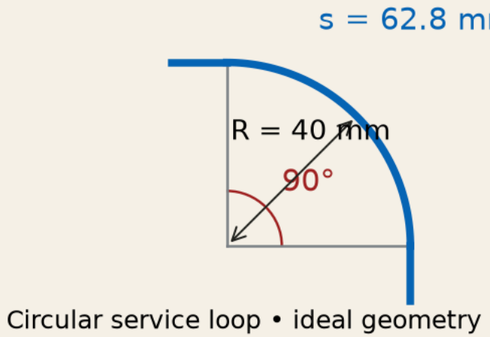
$$s = R\varphi = 40\frac{\pi}{2} = 62.8 \text{ mm}$$

3 Close interfaces with measurable criteria

Example acceptance criteria are at least 5 mm clearance throughout travel, no tensile connector loading and no jacket wear after the target cycles. These are example design targets, not universal standards. In an enclosure, evaluate seal continuity, cable entries and heat rejection together. An inaccessible fuse or a bearing that cannot be removed is also an integration failure.



Service-loop bend radius and arc length; the geometry below is idealized for the numerical example.



Circular service loop • ideal geometry

Assumptions and physical limits

An enclosure’s appearance does not establish an IP rating; verify the required class under the applicable tests and product conditions. Geometric sweeps do not replace cable fatigue testing.

Finite-element analysis and prototype tests

A colored result map is not reliable evidence without appropriate loads and boundary conditions.

1 From the physical problem to a discrete model

Geometry, constitutive model, loads and supports come first. In small-strain linear elasticity, global K is stiffness [N/m], u contains nodal displacements [m] and f nodal forces [N]. Assemble K from element contributions. Unrestrained rigid-body motion can make it singular; excessive constraints artificially stiffen the structure. A node displacement deforms neighbouring elements, so K is not merely a list of independent spring constants. Its off-diagonal terms describe how motion at one location is coupled to forces at other locations.

$$K\mathbf{u} = \mathbf{f}$$

2 Independent analytical check

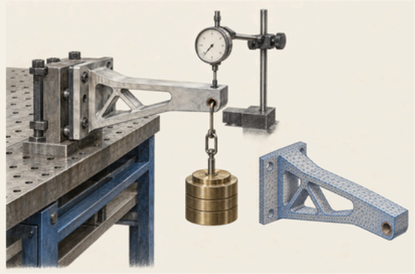
Beam theory provides an initial check for a long slender cantilever bracket. With F = 20 N, L = 0.20 m, E = 70 GPa, b = 0.03 m and h = 0.01 m, I is 2.5×10^{-9} m⁴ and tip deflection is 0.305 mm. This is an analytical reference, not an FEA result. Short thick beams, shear deformation or compliant mounting can produce different results.

$$I = \frac{bh^3}{12}, \qquad \delta_{\text{beam}} = \frac{FL^3}{3EI}$$

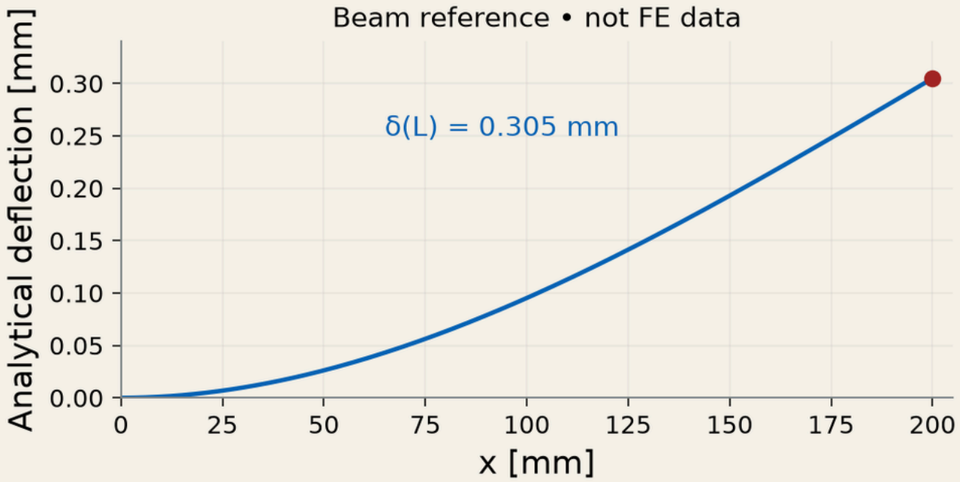
3 Convergence and measurement are different checks

Mesh refinement tracks a chosen output of the same physical model, such as tip displacement. A singular stress peak at a sharp corner may grow under refinement and fail as a convergence metric. Reproduce the loading on a prototype using a load cell and displacement gauge. Before blaming the mesh, examine fixture compliance, actual thickness and material uncertainty.

$$\varepsilon_{\text{rel}} = \frac{|\delta_{\text{FE}} - \delta_{\text{ref}}|}{|\delta_{\text{ref}}|}$$



The test apparatus is illustrative. The analytical deflection curve is computed from the stated F, L, E, b and h values.



Assumptions and physical limits

This page makes no claim of an actual FE solution or measured result. The diagram shows the verification workflow; the curve is analytical beam theory only. Preserve solver and test records when those data are produced.

From physical robot to mathematical model

A model combines chosen variables, physical laws and explicit assumptions to answer a question.

1 Configuration, state and input

Configuration q specifies geometric arrangement. Position alone cannot determine the future of a second-order mechanical model; velocity is also needed. Thus a fixed-base arm commonly uses state x = (q, q̇). Input u depends on the physical interface: joint torque, wheel speed and motor voltage define different models. Sensor output need not reveal the entire state.

$$\dot{\mathbf{x}} = f(\mathbf{x}, \mathbf{u}), \qquad \mathbf{y} = h(\mathbf{x})$$

2 Kinematics and dynamics ask different questions

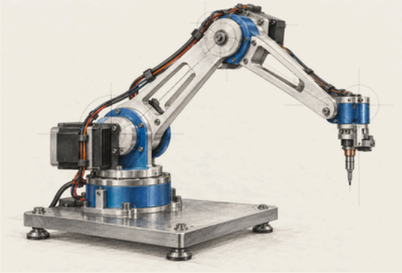
Kinematics computes tool pose from joint angles or motion from velocities without solving the required forces. Dynamics relates force to acceleration through mass and inertia. Flexibility, friction, cable compliance and drive delay may be omitted initially, but those omissions must be checked against measured behavior.

$$\mathbf{p}_E = f_K(\mathbf{q}), \qquad M(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{h}(\mathbf{q}, \dot{\mathbf{q}}) = \boldsymbol{\tau}$$

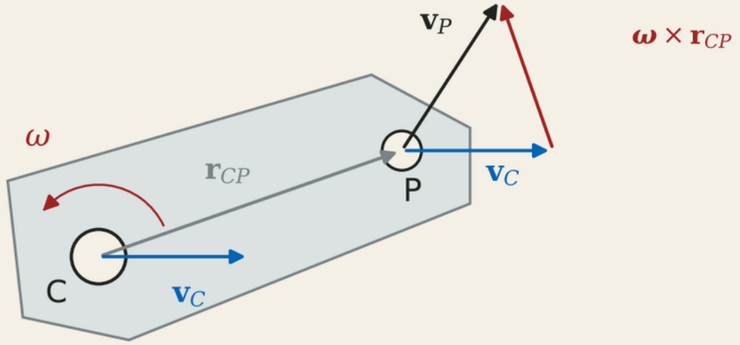
3 Different point velocities on one rigid body

Velocity at point P of a vehicle or robot link combines reference-point velocity at C with the rotational contribution. Express all vectors in the same frame. A sensor 0.4 m ahead of C has 0.2 m/s relative lateral velocity when the body rotates at 0.5 rad/s. Changing reference point changes coordinate equations, not physical motion. The vector rCP points from C to P; reversing it reverses the rotational contribution. In pure translation, ω=0 and all points share one velocity. In rotation, points farther from the axis move faster; the relation covers both cases.

$$\mathbf{v}_P = \mathbf{v}_C + \boldsymbol{\omega} \times \mathbf{r}_{CP}$$



A physical robot is reduced to a rigid-link model. The diagram shows velocity differences between two points of one body.



All vectors expressed in one frame

Assumptions and interpretation

General 3D configuration includes orientation. A rotation matrix or unit quaternion uses redundant coordinates; rotational freedom remains three-dimensional. A finite ICR applies to planar motion with nonzero angular rate; a general spatial screw motion need not have a stationary rotation center.

A shared modeling language for robots

Equations may share a form while states, inputs, frames and contact constraints differ by platform.

1 Coordinate frames are part of the model

The world frame observes motion externally; the body frame moves with the robot. Use world Z upward, body x forward and body y left in this lecture. R maps body-frame vectors to world axes. An IMU may measure in body axes while mapped position uses world axes; the numerical representations cannot be added directly. The columns of R are the body unit axes expressed in world coordinates. Orthogonality preserves lengths and angles, while determinant +1 excludes reflection. Consequently, nine arbitrary numbers cannot be used as a valid orientation matrix.

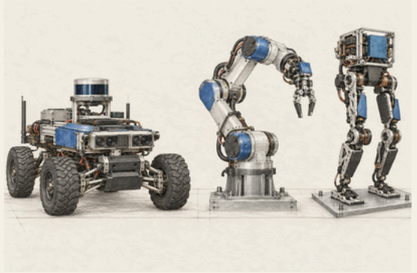
$$^{W}\mathbf{v} = R\,^{B}\mathbf{v}, \qquad R^{\mathsf{T}}R = I, \quad \det R = 1$$

2 Preserve the physical meaning of inputs

A mobile robot may use speed commands u = (v, r); an arm may use joint torques u = τ. Humanoid joint motors do not directly command pelvis position; ground reactions mediate motion. Four drone rotors produce collective thrust and three moments; treating two horizontal forces as independent extra inputs changes standard quadrotor physics.

3 A measurable, testable model contract

Specify initial state, input units, sample timing and predicted output for every model. Is vehicle steering measured at the road wheel or steering wheel? Is arm torque at the motor shaft or joint output? Is drone thrust in newtons or a normalized command? Reusing a symbol does not make these interfaces equivalent.



Physical views of three mechanical platforms; the table includes the aerial platform in the state–input comparison.

Platform	Example state	Input	Key constraint
Mobile robot	X, Y, ψ	v, r	Rolling
Robot arm	q, $\dot{q}$	τ	Fixed base
Humanoid	Base, q, velocities	Joint τ	Contact mode
Quadrotor	p, v, R, ω	$f_1 \ldots f_4$	Thrust direction

Assumptions and interpretation

The table gives model choices, not one mandatory state vector per platform. Assuming an ideal velocity controller permits a simpler model. Studying current or torque limits requires restoring controller and actuator dynamics.

Differential-drive kinematic model

The sum of two driven wheel speeds determines translation; their difference determines yaw.

1 Rolling and no lateral slip

Two equal-radius driven wheels share one axle; a passive caster supplies support without adding a fixed rolling direction. Let r_v be wheel radius, b wheel-center separation, and ωR, ωL wheel angular speeds positive forward. The reference is the axle midpoint. No lateral slip constrains its velocity to the body longitudinal direction. The mean wheel speed determines midpoint translation; their difference determines rotation. Dividing the peripheral-speed difference by b explains why a wider wheel separation produces a smaller yaw rate for the same speed difference.

$$v = \frac{r_w}{2}(\omega_R + \omega_L), \qquad r = \frac{r_w}{b}(\omega_R - \omega_L)$$

2 Motion in world coordinates

For body heading ψ, rotate velocity into world axes. The lateral constraint does not confine configuration to a surface f(X,Y,ψ) = 0; it restricts instantaneous velocities. The robot cannot command immediate sideways motion but may reach a laterally displaced pose by maneuvering. Instantaneous velocity freedom and finite-time reachability are different concepts.

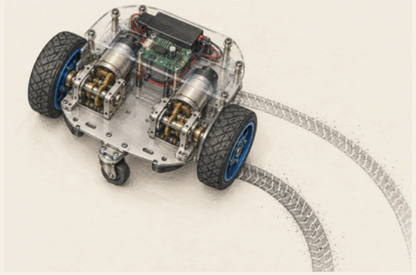
$$\dot{X} = v\cos\psi, \quad \dot{Y} = v\sin\psi, \quad \dot{\psi} = r$$

$$-\dot{X}\sin\psi + \dot{Y}\cos\psi = 0$$

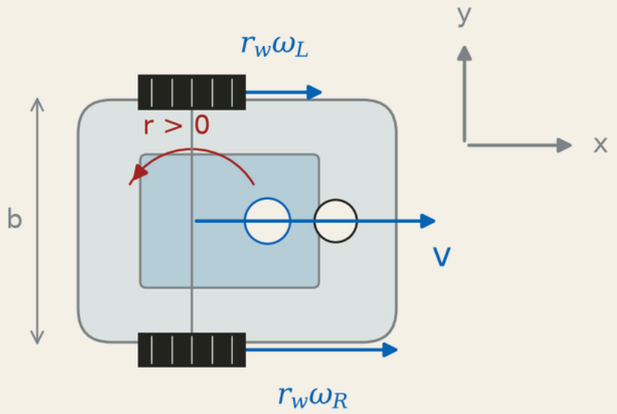
3 Inverse velocity mapping

Desired (v,r) directly determines the wheel speeds. With v = 0 and r ≠ 0 the wheels counter-rotate, permitting an in-place turn. Real drives impose wheel-speed, acceleration and torque limits. Turning a four-fixed-wheel skid-steer typically requires lateral slip; its physics differs from the ideal two-wheel model.

$$\omega_R = \frac{v + br/2}{r_w}, \qquad \omega_L = \frac{v - br/2}{r_w}$$



Physical two-wheel robot and exact axle geometry. Wheel directions, velocity arrows and positive yaw use one consistent convention.



Assumptions and interpretation

Assume level ground, a rigid body, equal effective radii and no-slip rolling. Encoder-derived speed equals ground speed only under these assumptions. Wear, deformation and slip create systematic or variable odometry error and require additional modeling.

Differential drive: a computed trajectory

Constant wheel speeds produce a straight path or a circular arc of constant curvature.

1 From wheel speeds to body motion

Let r_v = 0.10 m, b = 0.50 m, ωR = 12 rad/s and ωL = 8 rad/s. Their sum gives v = 1.00 m/s and difference gives r = 0.80 rad/s. Axle-midpoint turning radius is R = v/r = 1.25 m. Inner left and outer right wheel radii are 1.00 m and 1.50 m, consistent with circumferential speeds 0.8 and 1.2 m/s.

$$R_L = R - b/2, \qquad R_R = R + b/2$$

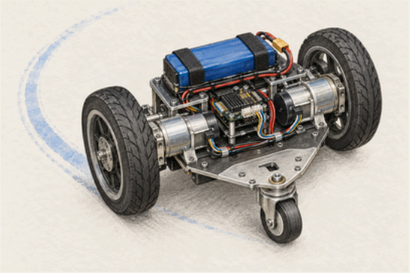
2 Integrate the continuous model exactly

Start at $(X_0, Y_0, \psi_0) = (0,0,0)$. With constant r, $\psi = rt$, so the cosine and sine velocities can be integrated over time. This accounts for changing heading throughout the interval. After one second, position is (0.897,0.379) m and heading 0.8 rad. These are ideal-model predictions, not measurements. X and Y are displacement components, not independent copies of travelled distance vt. Changing heading continuously rotates these velocity components during integration. A circular path thus shows that constant wheel speeds do not imply a constant world-velocity vector.

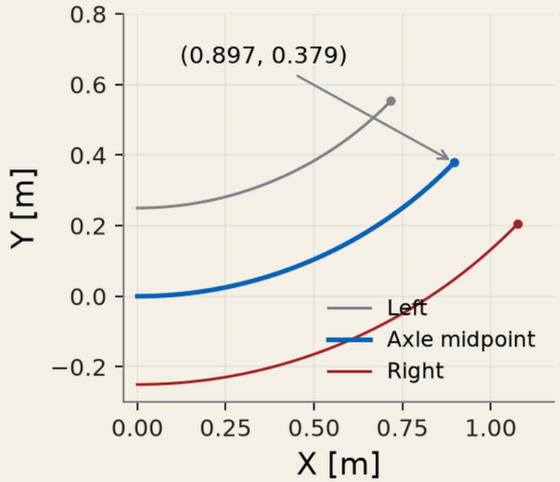
$$X(t) = \tfrac{v}{r}\sin(rt), \quad Y(t) = \tfrac{v}{r}[1 - \cos(rt)], \quad \psi(t) = rt$$

3 Check the physical result

The faster right wheel must turn the vehicle left; r > 0 and Y > 0 confirm that. Path length vt = 1 m exceeds the straight distance between endpoints. As r → 0, sin(rt)/r → t and Y → 0, recovering straight motion continuously. Numerical implementations use the limiting form near zero yaw rate.



The plot comes from the analytical solution. Thin curves track inner/outer wheels; the bold curve tracks the axle midpoint.



Assumptions and interpretation

The example assumes constant realized wheel speeds. Motor acceleration, saturation and contact slip change the startup transient. Validate odometry using timestamped heading, inner/outer wheel distances and external position measurements, not only the final point.

Autonomous vehicle: kinematic bicycle model

Steering determines path curvature through the no-slip rolling constraint.

1 Choose the reference point

Represent the front and rear wheel pairs of a planar rigid vehicle by two equivalent wheels. (X, Y) locates the rear-axle midpoint in the world; ψ is body heading relative to the X axis. L is wheelbase [m], δ is equivalent front-wheel steering angle [rad], and v is rear-axle longitudinal speed [m/s]. δ is not the steering-wheel angle.

$$\mathbf{x} = (X, Y, \psi)^{\mathsf{T}}, \qquad \mathbf{u} = (v, \delta)^{\mathsf{T}}$$

2 From rolling constraint to equations

With no lateral slip, rear-axle velocity is parallel to the body. The front axle lies L ahead, so its body-frame velocity is (v, Lψ̇). Its velocity component normal to the steered wheel must vanish. This directly relates steering to yaw rate, without a small-angle approximation. The zero constraint concerns velocity sideways to the steered wheel, not sideways to the body. The front point can therefore have nonzero Lψ̇. Steering aligns its rolling direction with the resultant velocity so that lateral wheel slip is absent.

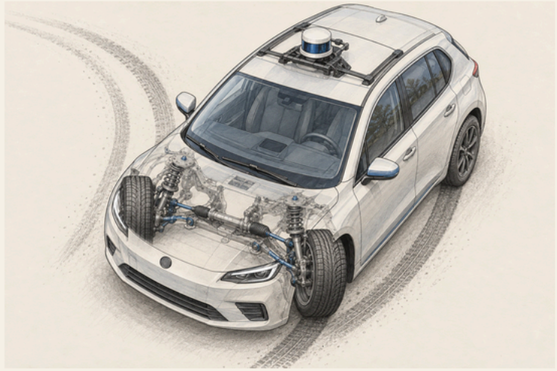
$$-v\sin\delta + L\dot{\psi}\cos\delta = 0$$

$$\dot{X} = v\cos\psi, \quad \dot{Y} = v\sin\psi, \quad \dot{\psi} = \tfrac{v}{L}\tan\delta$$

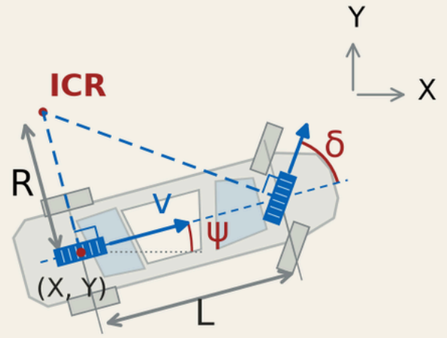
3 Geometry and a numerical check

Normals to the wheel planes intersect at the instantaneous center of rotation. For the rear-axle circular path, Rψ̇ = v. With L = 2.50 m, δ = 10° and v = 5.00 m/s, R = 14.18 m, curvature is 0.0705 m⁻¹ and ψ̇ = 0.3527 rad/s. Doubling speed at the same steering angle leaves geometric radius unchanged and doubles yaw rate.

$$R = \frac{L}{\tan\delta}, \qquad \kappa = \frac{\tan\delta}{L}, \qquad \dot{\psi} = v\kappa$$



Physical vehicle above; exact geometry of the two equivalent wheels below. ICR: instantaneous center of rotation. δ is enlarged for legibility.



Model scope

Assume level ground, a rigid body and negligible tire slip. Forces, mass, tire grip and steering-actuator lag are not computed here. If speed and steering are not applied instantaneously, add v and δ as states and use acceleration and steering rate as inputs. Choosing the center of mass as reference also changes the position equations.

S61 · CommonRoad Vehicle Models, §5, Fig. 1, eq. (6); original derivation and numerical example.

Geometric derivation of the bicycle model

Lateral no-slip conditions at front and rear wheels

1 Choose the rear axle as reference

Use body x forward and y left. Rear-axle-center velocity is (v,0), with yaw rate r. The front axle is L meters ahead, so rigid-body kinematics gives its velocity as (v,Lr). The front-wheel heading relative to the body is δ.

$$\mathbf{v}_f = \mathbf{v}_r + \boldsymbol{\omega} \times (L, 0, 0)^T = (v, Lr, 0)^T$$

2 Project onto the front-wheel normal

The front-wheel lateral normal is (−sinδ,cosδ). No slip requires zero velocity along it. Expanding the dot product and solving for r yields the tangent relation. Here v is rear-axle longitudinal speed, not the magnitude of CoM velocity. The normal–velocity dot product measures the component sideways to the wheel. Setting it to zero permits motion only along the wheel plane. The tangent term is therefore not a memorized vehicle formula; it is the ratio of two perpendicular velocity components.

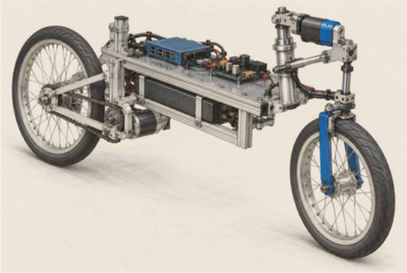
$$-v\sin\delta + Lr\cos\delta = 0 \quad \Rightarrow \quad r = \frac{v}{L}\tan\delta$$

3 Transform to world coordinates

With body heading ψ, rear-axle world velocities are v cosψ and v sinψ. For a CoM lr ahead of the rear axle, body velocity is (v,lr r), giving β=atan(lr tanδ/L). With L=2.5 m, v=5 m/s and δ=10°, r=0.3527 rad/s.

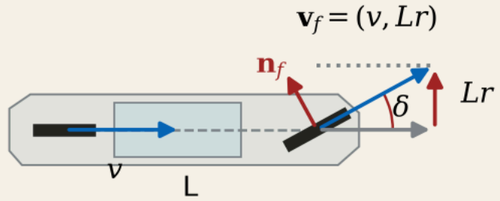
$$\dot{X} = v\cos\psi, \quad \dot{Y} = v\sin\psi, \quad \dot{\psi} = r$$

$$\beta = \arctan\left(\frac{l_r}{L}\tan\delta\right)$$



The exact diagram shows both front-velocity components and the wheel normal in the same body frame.

$$\mathbf{n}_f^T \mathbf{v}_f = -v\sin\delta + Lr\cos\delta = 0$$



Assumptions and physical limits

Assume pure rolling, level ground and rigid geometry. Changing speed definition for a CoM-based model requires transforming the equations consistently.

Bicycle model at the center of mass

The same rigid motion has different position and velocity equations at different reference points.

1 Heading and direction of travel

Let the center of mass C lie ℓ_r ahead of the rear axle and ℓf behind the front axle, with $L = \ell f + \ell_r$. Body heading is ψ; velocity at C makes angle β with the body. Here v is the magnitude of center-of-mass velocity. Body sideslip angle $\beta = \operatorname{atan2}(v_y, v_x)$ can be nonzero geometrically even when both wheels roll without lateral slip. The rear axle and centre of mass belong to one rigid vehicle but have different velocity vectors during turning. β captures that geometric difference. Using a centre-of-mass reference does not itself introduce tyre slip; first specify which point's velocity is being described.

$$v_x = v\cos\beta, \quad v_y = v\sin\beta, \quad \tan\beta = \frac{\ell_r}{L}\tan\delta$$

2 Derive from rigid-body velocity

Rear-axle lateral velocity is zero. Since C lies ℓ_r ahead, $v_y = \ell_r\dot{\psi}$. Front-axle lateral velocity is $v_y + \ell f\dot{\psi} = L\dot{\psi} = v_x\tan\delta$. These constraints determine β and $\dot{\psi}$. Global velocity points at $\psi + \beta$; using ψ alone in the position update would be incorrect.

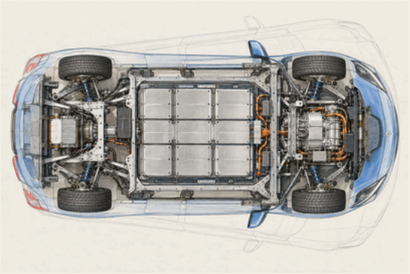
$$\beta = \arctan\left(\frac{\ell_r}{L}\tan\delta\right), \quad \dot{\psi} = \frac{v\cos\beta}{L}\tan\delta$$

$$\dot{X}_C = v\cos(\psi+\beta), \quad \dot{Y}_C = v\sin(\psi+\beta)$$

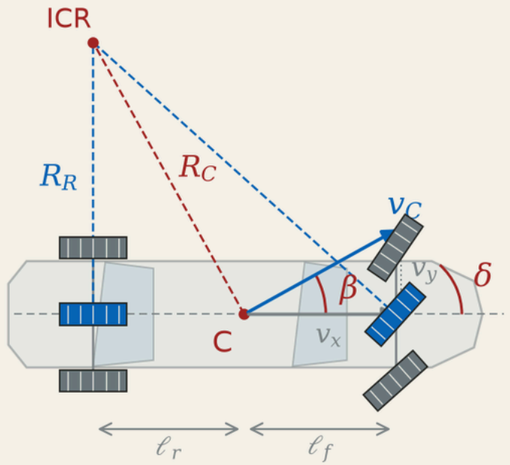
3 Same vehicle, consistent speed definition

With L = 2.5 m, ℓ_r = 1.25 m and $\delta = 10°$, $\beta = 5.04°$. A rear-axle speed of 5 m/s gives center-of-mass speed 5.019 m/s; both models predict $\dot{\psi} = 0.3527$ rad/s. Supplying the same numerical speed under different definitions creates a small systematic inconsistency.

$$X_C = X_R + \ell_r\cos\psi, \qquad Y_C = Y_R + \ell_r\sin\psi$$



The cutaway establishes the physical reference; the redraw shows velocity components at C and exact turning geometry.



Assumptions and interpretation

Planar, no-slip model with front steering. With rear steering, $\tan\beta = (\ell f \tan\delta_r + \ell_r \tan\delta f)/L$ and $\dot{\psi} = v\cos\beta(\tan\delta f - \tan\delta_r)/L$. β is exaggerated in the diagram. Geometric body angle β and elastic tire slip angle α are different quantities.

Ackermann steering geometry

No-slip rolling in a turn requires different steering angles at the inner and outer front wheels.

1 A common instantaneous center

In planar turning, each contact-point velocity is perpendicular to the line joining it to the ICR. With unsteered rear wheels, the ICR lies on the rear-axle line. R is the rear-axle midpoint radius, B track width and L wheelbase. The inner front wheel follows a tighter turn and requires a larger steering angle. Extend the line normal to each wheel's rolling direction toward the ICR. That line is a radius; actual motion is tangent to it. Reading the drawing this way explains why a larger steering angle corresponds to a smaller path radius.

$$\tan\delta_{\text{in}} = \frac{L}{R - B/2}, \qquad \tan\delta_{\text{out}} = \frac{L}{R + B/2}$$

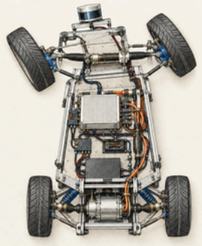
2 Equivalent wheel and physical linkage

The bicycle-model angle δ is defined by a virtual centerline wheel, not by the arithmetic mean of the two front-wheel angles. A trapezoidal steering linkage approximates the Ackermann relation over a working range. A real linkage need not satisfy the exact geometry at every angle; compliance and tire forces modify its behavior.

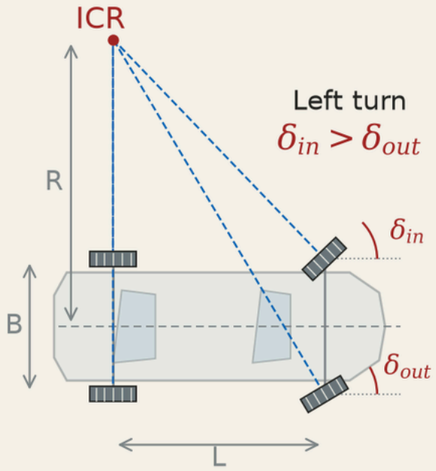
$$\cot\delta_{\text{out}} - \cot\delta_{\text{in}} = \frac{B}{L}, \qquad \delta = \arctan\frac{L}{R}$$

3 Compute angles using inverse tangent

For L = 2.5 m, B = 1.6 m and R = 8 m, inner, outer and equivalent steering angles are 19.15°, 15.86° and 17.35°. The approximation δ ≈ L/R requires small angles in radians. Inner/outer labels are independent of turn direction; left/right roles swap when the turn reverses.



Geometry is recomputed from the source. Inner and outer wheel angles differ; B and L denote different dimensions.



Assumptions and interpretation

The diagram shows a left turn; all wheel normals intersect at the same ICR. Assume R > B/2. This is a low-slip geometric condition, not a complete steering design criterion at high lateral acceleration. Steering-wheel ratio and actuator limits require separate models.

Kinematic model: validity and discretization

Correct geometry does not guarantee that available forces can realize the motion.

1 No slip is a modeling assumption

Kinematic models are often useful for parking and low-lateral-acceleration tracking. Yet a vehicle can slip on ice even at low speed. Lateral acceleration demand grows as v²κ; tire forces, load transfer and body dynamics matter in faster turns. There is no universal speed threshold: validity depends on surface, tires, curvature and required prediction accuracy.

$$a_y \simeq v^2\kappa = \frac{v^2}{L}\tan\delta$$

2 From continuous dynamics to a time step

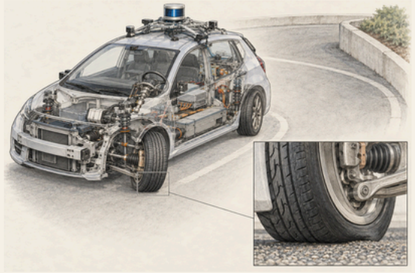
Forward Euler evaluates derivatives at the start of each step, here at the rear axle. Even with constant v and δ, heading changes, so the straight Euler step differs from the true circular arc. Smaller steps reduce integration error but do not remove tire-model mismatch. Replacing tanδ with δ is a separate small-angle approximation. Advancing along the initial heading gives an outward position error relative to the curved path. This numerical error differs from physical tyre-slip error. Reducing Δt addresses the former; evaluating an unsuitable model more frequently does not make its physics correct.

$$X_{k+1} = X_k + \Delta t\, v_k \cos\psi_k$$

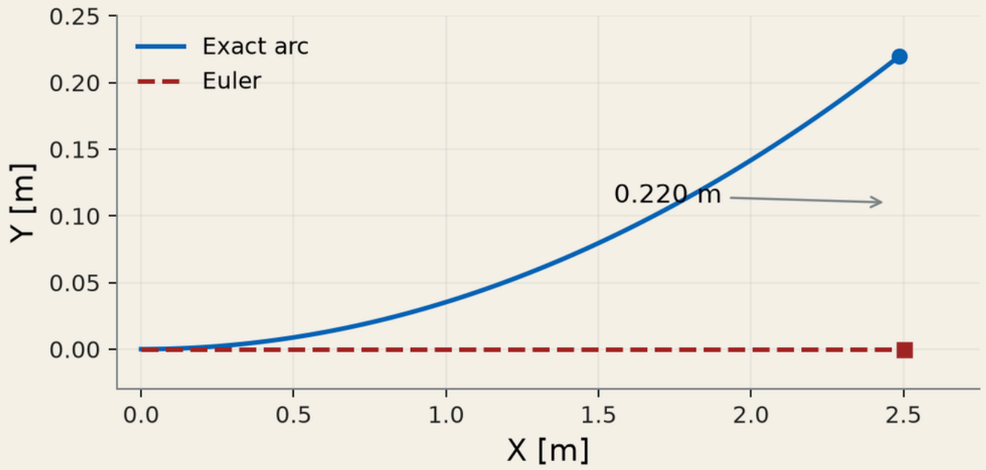
$$Y_{k+1} = Y_k + \Delta t\, v_k \sin\psi_k, \quad \psi_{k+1} = \psi_k + \Delta t \frac{v_k}{L}\tan\delta_k$$

3 A one-step error example

Start from (0, 0, 0) with L = 2.5 m, v = 5 m/s, δ = 10° and Δt = 0.5 s. Euler gives (2.500, 0) m; the exact constant-input arc gives (2.487, 0.220) m. Both give heading 0.1763 rad. The lateral position error comes from the straight-line position update.



The physical inset shows tire contact; the plot compares Euler with the exact arc under identical inputs.



Assumptions and interpretation

The plot compares two integrations of the same model, not experimental data. For constant r = v tanδ/L, ΔX = (v/r)[sin(ψ+rΔt) − sinψ] and ΔY = (v/r)[cosψ − cos(ψ+rΔt)]. The limit r → 0 gives straight motion. Verify delays and steering-rate limits separately.

Tire contact and longitudinal slip

Traction arises from rubber deformation and local friction limits within the contact patch.

1 The tread-block model

A loaded tire contacts the road over a finite patch. A tread element initially adheres to the surface and shears elastically as the carcass moves. It starts sliding when local shear reaches the friction limit. Even at small overall slip, one part of the patch may adhere while another slides; partial adhesion and complete sliding are different regimes. Total force sums local shear stress times small contact areas, giving Pa×m²=N. Normal pressure p and friction coefficient determine the local bound. Contact-area size alone is therefore insufficient to infer the available force.

$$|\tau_x(\xi)| \le \mu p(\xi), \qquad F_x = \int_A \tau_x \, dA$$

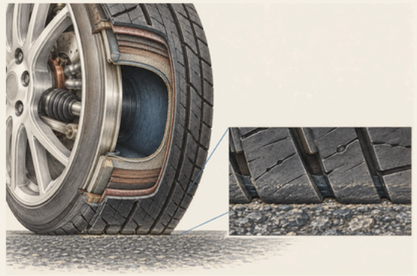
2 Define a slip ratio

For forward motion $v_x > 0$, use $s = (R_e\omega - v_x)/v_x$. R_e is effective rolling radius, ω wheel angular speed, and v_x longitudinal wheel-center speed. Positive s denotes traction and negative s braking. This dimensionless ratio differs from the circumferential speed difference used intuitively in the source. Check denominator and sign conventions across software.

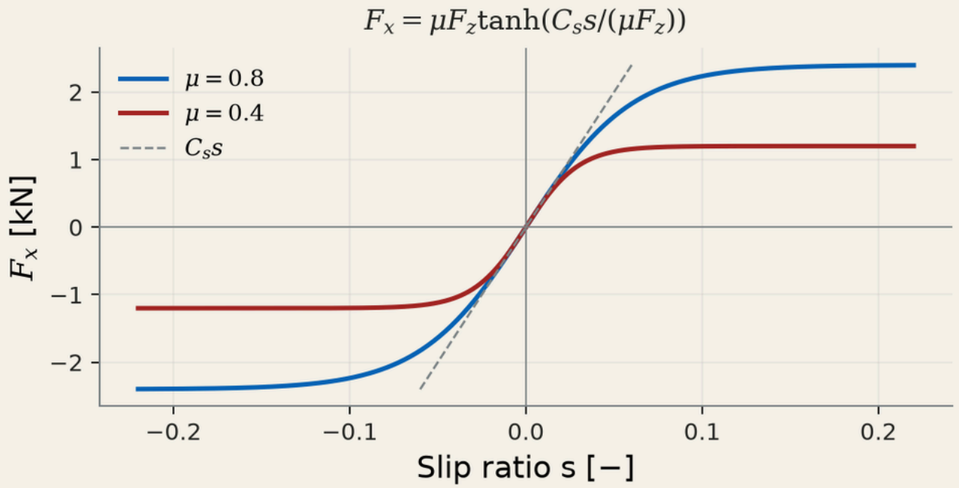
$$s = \frac{R_e\omega - v_x}{v_x}, \qquad F_x \simeq C_s s \quad (|s| \text{ small})$$

3 Linear region and saturation

For R_e = 0.30 m, ω = 35 rad/s and v_x = 10 m/s, s = 0.05. With C_s = 40 kN, the linear estimate is 2 kN. For Fz = 3 kN and μ = 0.8, a simple friction bound is 2.4 kN. The linear law cannot grow indefinitely; post-peak behavior depends on measured tire and surface properties.



Contact cutaway is qualitative; force curves use a quantitative saturation model. Initial stiffness and friction bound are separate parameters.



Assumptions and interpretation

τ_x [Pa] is local shear stress, p [Pa] contact pressure and C_s [N] slip stiffness. Curves are explicitly defined teaching models, not measurements. The ratio becomes singular as $v_x \to 0$; standstill needs a regularized contact model. Unchanged initial stiffness across friction levels is not a universal tire law.

Tire slip angle and lateral force

A tire produces lateral force when wheel orientation differs from its direction of travel.

1 Fix the sign convention first

Choose body x forward, y left, and positive yaw rate r = ψ̇ counterclockwise. Define α as wheel heading minus wheel-center velocity direction. Positive α produces positive force to the wheel's left. Some sources reverse α and write Fy = −Cα; equations must be compared together with their definitions. The front velocity contains vy+ℓfr and the rear velocity vy−ℓrr because the axles lie on opposite sides of the centre of mass. Their rotational contributions have opposite signs. No rear steering term appears because the rear wheel is unsteered.

$$\alpha_f = \delta - \arctan\frac{v_y + \ell_f r}{v_x}, \qquad \alpha_r = -\arctan\frac{v_y - \ell_r r}{v_x}$$

2 Linear law at small slip angle

For small lateral tread deformation, axle force is approximately proportional to slip angle. Here Cf and Cr are total axle cornering stiffnesses [N/rad], not single-tire values. This avoids a factor-of-two error. At small angles, replace arctangent with its argument; project front tire force to the body axis using cosδ.

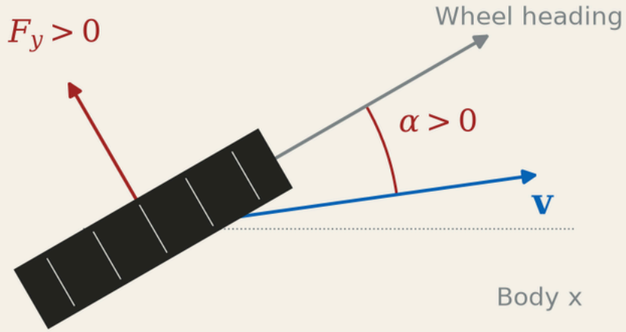
$$F_{yf} \simeq C_f\left(\delta - \frac{v_y + \ell_f r}{v_x}\right)$$

$$F_{yr} \simeq -C_r\frac{v_y - \ell_r r}{v_x}$$

3 Check force and units

For axle stiffness C = 80 kN/rad and α = 2° = 0.03491 rad, Fy ≈ 2.79 kN. Multiplying 80 000 directly by the angle in degrees gives a factor-57.3 error. At larger slip the linear model may exceed available grip; load, temperature, pressure and road surface also affect the curve.

The cutaway illustrates deformation. The exact vector diagram distinguishes wheel direction, velocity direction, α and force.



$$\alpha = \delta - \operatorname{atan2}(v_y, v_x)$$

Assumptions and interpretation

α is the tire heading–velocity difference; β is the angle between body heading and center-of-mass velocity. The kinematic bicycle assumes αf = αr = 0 while geometric β may remain nonzero. Angles are exaggerated in the diagram; positive force agrees with the stated α definition.

Combined braking and cornering: grip limit

Longitudinal and lateral forces share the finite capacity of the same tire–road contact.

1 The friction-circle approximation

For fixed normal load Fz and isotropic friction coefficient μ, horizontal force magnitude is bounded by μFz. Points inside the circle are feasible under this simple model; points outside are not. This inequality is not a force-generation law: a tire model is still needed to predict force from slip. The circle shows that a tyre does not have two independent force reserves. Longitudinal and lateral force share one contact capacity. Increasing one reduces the bound on the other; independent box limits can admit physically impossible commands near their corners.

$$F_x^2 + F_y^2 \le (\mu F_z)^2, \qquad |F_y| \le \sqrt{(\mu F_z)^2 - F_x^2}$$

2 Braking consumes lateral capacity

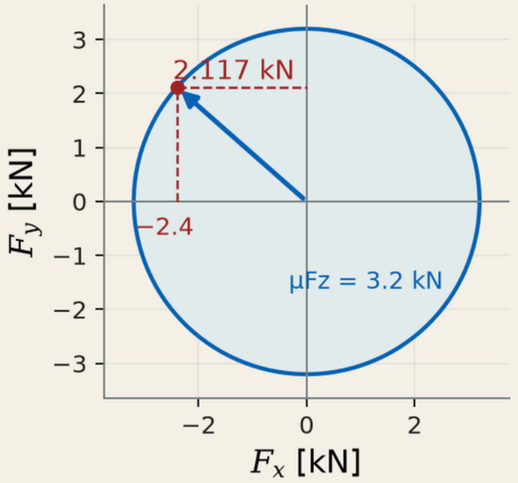
With Fz = 4 kN and μ = 0.8, the total bound is 3.2 kN. Requesting 2.4 kN braking leaves at most 2.117 kN lateral capacity. Subtracting magnitudes linearly would incorrectly give 0.8 kN; orthogonal components combine through squared magnitudes. A saturated tire cannot add braking while retaining the same lateral force.

$$|F_y|_{\max} = \sqrt{3.2^2 - 2.4^2} = 2.117 \text{ kN}$$

3 Connecting planning and control

On level ground, total normal load approximately mg gives the coarse vehicle-level bound $a_x^2 + a_y^2 \le (\mu g)^2$. Check planned speed and curvature together using $a_y \approx v^2\kappa$. ESC changes yaw moment through wheel-specific braking forces; it cannot increase the tire's physical grip capacity.

The quantitative plot uses fixed Fz = 4 kN and μ = 0.8. Negative Fx denotes braking; positive Fy denotes leftward force.



Assumptions and interpretation

Real tire limits may be elliptical and μ need not be constant. Load transfer, axle distribution and aerodynamic forces change individual wheel capacity. The circle is a useful first combined-slip check, not a substitute for measured tire models and engineering margin.

Dynamic bicycle model: force and moment

Motion is computed from tire forces and inertia rather than imposed no-slip direction constraints.

1 Acceleration in rotating body axes

Express center-of-mass velocity in body axes as (v_x, v_y), with yaw rate $r = \dot{\psi}$. Because the axes rotate, lateral acceleration is not just $\dot{v}_y$; the transport term $r\,v_x$ must be added. It follows by differentiating the world-frame velocity vector. Vehicle mass m has units kg; Iz is yaw inertia about the center of mass [kg·m²]. During steady-radius turning, body lateral velocity may remain constant while world velocity changes direction. The term rvx carries this centripetal effect. Equating $\dot{v}y$ with ay would therefore misapply Newton's law even in a steady turn.

$$a_y = \dot{v}_y + r v_x, \qquad m(\dot{v}_y + r v_x) = F_{yf}\cos\delta + F_{yr}$$

2 Moment arms of tire forces

The front axle lies ℓf ahead of C and the rear axle ℓ_r behind. A positive front lateral force produces positive yaw moment; a positive rear force produces the opposite moment. With small δ, cosδ ≈ 1 gives the equations below. Only lateral tire forces are retained; longitudinal speed is an externally maintained operating parameter.

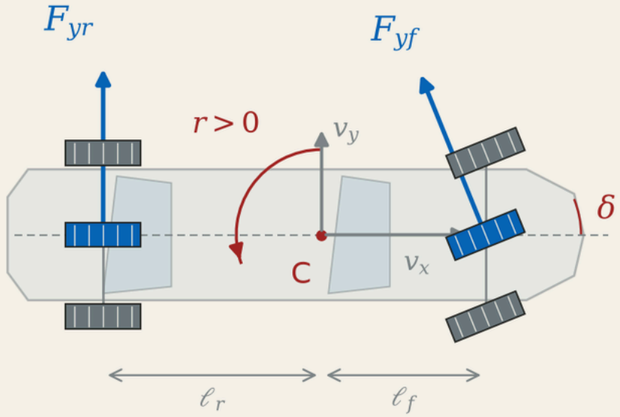
$$\dot{v}_y = \frac{F_{yf} + F_{yr}}{m} - v_x r$$

$$\dot{r} = \frac{\ell_f F_{yf} - \ell_r F_{yr}}{I_z}$$

3 Check signs with numbers

Let m = 1500 kg, Iz = 2500 kg·m², ℓf = 1.2 m, ℓ_r = 1.6 m and Cf = C_r = 80 kN/rad. At v_x = 15 m/s, v_y = 0.2 m/s, r = 0.1 rad/s and δ = 0.03 rad, the linear tire model gives Fyf = 693 N and Fyr = −213 N. Thus $\dot{v}_y$ = −1.18 m/s² and $\dot{r}$ = 0.469 rad/s², while physical lateral acceleration is ay = 0.32 m/s².

The free-body diagram preserves signs about C. Blue arrows are forces; the dashed line is the body axis. Angles and forces are schematic.



Assumptions and interpretation

Assume a planar rigid body, small steering/slip angles, fixed positive v_x and linear tires. Load transfer, roll, pitch and combined slip are omitted. General 3D rotation requires τ = Iω̇ + ω × (Iω); planar rotation about one vertical axis permits the simpler Izṙ relation.

Vehicle acceleration in a rotating frame

Origin of the speed × yaw term in lateral acceleration

1 A vector derivative differs from coordinate derivatives

Write world velocity as vW=R vB. The product rule differentiates both R and vB. Using Ṙ=R[ω]× and expressing world acceleration in the body frame gives aB=v̇B+ω×vB. The extra term accounts for rotating axes. The term ṘvB describes change caused by rotating axes even when the components are constant. Rv̇B describes changes in the components themselves. Both contribute to one physical acceleration, explaining why constant body-velocity components do not imply zero acceleration.

$$\mathbf{a}_W = \dot{R}\mathbf{v}_B + R\dot{\mathbf{v}}_B, \quad \mathbf{a}_B = \dot{\mathbf{v}}_B + \boldsymbol{\omega} \times \mathbf{v}_B$$

2 Expand the planar cross product

For ω=(0,0,r) and vB=(vx,vy,0), the cross product is (−rvy,rvx,0). Newton’s law uses physical acceleration, not coordinate derivatives alone. Transform front lateral tire force with cosδ and use moment arm lf in yaw balance.

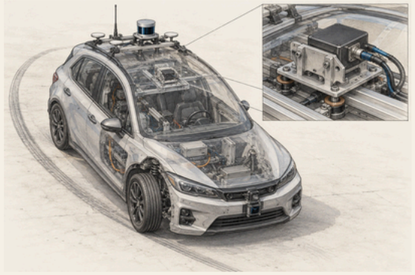
$$m(\dot{v}_y + r v_x) = F_{yf}\cos\delta + F_{yr}$$

$$I_z\dot{r} = l_f F_{yf}\cos\delta - l_r F_{yr}$$

3 Constant coordinates need not mean zero acceleration

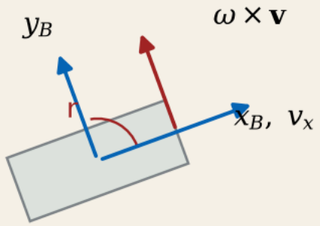
In a steady turn let vx=10 m/s, vy=0, r=0.2 rad/s and v̇y=0. Lateral acceleration remains 2 m/s². A 1000 kg vehicle needs 2000 N total lateral force. Constant yaw rate can require zero net yaw moment while lateral force remains nonzero.

$$a_y = 0 + 10(0.2) = 2 \text{ m/s}^2$$



Changing direction of the velocity vector creates acceleration even with constant body components.

$$a_y = \dot{v}_y + r v_x$$



Assumptions and physical limits

Signs follow x forward, y left, z up. These two balance equations do not replace tire laws or longitudinal dynamics.

State-space representation of vehicle dynamics

Substituting tire laws into force balance yields a linear model for control design.

1 Define states and input

Around fixed longitudinal speed V > 0, choose states z = $(v_y, r)^T$ and front-wheel angle δ as input. With fixed mass, geometry and axle stiffnesses, A(V) and B are constant. Changing V changes the matrix, so one time-invariant model does not cover all speeds. Heading ψ is integrated separately through $\dot{\psi} = r$.

$$\dot{\mathbf{z}} = A(V)\mathbf{z} + B\delta$$

$$\dot{v}_y = a_{11}v_y + a_{12}r + b_1\delta, \quad \dot{r} = a_{21}v_y + a_{22}r + b_2\delta$$

2 Read coefficients from physical terms

Negative diagonal terms reflect tire response to lateral velocity and yaw. The −V term in a_{12} comes from rotating coordinates, not tire stiffness. B describes steering effects on front-axle force and moment. Each row must have consistent units: m/s² and rad/s², respectively. The first row gives lateral-velocity rate and the second yaw-rate change. Each column describes one state's influence on both equations. Off-diagonal coefficients therefore quantify coupling between lateral motion and rotation, rather than being merely numerical entries.

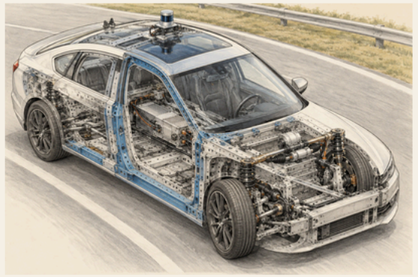
$$a_{11} = -\frac{C_f + C_r}{mV}, \quad a_{12} = \frac{C_r\ell_r - C_f\ell_f}{mV} - V$$

$$a_{21} = \frac{C_r\ell_r - C_f\ell_f}{I_z V}, \quad a_{22} = -\frac{C_f\ell_f^2 + C_r\ell_r^2}{I_z V}$$

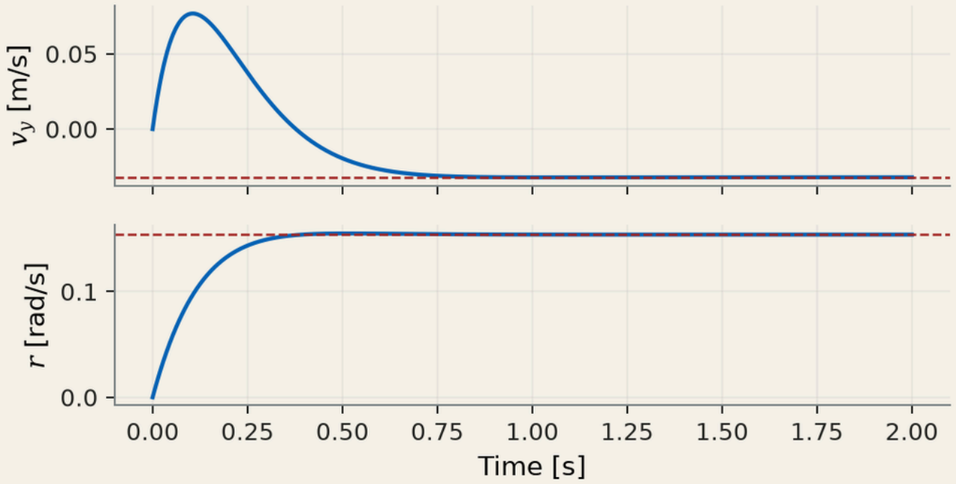
$$b_1 = \frac{C_f}{m}, \qquad b_2 = \frac{C_f\ell_f}{I_z}$$

3 Use it for control and estimation

Using the preceding parameters at V = 15 m/s gives $A_{11} = -7.111$, $A_{12} = -13.578$, $A_{21} = 0.853$ and $A_{22} = -8.533$. The plot computes the response from zero state to a 2° steering step. This linear model supports yaw-rate control or observer design; the plot is not a vehicle experiment.



Original computation: 2° steering step at V = 15 m/s. Panels have separate units; dashed lines indicate steady-state values.



Assumptions and interpretation

Adding global position through $\dot{X} = V\cos\psi - v_y\sin\psi$ and $\dot{Y} = V\sin\psi + v_y\cos\psi$ makes the combined model nonlinear. Terms 1/V become singular as V → 0; use a kinematic transition or validated low-speed regularization. The two-state form is equivalent to the source's (v_y, ψ, r) representation.

Vehicle models: detail and validity

Model selection means retaining the physics needed for the question, not maximizing equation count.

1 A different question at each level

A point model tracks position without necessarily representing body heading. A unicycle describes motion through forward speed and yaw rate. A kinematic bicycle adds steering and wheelbase. A dynamic bicycle adds mass, inertia and tire slip to predict lateral transients. A full vehicle model may include wheel rotation, suspension, roll and load transfer.

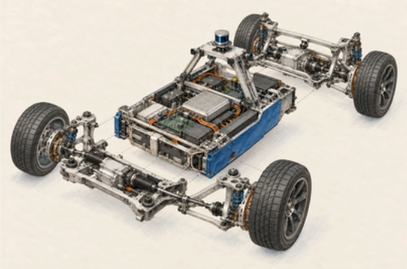
2 Detail costs parameters and validation

More detail is not automatically more accurate. Unknown cornering stiffness or incorrect inertia can undermine added dynamics. Estimate parameters from measurements and validate across intended speeds and maneuvers. A linear dynamic bicycle without tire saturation does not predict limit handling correctly merely because it is called dynamic. Before calculating RMSE, specify the output: position error has units of metres, yaw-rate error rad/s. Combining unlike units without scaling is meaningless. A small average error can also hide brief but substantial limit violations.

$$\mathbf{e}_k = \mathbf{y}_k^{\text{meas}} - \mathbf{y}_k^{\text{model}}, \qquad \text{RMSE} = \sqrt{\frac{1}{N}\sum_{k=1}^{N} e_k^2}$$

3 Compare with consistent interfaces

Match reference points, speed definitions, input units and timestamps when comparing models. Track yaw rate, lateral acceleration and computation time alongside position error. Low training error is insufficient; validate on different speeds, surfaces and payloads. Combined tire forces can become decisive when braking and cornering occur together.



Read physical components alongside the model table: each row makes additional behavior computable.

Model	Represented effect	Omission / requirement
Point model	Position / route	Heading, tires
Unicycle	Heading / speed	Steering geometry
Kinematic bicycle	δ, L / maneuver	Force, tire slip
Dynamic bicycle	v_y, r / transient	Roll, load transfer
Full vehicle	Suspension / contact	Parameter validation

Assumptions and interpretation

The table indicates possible applications, not universal speed limits. Planning, tracking and limit handling have different accuracy and computation requirements. Start simple and add detail when measured residuals reveal missing physics. Unjustified complexity adds uncertain parameters.

Robot-arm kinematic model

Forward kinematics maps joints to pose; inverse kinematics seeks joints that realize a desired pose.

1 Fix the frame convention first

Let {0} denote the base and {E} the tool frame. T_0E maps homogeneous point coordinates from the tool frame into the base frame. Rotation R is dimensionless and translation p is in meters. A rotation preserves length and has determinant +1. Pose therefore includes orientation as well as three position coordinates.

$$^0\bar{p} = T_{0E}\,{}^E\bar{p}, \qquad R^TR = I, \quad \det R = 1$$

2 Compose transforms in the correct order

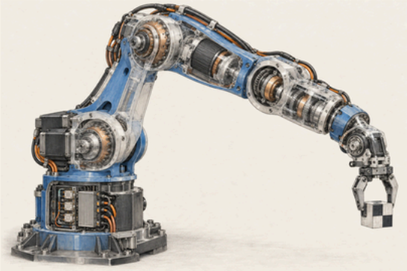
Each joint transform connects neighboring link frames. The rightmost transform acts first on a point, and matching intermediate indices check the order. Revolute coordinates are in radians and prismatic coordinates in meters. Combine fixed link geometry with joint motion in the correct order; rotations and translations generally do not commute. For example, T01T12 maps a point from frame two to frame one and then to the base. T12T01 is not the same chain. Compatible matrix dimensions alone do not guarantee that coordinate frames have been composed in a physically meaningful order.

$$T_{0E}(q) = T_{01}(q_1)T_{12}(q_2)\cdots T_{nE}(q_n)$$

3 Express a camera target in the arm base

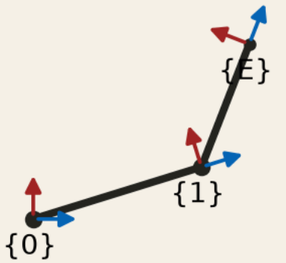
A point measured in the camera frame cannot be sent directly to arm inverse kinematics. Transform it into the base using known T_0C, then specify the grasp orientation separately. T(q) = Td may have zero, one or multiple solutions. Check any resulting joints against limits, self-collision, environmental collision and an accessible approach path.

$$^0\bar{p} = T_{0C}\,{}^C\bar{p}, \qquad T(q) = T_d$$



Physical arm architecture above; the base-to-tool frame chain below. Axes and transform directions follow one convention.

$$T_{0E} = T_{01}T_{1E}$$



Assumptions and physical limits

Kinematics describes motion geometry, not motor torque or link deformation. Calibration error can shift the physical tool pose even when the algebraic solution is exact.

Forward and inverse kinematics of a 2R arm

Two elbow postures can produce the same endpoint; verify every solution with forward kinematics.

1 Add the link vectors

Base x points right and y upward. q_1 is the first link angle from x; q_2 is the second link angle relative to the first. The second link's absolute angle is therefore q_1+q_2. Lengths L_1 and L_2 are in meters. Resolve each link vector into horizontal and vertical components to obtain the endpoint.

$$x = L_1\cos q_1 + L_2\cos(q_1 + q_2)$$
$$y = L_1\sin q_1 + L_2\sin(q_1 + q_2)$$

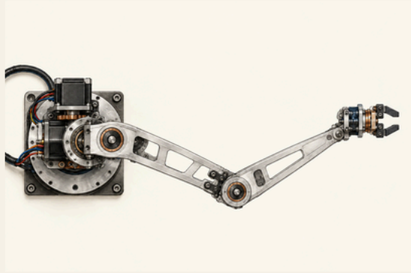
2 Cosine rule and the correct angle branch

The squared target distance is x^2+y^2. The cosine rule first gives $c_2 = \cos q_2$. A target with $|c_2|>1$ is unreachable. The two square-root signs select the elbow branches; atan2 preserves the quadrant. Obtain q_1 by subtracting the link-triangle angle from the target direction. Using atan2(y,x) rather than atan(y/x) preserves the correct half-plane when x is negative. Changing the square-root sign requires recomputing q1 as well as q2. Otherwise, joint angles from different inverse-kinematic branches can be mixed incorrectly.

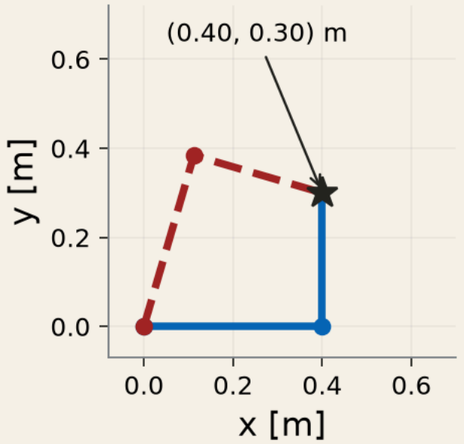
$$c_2 = \frac{x^2 + y^2 - L_1^2 - L_2^2}{2L_1L_2}, \quad q_2 = \operatorname{atan2}(\pm\sqrt{1 - c_2^2}, c_2)$$
$$q_1 = \operatorname{atan2}(y, x) - \operatorname{atan2}(L_2\sin q_2, L_1 + L_2\cos q_2)$$

3 Substitute both solutions

For $L_1 = 0.40$ m, $L_2 = 0.30$ m and target (0.40,0.30) m, c_2 is zero. Solutions are (0°,90°) and (73.74°,−90°). Both return the same position under forward kinematics, but the second-link orientations differ. This is position-only IK; imposing tool orientation as well makes arbitrary poses impossible with just two joints.



Both plotted arms use the same base, link lengths and target; axes are in meters.



Assumptions and physical limits

This is an ideal planar rigid 2R model without joint limits. Branches can merge at the boundary. Real branch selection also considers joint limits, cables, collisions and continuity from the previous posture.

Robot-arm velocity model

A Jacobian column is the endpoint velocity generated by unit speed at one joint alone.

1 Differentiate position

Endpoint position p(q) depends on time through the joint angles. The chain rule sums each joint contribution. A planar 2R position Jacobian is 2×2, with columns in meters per radian. Below, $s_{12} = \sin(q_1+q_2)$ and $c_{12} = \cos(q_1+q_2)$. Each column is tangent to the radius from its corresponding joint axis to the tool.

$$\dot{p} = J(q)\dot{q} = J_1\dot{q}_1 + J_2\dot{q}_2$$

$$J_1 = (-L_1 \sin q_1 - L_2 s_{12},\ L_1 \cos q_1 + L_2 c_{12})^T$$

$$J_2 = (-L_2 s_{12},\ L_2 c_{12})^T$$

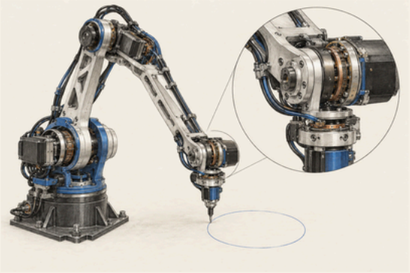
2 Evaluate the velocity contributions

For $L_1 = 0.40$ m, $L_2 = 0.30$ m and q = (0°,90°), $J_1 = (-0.30, 0.40)$ and $J_2 = (-0.30, 0)$. Moving only joint 1 at 1 rad/s gives endpoint velocity $(-0.30, 0.40)$ m/s. Moving both at 1 rad/s sums the contributions to $(-0.60, 0.40)$ m/s. The same joint speeds produce a different tool velocity at another posture.

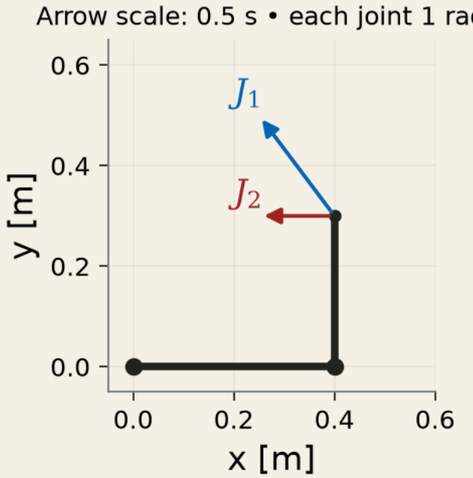
3 Force-torque mapping from virtual power

Equate power for tool force F exerted by the robot on the environment and the corresponding ideal joint torque τ. Equality for all virtual joint velocities gives $\tau = J^T F$. For F = (0,10) N, τ = (4,0) N·m. Choosing the environment's force on the robot reverses the equilibrium sign; gravity and friction must be added separately. The transpose follows from rearranging power: $F^T J\dot{q}$ becomes $(J^T F)^T \dot{q}$. Equality must hold for every virtual joint velocity, not just one selected motion. This explains why velocity and force mappings share the same geometric Jacobian.

$$F^T\dot{p} = \tau^T\dot{q} \quad \Rightarrow \quad \tau = J^T F$$



Blue and red arrows are computed tool velocities from individual unit joint speeds at the same posture.



Assumptions and physical limits

This J maps only 2D position velocity. A six-dimensional twist Jacobian also requires angular/linear ordering and the spatial/body frame convention.

Jacobian singularity

Losing an instantaneous motion direction is a rank loss in the velocity map, not a change in joint count.

1 When do the columns become dependent?

Expanding and simplifying the 2R Jacobian determinant gives $L_1L_2 \sin q_2$. At full extension or folding, $\sin q_2$ is zero and both columns become collinear. The endpoint can generate instantaneous velocity only perpendicular to the arm direction. At horizontal extension, finite joint speeds cannot create first-order horizontal tool velocity. A singularity does not mean that the robot cannot move at all; it loses an independent instantaneous motion direction. A finite manoeuvre that first bends the arm may restore other directions. This local Jacobian property is not a global workspace prohibition.

$$\det J = L_1 L_2 \sin q_2, \qquad q_2 = 0 \;\Rightarrow\; \operatorname{rank} J = 1$$

2 Map the joint-speed circle to an ellipse

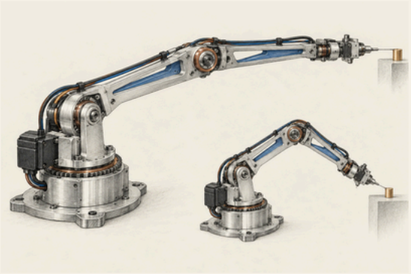
Map commands with joint-speed norm at most 1 rad/s through J. In the SVD, U gives task-velocity directions, V joint combinations, and singular values the gains. With $L_1 = 0.40$, $L_2 = 0.30$ m and $q_2 = 90°$, the determinant is 0.12 m². As q_2 approaches zero, the minor semiaxis vanishes and the ellipse collapses to a segment.

$$J = U\Sigma V^{T}, \qquad \|\dot{q}\|_2 \le 1$$

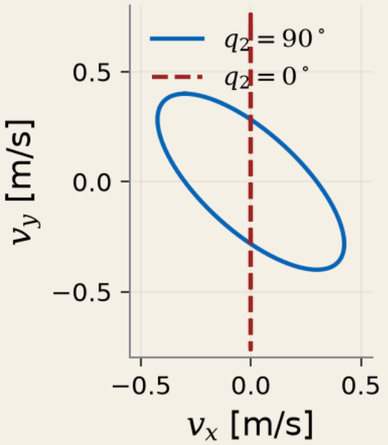
3 The cost of a damped inverse

Inverting a small singular value can demand very large joint speeds. Damping λ limits amplification but leaves task-velocity residual error. Choose λ consistently with task scaling and Jacobian units. Damping does not restore the missing physical motion direction; it provides a controlled approximation to an unattainable target.

$$\dot{q} = J^T(JJ^T + \lambda^2 I)^{-1} v_d$$



Both ellipses use the same speed norm and link lengths; full extension has zero horizontal semiaxis.



Assumptions and physical limits

The ellipse depends on the joint-speed norm assumption; independent motor-speed bounds produce a different polygon. Do not confuse instantaneous singularity with global reachability.

Robot-arm dynamic model

The same motion geometry requires different motor torques when mass distributions differ.

1 Start from kinetic and potential energy

Sum translational and rotational kinetic energy over the links; gravitational potential depends on center-of-mass height. The Lagrangian is $\mathcal{L} = T - V$. T and V are energies [J], and q contains joint coordinates. The Euler–Lagrange equation converts position and velocity dependence of energy into generalized force.

$$\mathcal{L} = T - V, \qquad \frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{q}_i} - \frac{\partial \mathcal{L}}{\partial q_i} = Q_i$$

2 Separate the physical contributions

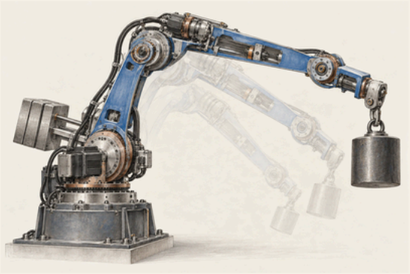
M(q) is the symmetric inertia matrix mapping acceleration to torque. c(q,q̇) contains centripetal and Coriolis velocity products; g(q) = ∂V/∂q contains gravitational torque. If Fext is the external tool force/wrench acting on the robot, add J^TFext on the right. One may write c = Cq̇, but C is not unique for a given c vector. Off-diagonal elements of M explain why accelerating one joint can require torque at another. Motors cannot therefore be sized as if they drove independent masses. The same acceleration command can demand different torque when configuration changes the inertia matrix.

$$M(q)\ddot{q} + c(q, \dot{q}) + g(q) = \tau + J^T F_{\text{ext}}$$

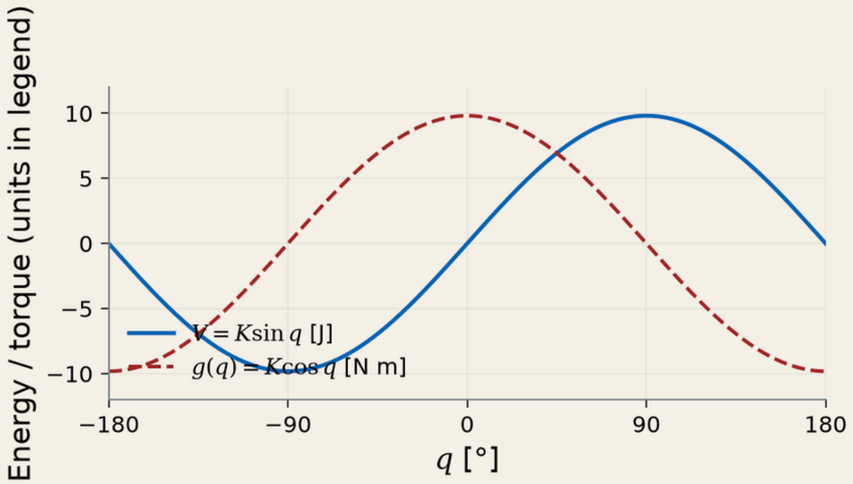
3 Check the derivation with one joint

For q measured from horizontal, choose T = Jq̇²/2 and V = K sin q; J is constant joint inertia and K the gravitational moment scale. The velocity derivative is Jq̇, its time derivative Jq̈, and the potential derivative K cos q. Thus Jq̈+K cos q = τ. Multi-joint forward dynamics likewise solves a linear system with M; no explicit matrix inverse is needed numerically.

$$J\ddot{q} + K\cos q = \tau$$



Arm with center-of-mass and gravity effects; the lower diagram computes one joint's energy and torque terms.



Assumptions and physical limits

The source example uses endpoint point masses; the next slide instead uses a uniform link and a separate payload. These inertia models are not interchangeable.

Lagrange derivation for a robot joint

From distributed mass to energy and then torque

1 Obtain inertia by integration

For a uniform link of length L and mass mℓ, dm=(mℓ/L)ds. Integrate squared distance from the axis times mass; a tip mass mp adds mpL². With L=0.4 m, mℓ=1 kg and mp=2 kg, J=0.37333 kg·m².

$$J = \int_0^L s^2 \frac{m_\ell}{L} ds + m_p L^2 = \frac{m_\ell L^2}{3} + m_p L^2$$

2 Energy and partial derivatives

Measure q from horizontal; mass-center heights are L sinq/2 and L sinq. Thus T=Jq̇²/2 and V=K sinq, with K=(mℓL/2+mpL)g=9.81 N·m. For the Lagrangian ℒ=T−V, the velocity derivative is Jq̇ and the position derivative is −K cosq. For partial derivatives, q and q̇ are treated as independent variables: hold q fixed when differentiating with respect to velocity. The subsequent time derivative follows the actual motion. Separating these operations explains why potential energy contributes nothing to the velocity derivative.

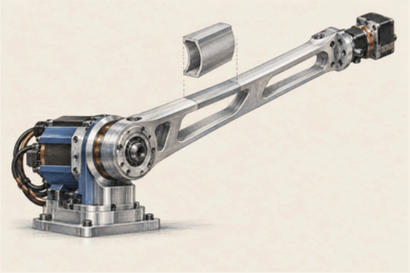
$$\frac{\partial \mathcal{L}}{\partial \dot{q}} = J\dot{q}, \quad \frac{\partial \mathcal{L}}{\partial q} = -K\cos q$$

3 Euler–Lagrange and dissipative torque

Differentiate the first expression in time and subtract the second to obtain Jq̈+K cosq. Generalized external force is motor torque minus viscous loss. At horizontal equilibrium q=0 with zero speed and acceleration, τ=9.81 N·m, providing a physical check.

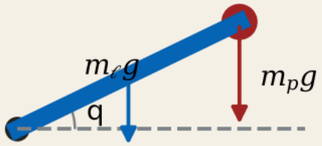
$$\frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{q}} - \frac{\partial \mathcal{L}}{\partial q} = \tau - b\dot{q}$$

$$J\ddot{q} + b\dot{q} + K\cos q = \tau$$



The uniform link and point payload make separate weight and inertia contributions.

$$V = (m_\ell L/2 + m_p L) g \sin q$$



$$J = m_\ell L^2/3 + m_p L^2$$

Assumptions and physical limits

Measuring q from vertical changes sine/cosine forms. Assume a rigid link, fixed axis and point tip mass; rotor inertia can be added separately.

Load and motion torque at one joint

Refer all torques to the same output shaft; mass, inertia and gravitational moment are distinct quantities.

1 Obtain inertia from the mass distribution

Let a uniform link of length L have mass mℓ and an endpoint point payload have mass mp. Link inertia about the joint is mℓL²/3; payload inertia is mpL². With L = 0.40 m, mℓ = 1 kg and mp = 2 kg, total J is 0.37333 kg·m². The link mass is not concentrated at its tip.

$$J = \frac{m_\ell L^2}{3} + m_p L^2 = 0.37333\ \mathrm{kg\,m^2}$$

2 Add gravity, acceleration and friction

Measure q positively upward from horizontal. Gravity moment arms are L/2 for the link and L for the payload, scaled by cos q. Viscous friction bq̇ opposes motion and appears positively in the required motor torque. At q = 0, q̇ = 1 rad/s, q̈ = 2 rad/s² and b = 0.10 N·m·s/rad, the contributions are 0.7467, 0.1000 and 9.8100 N·m.

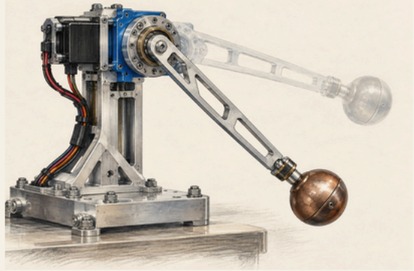
$$\tau = J\ddot{q} + b\dot{q} + \left(\frac{m_\ell L}{2} + m_p L\right) g \cos q$$

$$\tau = 10.6567\ \mathrm{N\,m}$$

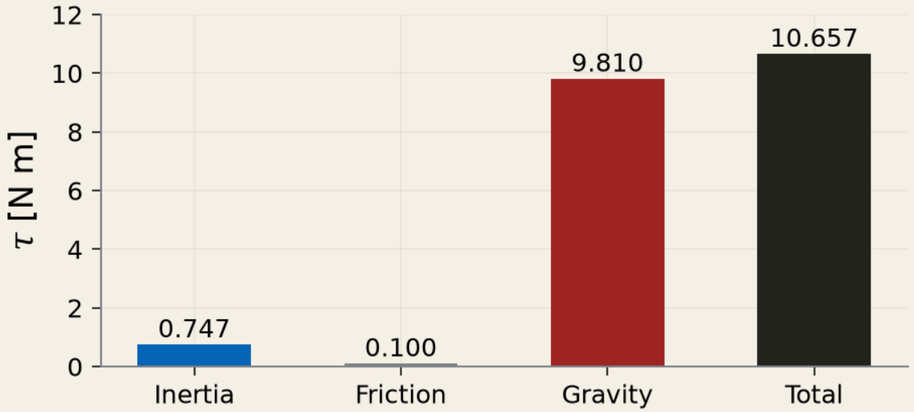
3 Small motion about equilibrium

Let K = 9.81 N·m. Holding a stationary angle q₀ requires τ₀ = K cos q₀. Substitute q = q₀+δq and expand cosine to first order to obtain the local model below. At horizontal q₀ = 0, first-order gravitational slope is zero; gravity torque is not zero, since the constant 9.81 N·m is still being balanced. δq and δτ are deviations from equilibrium, not the complete angle and torque. The equilibrium input τ0 supports gravity, while δτ drives the small motion. Otherwise, the missing constant term in the local equation may be mistaken for an absent physical load.

$$J\delta\ddot{q} + b\delta\dot{q} - K\sin q_0\,\delta q = \delta\tau$$



The three bars are output torque contributions at one operating point; their sum is the instantaneous required actuator torque.



Assumptions and physical limits

Motor/gear inertia, dry friction and link compliance are omitted here. For real sizing, refer them to the same shaft and include them; do not confuse peak and continuous torque.

Humanoid robot configuration

A humanoid body is not fixed to the world; base position and orientation are model variables.

1 Floating base and articulated joints

Attach a body frame to the pelvis. Its three position and three rotational freedoms add to the n articulated joints. Without contact constraints, configuration dimension is n+6. This base is not a physical six-axis actuator; it is a mathematical joint representing free motion of the whole robot relative to the world.

$$\mathcal{Q} = SE(3) \times \mathcal{Q}_j, \qquad \dim \mathcal{Q} = n + 6$$

2 Quaternion coordinates are not independent DOFs

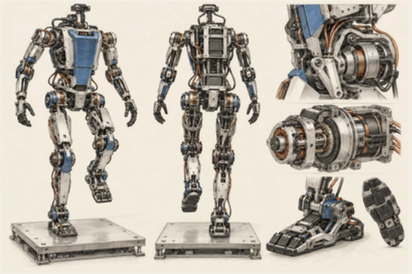
A unit quaternion Q stores orientation using four numbers, but its norm constraint leaves three independent rotational freedoms. A 30-joint example stores 3 position + 4 quaternion + 30 joint coordinates = 37 numbers, while physical configuration dimension is 36. Q and −Q represent the same orientation: the representation is nonsingular but double-covered.

$$q = (p_B, Q_B, q_j), \qquad \|Q_B\| = 1$$

3 Define velocity coordinates separately

Generalized velocity v contains three base linear velocities, three angular velocities and n joint speeds: 36 components for 30 joints. Quaternion derivative is not the angular-velocity vector; it requires the map N(q). With a foot fixed to the ground, the stored coordinates may remain unchanged while admissible velocities shrink to the contact-constrained subspace. Because N is rectangular, $\dot{q}$ and v need not even have the same length. Angular velocity has three components describing physical rotation; quaternion rate has four describing representation change. These objects must not be equated directly in the dynamics.

$$\dot{q} = N(q)v, \qquad N \in \mathbb{R}^{(n+7)\times(n+6)}$$



The pelvis adds three translations and three rotations to the joint tree; foot contact adds a separate constraint.

$$\|Q_B\| = 1$$

pB	QB	qj
3	4 → 3	30

37 coordinates • 36 independent DOFs

Assumptions and physical limits

Stored coordinates and independent physical dimension are different. Keep the world/body convention for velocity components consistent throughout the dynamics.

Floating-base dynamics and underactuation

Joint motors generate internal forces; external forces and moments change whole-robot momentum.

1 Place motor inputs using a selection matrix

Order generalized velocity as v = (vB,ωB,q̇j). Motors perform direct work only at the n joint coordinates. S = [0 I] selects joint speeds, and its transpose places motor torques in the full dynamics. For 30 joints, S is 30×36, τ has 30 entries and $S^T\tau$ has 36, with the first six entries zero. The zero block in S indicates that no six independent actuators directly connect the pelvis to the world. Foot contact influences those dynamics, but contact force is not simply an additional joint motor that can be commanded independently.

$$S = (0_{n\times 6}\ I_n), \qquad \dot{q} = N(q)v$$

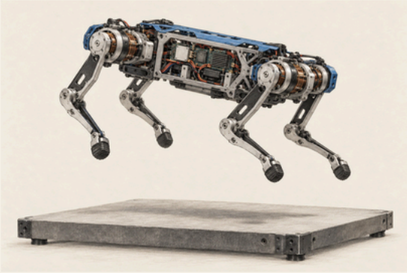
2 Contact applies external loading to the base

M is the 36×36 inertia matrix; h collects velocity and gravity terms. Contact forces λ become generalized forces through the transpose contact Jacobian. Depending on the contact model, λ may contain point forces or full contact wrenches. Every product must use consistent frames and power-conjugate velocity/force ordering.

$$M(q)\dot{v} + h(q,v) = S^T\tau + J_c^T\lambda$$

3 What can change during flight?

In flight, λ = 0. Motors can move limbs and alter body orientation through internal momentum exchange, but cannot change total horizontal CoM momentum without an external force. Underactuated does not automatically mean uncontrollable. Here it means that independent motor inputs are fewer than the unconstrained dynamic velocity coordinates.



Robot in flight and the zero base block of the input vector; the diagram shows direct motor actuation, not all possible motion.

$$M \in \mathbb{R}^{36\times 36}$$

$$S^T\tau = \begin{bmatrix} 0_6 \\ \tau_{30} \end{bmatrix}$$

Base: no direct motor input

30 joint torques

Assumptions and physical limits

Zero direct motor input in the first six rows does not mean zero base acceleration. Coupling in M, gravity and contact determine base motion.

Contacts change the dynamics

Flight, single support and double support are different constraint and force regimes of the same robot.

1 Velocity constraint for sticking contact

A foot contact point sticking to stationary ground must have zero world velocity. Jc maps generalized velocity v to contact velocity. Differentiation must account for the Jacobian changing with configuration, producing the $\dot{J}c\,v$ term in the acceleration constraint. Fixing the orientation of a flat sole requires more constraints than point contact. If the Jacobian were constant, only Jc $\dot{v}$ would remain. Joint motion generally changes how generalized velocities affect the contact point. The second product-rule term captures that change; generalized acceleration must compensate for it to keep contact acceleration zero.

$$J_c v = 0, \qquad J_c \dot{v} + \dot{J}_c v = 0$$

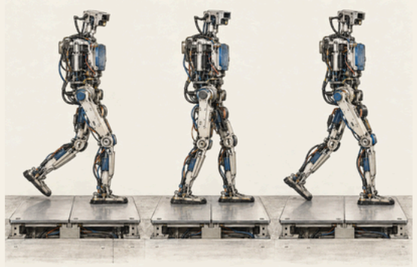
2 The ground can push, not pull

Normal force λn must be nonnegative. In the Coulomb model, tangential force magnitude is bounded by μλn. With λn = 300 N and μ = 0.50, the limit is 150 N. A solution demanding 170 N horizontally violates sticking even if it satisfies the algebraic dynamics. Negative normal force indicates that the contact should break.

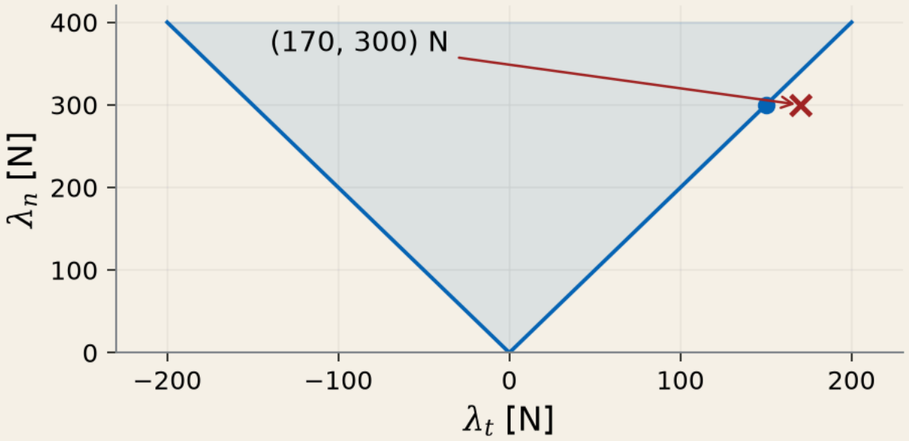
$$\lambda_n \geq 0, \qquad \sqrt{\lambda_x^2 + \lambda_y^2} \leq \mu \lambda_n$$

3 Touchdown is a separate event

At touchdown, velocity can change over a very short interval; a rigid impact model represents this through impulse. Forcing a continuous-contact equation to use one enormous acceleration is not the same model. Flight has no ground reaction, single support loads one foot, and double support loads both. Reassess constraints, feasible forces and load sharing in each mode.



A 2D cone section shows the normal–tangential force bound. The 170 N request lies outside the feasible region at 300 N normal load.



Assumptions and physical limits

The friction cone is an ideal local contact model. A finite sole also imposes CoP, torsional-moment and pressure-distribution limits; force magnitude alone is insufficient.

From contact constraints to dynamics

Solve contact force and acceleration together

1 From geometry to velocity and acceleration

Write persistent contact as φ(q)=0. With generalized-coordinate rate q̇=N(q)v, define Jc=(∂φ/∂q)N. One time derivative gives Jc v=0; a second gives Jc v̇+J̇c v=0. Omitting J̇c gives incorrect acceleration when geometry changes.

$$J_c\mathbf{v} = 0, \quad J_c\dot{\mathbf{v}} = -\dot{J}_c\mathbf{v}$$

2 Use the same unknowns in dynamics

M is the mass matrix, h collects velocity/gravity terms, S selects joints and λ is contact force. Dynamics contains unknown acceleration and contact force; the constraint restricts the same acceleration. The two rows form a block linear system. The minus sign comes from moving contact force to the left. The upper row enforces dynamics; the lower row enforces contact-compatible motion. λ is a reaction required for compatibility, not a prescribed input. Computing free acceleration and subsequently clamping motion to the ground is not equivalent to solving this coupled system.

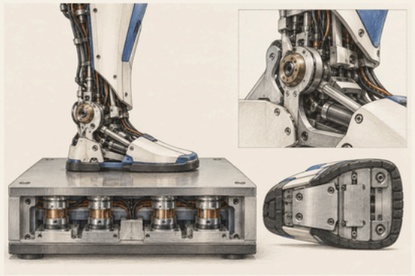
$$M\dot{\mathbf{v}} - J_c^T\boldsymbol{\lambda} = S^T\boldsymbol{\tau} - \mathbf{h}$$

$$J_c\dot{\mathbf{v}} = -\dot{J}_c\mathbf{v}$$

3 A vertical-mass example

For a mass m on the ground, take z upward and λ as upward force. The first row is m z̈−λ=−mg and the second z̈=0. Together they give λ=mg. If the solution needs λ<0, the ground would have to pull the body; the maintained-contact assumption must be rejected.

$$\ddot{z} = 0 \quad \Rightarrow \quad \lambda = mg = 588.6\ \text{N} \quad (m = 60\ \text{kg})$$



Solving both rows determines motion and ground reaction together.

Solve acceleration and contact force together

$$\begin{bmatrix} \mathsf{M} & -J_c^T \\ J_c & 0 \end{bmatrix} \begin{bmatrix} \dot{v} \\ \lambda \end{bmatrix} = \begin{bmatrix} S^T\tau - h \\ -\dot{J}_c v \end{bmatrix}$$

$$J_c v = 0 \implies J_c\dot{v} + \dot{J}_c v = 0$$

Assumptions and physical limits

The equality model assumes persistent sticking contact and independent constraints. Lift-off, sliding and impact need other modes; check friction feasibility separately.

Humanoid momentum and balance model

Despite many joints, whole-robot momentum balance is written in terms of external interactions.

1 Track the total center of mass

c is the mass-weighted mean position of all links. m is total mass, fi is each contact force on the robot, and g is the downward gravity vector. Internal motor and joint forces cancel in total momentum balance. CoM acceleration is therefore set by total external contact force plus weight.

$$c = \frac{\sum_k m_k c_k}{m}, \qquad m\ddot{c} = \sum_i f_i + m\mathbf{g}$$

2 Sum moments about the CoM

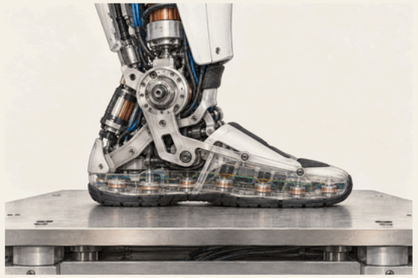
Lc is total angular momentum about the CoM. A force at contact position pi has lever arm pi−c; add any pure contact moment τi separately. The resultant gravitational force passes through the CoM, so gravity produces no net moment about that point. Express all vectors in the same world frame. hG contains three linear and three angular momentum components. Columns of AG describe how each generalized velocity contributes to that total. Joint motion can affect base orientation without changing total linear momentum in the absence of external force.

$$\dot{L}_c = \sum_i [(p_i - c) \times f_i + \tau_i]$$

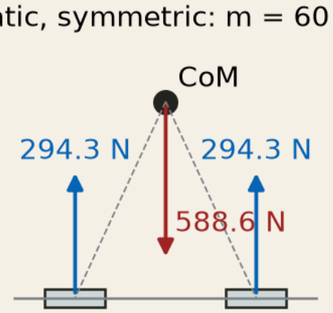
$$h_G = A_G(q)v$$

3 Load sharing requires additional conditions

A stationary 60 kg robot needs total vertical force 60×9.81 = 588.6 N. Symmetric geometry and sharing give 294.3 N per foot. The force-sum equation alone does not determine that allocation; moment balance, sole pressure and joint torque limits also matter. hG = AG(q)v collects the contributions of all generalized velocities to total momentum.



Foot forces and moment arms about the CoM; equal load sharing belongs to the symmetric static example.



Assumptions and physical limits

Centroidal dynamics does not replace full joint dynamics in every case. Feasible total force and momentum do not guarantee reachable feet or sufficient joint torque.

ZMP and the linear inverted pendulum

During motion, the projected center of mass and center of pressure need not coincide.

1 Define the center of pressure

On horizontal ground, nonnegative normal forces make the center of pressure a force-weighted average of contact positions inside their convex hull. This definition is unavailable when total normal force is zero. On a planar surface, the ZMP is where the net contact wrench has zero horizontal moments; under the corresponding contact assumptions it coincides with CoP.

$$p_{\mathrm{CoP}} = \frac{\sum_i p_i f_{zi}}{\sum_i f_{zi}}$$

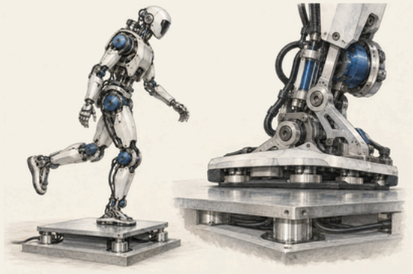
2 Simplify momentum balance

Assume constant CoM height h, zero vertical acceleration and negligible angular-momentum rate about the CoM. Horizontal force is m c̈x and vertical force mg. Zero moment about the CoM relates pressure-center location to acceleration. Mass cancels, but the model's simplicity comes from these strong assumptions. Forward acceleration requires a forward ground force acting below the CoM. That force creates a moment; shifting the pressure centre backward lets the vertical force balance it. The minus sign follows from moment balance, not an arbitrary direction convention.

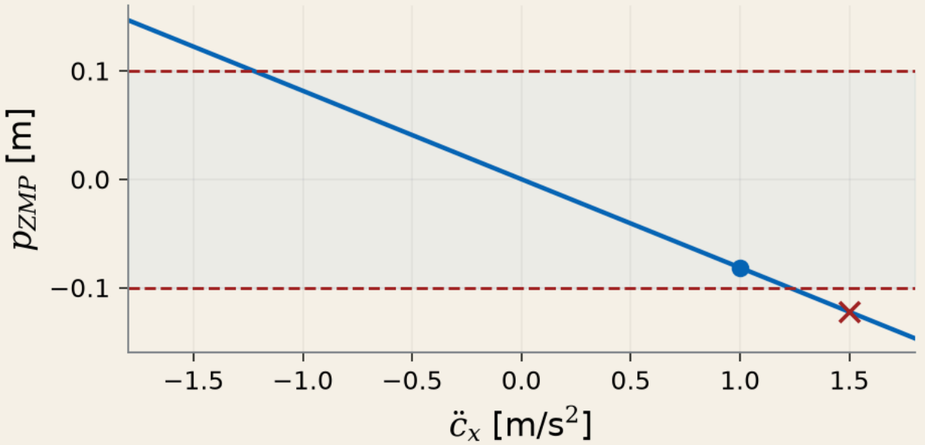
$$p_{\mathrm{ZMP}} = c_x - \frac{h}{g}\ddot{c}_x, \qquad \ddot{c}_x = \frac{g}{h}(c_x - p_{\mathrm{ZMP}})$$

3 Compare acceleration demand with support

With h = 0.80 m, cx = 0 and forward acceleration 1 m/s², ZMP is −0.08155 m, behind the CoM projection. It lies inside a support interval [−0.10,0.10] m. A 1.5 m/s² request instead gives −0.1223 m and exceeds the interval, so that request is infeasible under the constant-height, zero-angular-momentum-rate strategy.



Blue is the computed acceleration–ZMP relation; red lines are support limits. Forward acceleration shifts ZMP backward.



Assumptions and physical limits

ZMP inside support is not sufficient for full-robot feasibility: check friction, joint torque, reach and contact moment limits too. The model is invalid in flight.

Momentum and the ZMP relation

From general force balance to a constant-height approximation

1 Why do internal forces cancel in the sum?

Sum Newton’s equation over all parts. Equal and opposite internal action–reaction forces cancel in total linear momentum. External contact forces and gravity remain. CoM acceleration equals their sum divided by total mass.

$$m\ddot{\mathbf{c}} = \sum_i \mathbf{f}_i + m\mathbf{g}$$

2 Expand the moment about the CoM

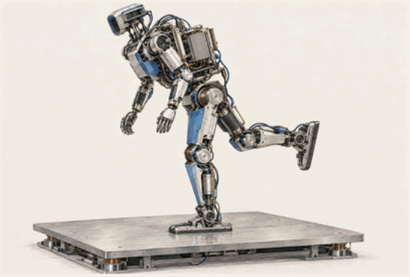
On level ground take contact p=(px,0,0) and CoM c=(cx,0,h). For force (Fx,0,Fz), the y component of (p−c)×F is (cx−px)Fz−hFx. This equals L̇y, the angular-momentum rate about the CoM; free contact moment is assumed zero here.

$$\dot{L}_y = (c_x - p_x)F_z - hF_x$$

3 Now apply the LIPM assumptions

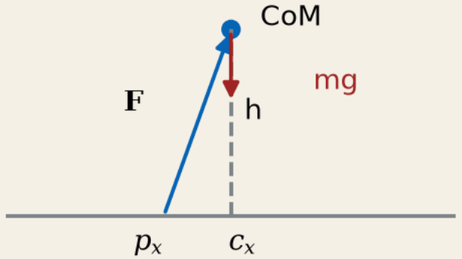
Constant height gives c̈z=0 and Fz=mg. Set angular-momentum rate to zero and use Fx=m c̈x to obtain px=cx−h c̈x/g. For h=0.8 m and c̈x=1 m/s², center of pressure lies 0.08155 m behind the CoM projection. If L̇y is nonzero, the general moment equation adds a contribution −L̇y/Fz to px. Rapid arm or torso rotation therefore changes the simplified inverted-pendulum assumption. This distinguishes the full momentum equation from its reduced balance criterion.

$$p_x = c_x - \frac{h}{g}\ddot{c}_x = c_x - 0.08155\ \text{m}$$



Moment arms are measured about the CoM; forward acceleration moves the center of pressure backward.

$$\dot{L}_y = (c_x - p_x)F_z - hF_x = 0$$



Assumptions and physical limits

The final relation is not a universal humanoid law: it requires constant CoM height, planar contact and negligible angular-momentum rate.

Quadrotor: rigid-body dynamics

Thrust and moments from four rotors drive a body with six degrees of freedom.

1 Frames and state variables

Choose world Z upward and body x forward, y left, z upward. R ∈ SO(3) rotates body vectors into world coordinates. Position p and velocity v use world axes; angular velocity ω uses body axes. Independent state dimension is 12: three position, three velocity, three orientation and three angular-rate variables. The nine entries of R are not nine independent rotation variables.

$$\dot{\mathbf{p}} = \mathbf{v}, \qquad \dot{R} = R[\boldsymbol{\omega}]_\times$$

2 Newton–Euler equations

Collective thrust T acts along body z and is rotated into the world by R. Gravity opposes world Z. I is the constant body-frame inertia tensor about the center of mass; τ is net rotor-generated body moment. The term ω × Iω comes from angular-momentum transport in rotating axes and cannot generally be omitted in 3D motion. R e3 is a unit direction; multiplying by T gives a force in newtons. Increasing thrust changes its magnitude, while rotating the body changes its direction. This distinction explains why translational control depends on attitude control.

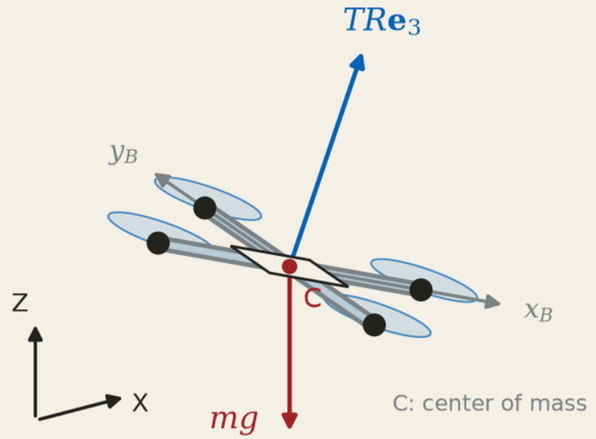
$$m\dot{\mathbf{v}} = -mg\mathbf{e}_3 + TR\mathbf{e}_3$$

$$I\dot{\boldsymbol{\omega}} + \boldsymbol{\omega} \times (I\boldsymbol{\omega}) = \boldsymbol{\tau}$$

3 Tilt the body to accelerate horizontally

A standard quadrotor has no independent body-x or body-y thrust channels. Horizontal acceleration requires tilting the collective thrust vector, coupling attitude and translation. At level stationary hover, R = I, ω = 0 and T = mg. For m = 1.2 kg, collective thrust is 11.772 N, or 2.943 N per rotor in a symmetric vehicle.

Physical drone and force diagram are distinct: thrust follows body z while weight remains world-vertical.



Assumptions and interpretation

$e_3 = (0,0,1)^T$ and $[\omega]\times a = \omega \times a$. Assume a rigid body, fixed mass, parallel rotor axes and instantaneous thrust. Aerodynamic drag, ground effect, rotor inertia and motor lag are omitted. Euler-angle rates do not generally equal components of ω.

Newton–Euler derivation for a drone

From body thrust to world acceleration and gyroscopic torque

1 Express the thrust axis in the world

Use world z upward and body e3 along thrust; R maps body to world. Total thrust in world coordinates is TRe3. Add gravity −mg e3. The third column of R therefore gives the drone’s world thrust direction.

$$m\ddot{\mathbf{p}} = TR\mathbf{e}_3 - mg\mathbf{e}_3, \quad \dot{R} = R[\boldsymbol{\omega}]_\times$$

2 Angular momentum is also a rotating vector

Let H=Iω in body axes. Its inertial derivative expressed in body coordinates is Iω̇+ω×Iω. Thus τ is not merely Iω̇; existing rotation and directional inertia create an additional moment. Diagonal I assumes principal inertia axes. Iω is angular momentum and need not be parallel to ω when principal inertias differ. Their cross product can therefore be nonzero. For spherical inertia, I is a scalar multiple of the identity, making the vectors parallel and eliminating this contribution.

$$\boldsymbol{\tau} = I\dot{\boldsymbol{\omega}} + \boldsymbol{\omega} \times I\boldsymbol{\omega}$$

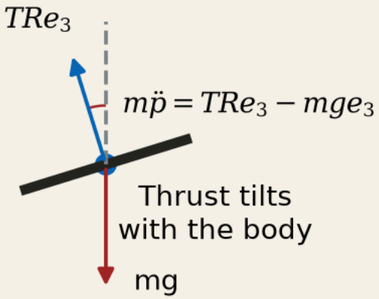
3 Numerical cross product and hover

For I=(0.02,0.025,0.04) kg·m² and ω=(1,2,0) rad/s, Iω=(0.02,0.05,0), giving a gyroscopic z moment of 0.01 N·m. Near zero yaw, small roll φ and pitch θ give Re3≈(θ,−φ,1); tilting produces horizontal acceleration.

$$\boldsymbol{\omega} \times I\boldsymbol{\omega} = (0, 0, 0.01)^T \ \mathrm{N\,m}$$

$$\ddot{p}_x \approx g\theta, \quad \ddot{p}_y \approx -g\phi \qquad (T \approx mg)$$

Thrust tilts with the body while gravity remains along world vertical.



Assumptions and physical limits

ω is body angular velocity, not generally Euler-angle rates. Small angles and T≈mg apply only to the final hover approximation.

Quadrotor: rotor-force allocation

Desired collective thrust and three moments map to four forces through explicitly defined rotor geometry.

1 Numbering and moment signs

In a plus layout, rotor 1 is forward (+x), 2 left (+y), 3 aft (−x), and 4 right (−y). Each produces force fi along +z, at distance ℓ from the center. The cross product $r_i \times f_i e_3$ determines roll and pitch moments. Choose positive body reaction yaw moments for rotors 1 and 3, negative for 2 and 4.

$$T = f_1 + f_2 + f_3 + f_4$$

$$\tau_x = \ell(f_2 - f_4), \quad \tau_y = \ell(f_3 - f_1), \quad \tau_z = \kappa(f_1 - f_2 + f_3 - f_4)$$

2 Rotor model and inverse mapping

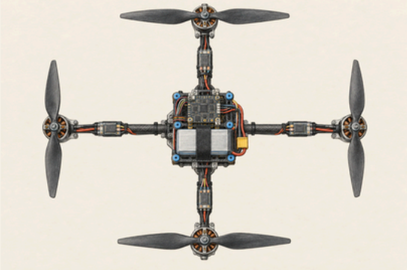
Use the ideal static rotor relations $f_i = k_f \Omega_i^2$ and reaction-moment magnitude $k_m \Omega_i^2$, giving $\kappa = k_m/k_f$ [m]. Rotor speed Ω_i [rad/s] differs from body angular velocity ω. The allocation matrix is invertible, but computed rotor forces must separately satisfy physical bounds.

$$f_1 = \frac{T}{4} - \frac{\tau_y}{2\ell} + \frac{\tau_z}{4\kappa}, \quad f_3 = \frac{T}{4} + \frac{\tau_y}{2\ell} + \frac{\tau_z}{4\kappa}$$

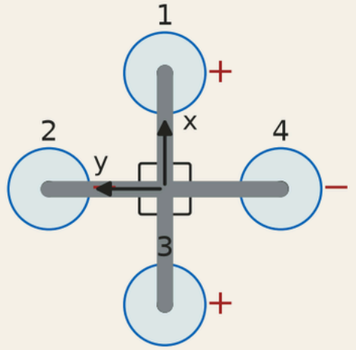
$$f_2 = \frac{T}{4} + \frac{\tau_x}{2\ell} - \frac{\tau_z}{4\kappa}, \quad f_4 = \frac{T}{4} - \frac{\tau_x}{2\ell} - \frac{\tau_z}{4\kappa}$$

3 Roll moment at fixed collective thrust

Let T = 11.772 N, ℓ = 0.20 m, τ_x = 0.08 N·m and other moments zero. Then $f_1 = f_3$ = 2.943 N, f_2 = 3.143 N and f_4 = 2.743 N. Equal and opposite changes preserve collective thrust and create moment 0.20 × 0.40 = 0.08 N·m. Equal and opposite changes in opposite rotors create the requested moment while preserving total thrust. Increasing just one rotor also changes total thrust and yaw moment. Allocation solves these coupled effects together to obtain the desired force–moment combination.



The diagram labels rotor positions and signs of yaw reaction moment on the body; these signs are not propeller spin arrows.



Reaction moment: ±τz

$$[T, \tau_x, \tau_y, \tau_z]^{\mathsf{T}} = G\mathbf{f}$$

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & \ell & 0 & -\ell \\ -\ell & 0 & \ell & 0 \\ \kappa & -\kappa & \kappa & -\kappa \end{bmatrix}$$

$$0 \le f_i \le f_{\max}$$

Assumptions and interpretation

Ordinary unidirectional propellers require 0 ≤ fi ≤ fmax. Negative or excessive results indicate an infeasible requested wrench; solve constrained allocation. An X layout changes moment arms and the matrix. Changing rotor order or reaction-moment convention requires matching mixer signs.

S65 · Drake QuadrotorPlant, rotor moment cross products; independently derived plus-layout mixer.

Building the rotor allocation matrix

Each column is one rotor’s force and moment contribution

1 From position to moment sign

For a plus layout use rotors 1:(ℓ,0), 2:(0,ℓ), 3:(−ℓ,0), 4:(0,−ℓ). Each produces fi along body +z. Since ri×fi e3=(yi fi,−xi fi,0), roll/pitch signs follow directly from position. Choose yaw reaction signs (+,−,+,−). Read one column of A by applying 1 N at its rotor and zero at the others. Its entries give total thrust and roll, pitch and yaw moments. Adding the weighted columns is simply superposition of forces and moments.

$$A_i = (1,\ y_i,\ -x_i,\ s_i\kappa)^T, \quad \mathbf{w} = A\mathbf{f}$$

2 Sum and difference equations

Total thrust is the sum of four forces. Here roll follows signed y positions and pitch signed x positions. κ is yaw moment divided by thrust, with units of meters. Renumbering rotors or changing body axes requires rebuilding the signs.

$$T = f_1 + f_2 + f_3 + f_4, \quad \tau_x = \ell(f_2 - f_4)$$

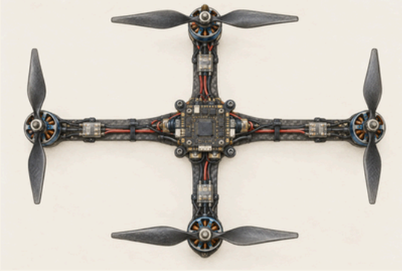
$$\tau_y = \ell(f_3 - f_1), \quad \tau_z = \kappa(f_1 - f_2 + f_3 - f_4)$$

3 Check feasibility before applying the solution

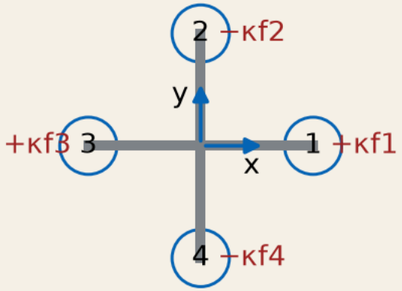
The first two inverse expressions are below; the opposite rotors use opposite roll/pitch terms. For ℓ=0.2 m, increasing only rotor 2 by 0.2 N adds 0.2 N thrust and 0.04 N·m roll, with yaw change −0.2κ. Negative or excessive fi makes the solution physically infeasible.

$$f_1 = T/4 - \tau_y/(2\ell) + \tau_z/(4\kappa)$$

$$f_2 = T/4 + \tau_x/(2\ell) - \tau_z/(4\kappa)$$



Rotor numbering, force directions and torque signs match the main drone page.



Top view; thrust along +z

Assumptions and physical limits

With fixed thrust coefficient use fi=kfΩi²; negative thrust cannot be converted by a real square root. Do not copy this matrix unchanged to an X layout.

Drake QuadrotorPlant force/torque implementation; MIT Underactuated; original allocation derivation.

Quadrotor: the model near hover

Small attitude deviations generate horizontal acceleration on the scale of gravity.

1 Subtract equilibrium and retain small deviations

Use equilibrium at fixed position, zero velocity, level attitude and $T_0 = mg$. With $R = R_z(\psi)R_y(\theta)R_x(\varphi)$, expansion about $\psi_0 = 0$ gives $Re_3 \approx (\theta, -\varphi, 1)^T$. Let $\Delta T = T - mg$ and φ, θ, ψ be small rotations in radians. Products of small angle and thrust perturbation are second order and are discarded. Write $T=mg+\Delta T$ during linearization. The product of mg and tilt remains first order, while the product of ΔT and tilt is second order and is discarded. This produces coefficient g in horizontal acceleration; small-angle reasoning does not justify deleting arbitrary products.

$$\ddot{X} \simeq g\theta, \quad \ddot{Y} \simeq -g\phi, \quad \ddot{Z} \simeq \frac{\Delta T}{m}$$

2 Local state space and control

Local state contains positions, Euler angles and their rates; inputs are ΔT, τx, τy, τz. With principal body axes and small angular rates, rotational dynamics reduce to the double integrators below. Changing horizontal position first requires generating φ or θ. Underactuation does not necessarily imply uncontrollability near this equilibrium.

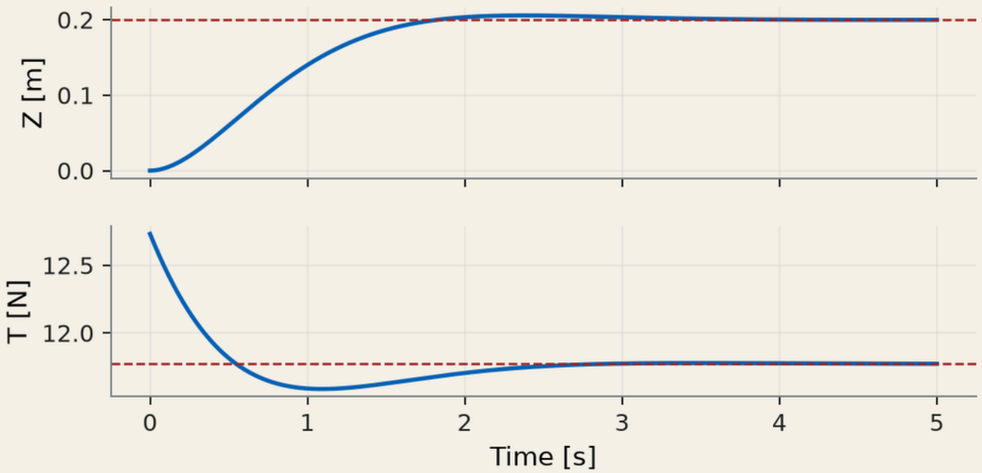
$$I_x\ddot{\phi} \simeq \tau_x, \quad I_y\ddot{\theta} \simeq \tau_y, \quad I_z\ddot{\psi} \simeq \tau_z$$

$$\Delta T = m[K_p(Z_d - Z) - K_d\dot{Z}]$$

3 Small-angle and altitude examples

At $\theta = 2°$, linear horizontal acceleration is $9.81 \times 0.03491 = 0.3424$ m/s²; exact altitude-maintaining acceleration is $g \tan\theta = 0.3426$ m/s². Vertical gains $K_p = 4$ s^{-2} and $K_d = 3$ s^{-1} give $\ddot{Z} + 3\dot{Z} + 4(Z - Z_d) = 0$. The plot is the ideal response to a 0.2 m altitude step, assuming an ideal inner attitude loop and unsaturated thrust.

Computed altitude response: m = 1.2 kg, Kp = 4 s⁻², Kd = 3 s⁻¹. This is not experimental data.



Assumptions and interpretation

This is a local model: large tilt, aggressive maneuvers and motor delay need the nonlinear model. Use radians; Euler rates and body angular rates agree only approximately near small angles. The PD example is a theoretical calculation with explicit assumptions, not a tuned flight controller for a real vehicle.

Comparing robot platform models

The mathematical language stays common; base, contact and actuation assumptions change by platform.

1 Parts of the common equation

M is generalized inertia, h collects velocity and gravity effects, B maps inputs to generalized forces, and λ contains contact forces. This structure organizes modeling; it does not give every robot the same M or input units. A(q)v = 0 separately describes admissible instantaneous velocities. Define states, inputs and contacts before solving numerically. The shared notation helps compare where each model places physical effects. A fixed arm's support reactions may disappear in reduced coordinates, while humanoid contact forces remain explicit unknowns. A force absent from a reduced equation is not necessarily physically absent.

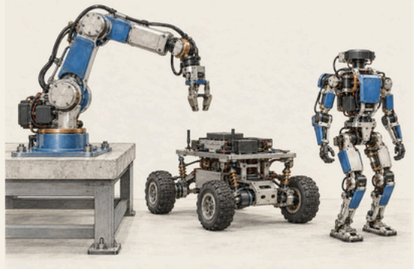
$$M\dot{v} + h = B(q)u + J_c^T\lambda, \qquad A(q)v = 0$$

2 Fixed-base arm and wheeled vehicle

An arm has a world-fixed base, joint-angle coordinates and usually joint-torque inputs. A rigid-link model omits load deformation. A vehicle body moves through the world, with rolling contact and lateral no-slip assumptions imposing velocity constraints. Do not confuse steering/speed inputs in a kinematic model with force/torque inputs in a dynamic model.

3 Humanoid and drone

A humanoid combines a floating base with many joints and switching ground contacts. A drone applies four rotor thrust inputs to a free rigid body; a conventional quadrotor has four independent inputs despite six rigid-body freedoms. Contact slip in a humanoid, large tire slip in a vehicle, arm compliance and drone aerodynamics illustrate limitations of the corresponding simple models.



The table compares inputs and constraints under common headings; one platform model is not forced onto the others.

Platform	Input	Constraint / limit
Arm	τ [N m]	Fixed base
Vehicle	δ [rad], v [m/s]	Pure rolling
Humanoid	τ [N m]	Contact modes
Quadrotor	f_i [N]	4 inputs / 6 DOFs

Assumptions and physical limits

Evaluate models against their task: parking position, grasp pose, walking contact feasibility or flight acceleration/attitude. Do not combine unlike units into one unscaled error norm.

The actuation problem

Desired motion and independently available force directions are different things.

1 Write the input map

u may contain motor currents, torques or rotor thrusts; B maps them into generalized forces. Actuation rank counts independent force directions. B can vary with configuration. Counting motors alone is insufficient: mechanical connections can make several motors act in the same generalized direction. A column of B shows the generalized forces caused by increasing one input by one unit. Parallel columns do not add independent force directions. Rank counts the number of independent directions represented by these input effects.

$$M\dot{\nu} + h = B(q)u, \qquad \operatorname{rank} B \le \dim \nu$$

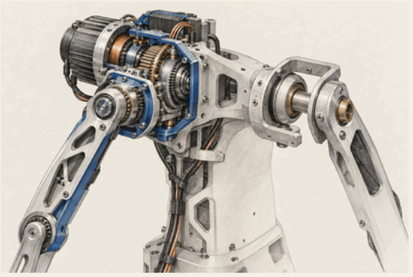
2 Full and underactuation

A fixed-base 6R arm with six independently driven joints can be fully actuated under an ideal torque-input model. A conventional quadrotor has four thrust combinations acting on six rigid-body freedoms and must tilt to accelerate horizontally. This instantaneous input deficit does not imply that position and attitude cannot be controlled through appropriate motion.

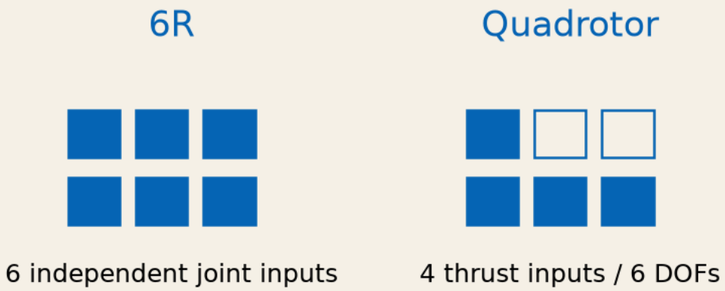
3 Full rank does not remove limits

Torque, speed, voltage, temperature and bandwidth limits restrict feasible inputs. Producing 2 N·m at 10 rad/s requires 20 W of mechanical power, but not every motor rated at 20 W can achieve that point. Power is the torque–speed product, not a single sizing number that replaces the operating envelope.

$$P = \tau^{T}\dot{q}, \qquad 2\ \mathrm{N\,m} \times 10\ \mathrm{rad/s} = 20\ \mathrm{W}$$



Input directions contrast independent joint actuation with four-input quadrotor rigid-body motion.



Assumptions and physical limits

Actuation rank and local/linear controllability are different analyses. Contacts and saturation alter available motion; keep modeling assumptions explicit when comparing them.

Robot actuators

Servo denotes a feedback architecture, not a separate electromagnetic motor principle.

1 Electromechanical options

A brushed DC motor uses mechanical commutation; BLDC/PMSM drives use electronic commutation. A stepper creates discrete equilibrium positions through phase excitation, and missed steps may go undetected in open loop. A linear motor generates force directly; a motor–screw assembly converts rotation to translation. A solenoid usually pulls over a limited stroke; precise continuous positioning requires additional feedback.

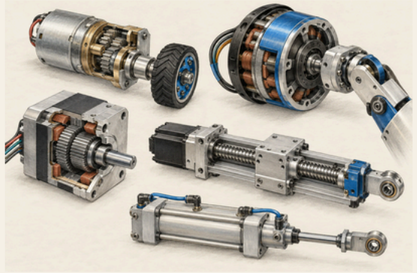
2 Fluid power and mechanical output

In hydraulics, pressure difference acting on piston area generates force and flow determines piston speed. Pneumatic gas compressibility adds significant compliance. Leakage, hoses, valves, compressors and pumps belong to the full architecture. Similarly, compare electric systems including drive, transmission and cooling rather than only bare motor mass. Pressure and flow play different roles: pressure difference sets load force, while flow sets speed. Increasing piston area raises force at fixed pressure but reduces speed at fixed flow. This trade-off follows a power balance analogous to gearbox torque–speed conversion.

$$F \approx \Delta p A, \qquad v \approx Q/A, \qquad P \approx \Delta p Q = F v$$

3 Select against requirements

A pneumatic gripper may suit binary grasping, an encoder-based electric servo precise continuous joint motion, and hydraulics heavy impact-prone work with a suitable power unit. These are conditional choices, not a universal ranking. For Δp = 1 MPa and A = 2 cm², ideal force is 200 N; at 0.05 m/s, mechanical output is 10 W and flow is 0.6 L/min. Losses increase required input power.



Distinct actuators and rotary, linear and fluid power conversions; servo control can be built around several of these mechanisms.

Rotary	Linear	Fluid
$\tau\omega$	Fv	ΔpQ
N m × rad/s	N × m/s	Pa × m³/s

Each represents power [W]

Assumptions and physical limits

Efficiency, force density and precision depend on product, temperature and operating point. Avoid numerical technology rankings without source data; derive criteria from the task.

How does an electric motor produce motion?

Permanent-magnet brushed DC motor: a coupled model of electromagnetic torque and back EMF.

1 From current to force and torque

A current-carrying conductor in a magnetic field experiences Lorentz force. Forces on opposite rotor conductors form a couple, producing shaft torque. With constant magnet flux and no saturation, winding geometry is collected in the motor constant. The commutator mechanically switches the appropriate winding-current direction as the rotor turns.

$$\mathbf{F} = i\boldsymbol{\ell} \times \mathbf{B}, \qquad \tau_e = k_t i$$

2 Electrical and mechanical equations couple

Voltage induced in the rotating winding opposes the applied voltage: back EMF. Terminal voltage equals resistive drop, inductive voltage and back EMF. Part of the electromagnetic torque accelerates inertia; the rest balances load and friction. Current therefore directly relates to torque, whereas voltage alone does not determine speed. At startup, low speed gives little back-EMF, so resistance, inductance and the drive mainly limit current. Back-EMF increases as the rotor accelerates. If added load reduces speed, current can increase at the same voltage through this electromechanical interaction.

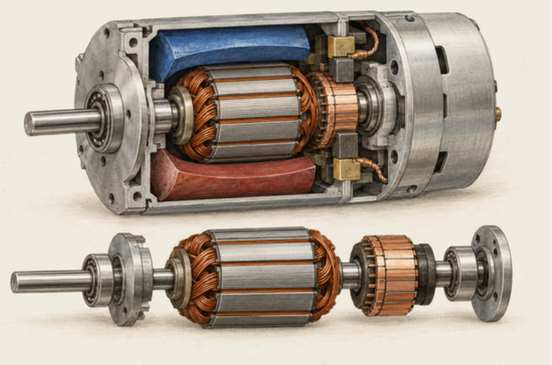
$$V = Ri + L_a \frac{di}{dt} + k_e \omega$$

$$J\dot{\omega} = k_t i - \tau_L - b\omega$$

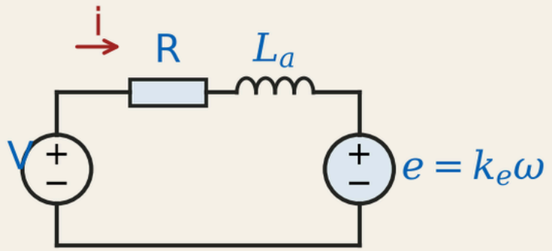
3 Check the model using power balance

Let k_t = 0.05 N·m/A, k_e = 0.05 V·s/rad and R = 0.8 Ω. At steady i = 6 A and ω = 200 rad/s, back EMF is 10 V, required terminal voltage is 14.8 V and torque is 0.30 N·m. Electrical input is 88.8 W; converted mechanical power is 60 W and copper loss is 28.8 W. Mechanical losses further reduce shaft output.

$$Vi = Ri^2 + \frac{d}{dt}\left(\frac{1}{2} L_a i^2\right) + \tau_e \omega$$



The cutaway shows magnets, wound rotor, commutator and brushes. The equivalent circuit below quantifies the same energy conversion.



Energy conversion

$$e\,i = \tau_e \omega \qquad P_{\mathrm{Cu}} = Ri^2$$

Symbols and model limits

R [Ω] is winding resistance, L_a [H] inductance, J [kg·m²] shaft-referred inertia, b [N·m·s/rad] viscous friction, and τL [N·m] opposing load torque. With consistent SI definitions, k_t and k_e are numerically equal. Brush drop, iron loss, temperature-dependent resistance and saturation are omitted; stall torque is not a continuous operating rating.

Electrical and mechanical motor equations

Voltage changes current, current torque, and torque acceleration

1 From Kirchhoff to current dynamics

Applied armature voltage equals resistive, inductive and back-EMF drops. R is in ohms, L in henries and ke in V·s/rad. Inductance prevents an instantaneous current jump under finite voltage. As the rotor spins, back EMF opposes part of the applied voltage.

$$L\dot{i} = V - Ri - k_e\omega$$

2 From Newton to rotor acceleration

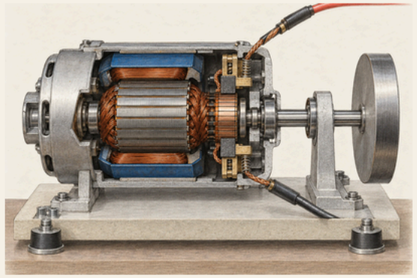
Electromagnetic torque is kt i. Subtract load torque τL and viscous loss bω to obtain Jω̇. Current and speed are therefore coupled states. The current equation represents electrical storage and the speed equation mechanical storage; a steady torque–speed line cannot describe the transient alone.

$$J\dot{\omega} = k_t i - b\omega - \tau_L$$

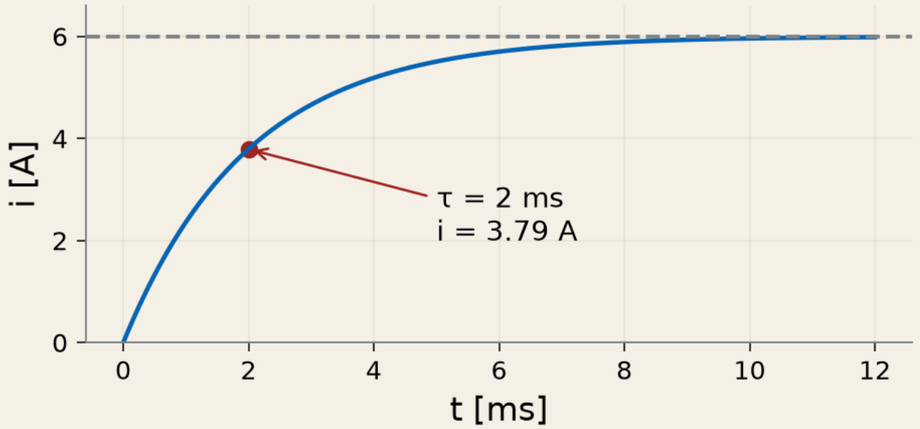
3 Solve a locked-rotor voltage step

With ω=0 and i(0)=0, the equation becomes Li+Ri=V. Its homogeneous solution is exponential and particular solution V/R; the initial condition sets the coefficient. For R=2 Ω, L=4 mH and V=12 V, τe=2 ms, final current is 6 A and current at 2 ms is 3.793 A. Check that current starts at zero and approaches V/R. Its initial slope is V/L, so greater inductance slows the initial rise. Reaching approximately 63.2% of the final value at one time constant is a general property of this exponential response.

$$i(t) = \frac{V}{R}(1 - e^{-t/\tau_e}), \quad \tau_e = L/R$$



Initial current slope is set by V/L; final current by V/R.



Assumptions and physical limits

The locked-rotor calculation is a modeling example, not an experiment to perform without current/thermal limits. A free rotor does not retain ω=0.

Motor constants and joint loading

Motor-shaft quantities in a data sheet are not the same as robot-joint quantities after a gearbox.

1 Convert constant definitions and units

Let kt be torque constant [N·m/A] and ke back-EMF constant [V·s/rad]. If speed constant kv is in rpm/V, convert rpm to radians per second, giving ke = 60/(2πkv) for the DC equivalent. kt and ke can be numerically equal with compatible SI conventions; differing phase/line, peak/RMS or electrical/mechanical definitions invalidate a direct substitution. A larger kv gives greater approximate no-load speed per volt but a smaller torque constant under consistent SI conventions. High kv therefore does not mean a more powerful motor by itself. Compare the current needed for the required torque as well.

$$\omega = \frac{2\pi n}{60}, \qquad k_e = \frac{60}{2\pi k_v}$$

2 Refer torque and speed to the same shaft

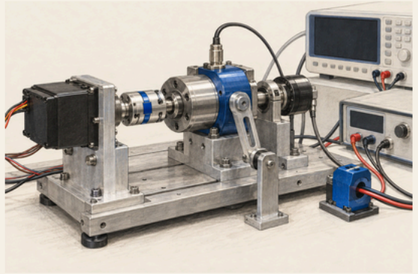
Let positive reduction ratio N = ωm/ωout and η denote motor-to-load efficiency. Motor torque is kt i and output torque approximately ηNkt i. With N = 20, η = 0.85, kt = 0.08 N·m/A and i = 2 A, motor torque is 0.16 N·m and output torque 2.72 N·m. An output speed of 2 rad/s requires 40 rad/s at the motor.

$$\tau_{out} = \eta N k_t i, \qquad \omega_m = N\omega_{out}$$

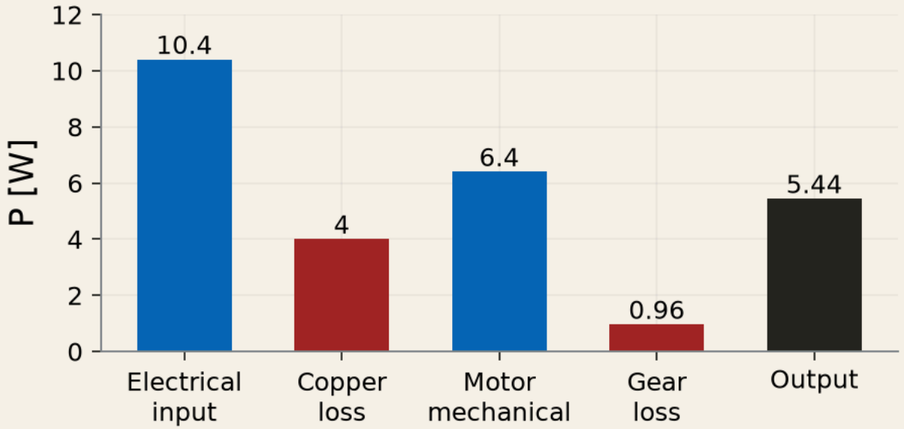
3 Cross-check power and voltage

Motor mechanical power is 0.16×40 = 6.40 W and output power 2.72×2 = 5.44 W, leaving 0.96 W transmission loss. With R = 1 Ω, ke = 0.08 and steady current, required voltage Ri+keω is 5.2 V. Of the 10.4 W electrical input, 4 W is copper loss and 6.4 W ideal electromechanical conversion.

$$Vi = Ri^2 + k_e\omega_m i$$



Torque, speed and power are shown separately at motor and output; power differences belong to distinct loss paths.



Assumptions and physical limits

This DC-equivalent loss model applies at a steady operating point. Real sizing also considers heating, inductive transients, drive loss and speed-dependent efficiency.

Torque–speed–power envelope

Motor torque and speed are not independent unlimited commands; voltage and current constrain them together.

1 Start with steady electrical balance

In the ideal DC equivalent with steady current, V = Ri+keω. Increasing speed raises back-EMF and leaves less voltage for current through resistance. Substitute i into τ = kt i to obtain the linear torque–speed relation. It applies at fixed applied voltage and constant parameters, not to every operating strategy of a controlled drive. The line slope −ktke/R measures how rapidly available torque falls with speed at fixed voltage. Raising voltage shifts the ideal line; raising allowable current increases thermal loading. These changes act on different operating constraints.

$$\tau = k_t \frac{V - k_e \omega}{R}$$

2 Stall and no-load speed limits

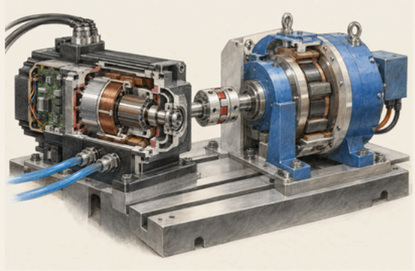
With V = 24 V, R = 2 Ω and kt = ke = 0.10, ideal no-load speed is 240 rad/s and stall torque 1.2 N·m. At stall, current is 12 A, copper loss 288 W, and mechanical power zero. Stall torque is therefore not a continuous rating. Real no-load current and friction shift the ideal endpoints.

$$\omega_0 \approx V/k_e, \qquad \tau_{stall} \approx k_t V/R$$

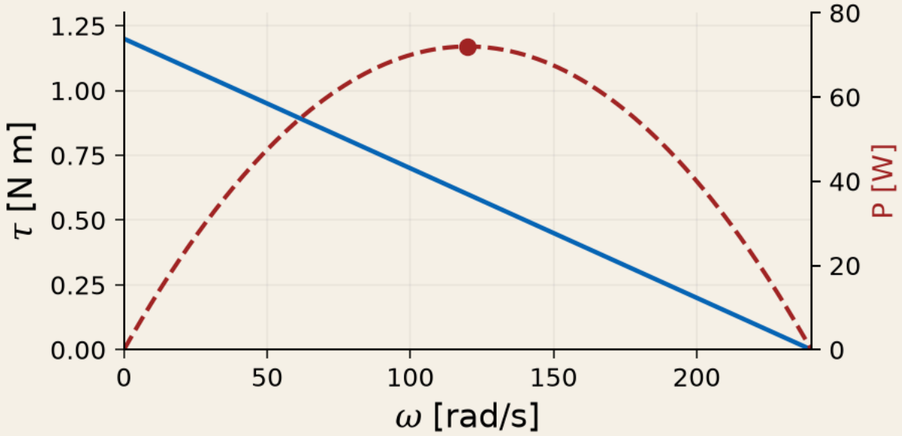
3 Power parabola and the real operating region

Multiplying the torque line by speed gives a parabolic power curve. Here, half no-load speed gives ω = 120 rad/s and τ = 0.60 N·m, or 72 W mechanical power. This is not a rated or continuously thermally safe point. Current limits cap torque, voltage limits cap high-speed capability, and temperature constrains duration.

$$P(\omega) = \frac{k_t}{R}(V\omega - k_e\omega^2)$$



Torque decreases linearly while power rises then falls; high torque at ω = 0 produces no mechanical power.



Assumptions and physical limits

The plot is a computed ideal DC model, not a commercial motor data sheet. Product selection requires actual continuous/peak current, voltage and thermal limits.

Continuous and peak motor operation

Heating depends on squared current and thermal time scales, not simply average torque.

1 Why copper loss grows quadratically

Winding resistance R produces heat I²R. If torque is proportional to current, doubling torque gives approximately four times the copper loss. Thermal capacitance stores short-term energy, while thermal resistance limits heat rejection. Allowed peak duration therefore depends on starting temperature and subsequent cooling time.

$$P_{cu} = I^2 R, \qquad C_{th}\dot{\Delta T} = P_{loss} - \frac{\Delta T}{R_{th}}$$

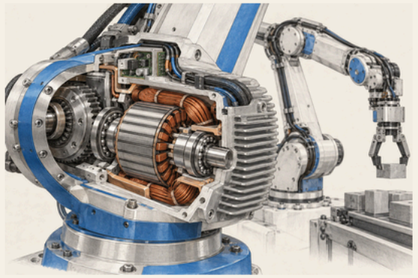
2 Compute RMS from the duty cycle

Consider a repeating 10-second cycle: 1 N·m for 2 seconds and 0.2 N·m for 8 seconds. Weight squared values by duration, divide by total time and take the square root, giving RMS torque 0.4817 N·m. The arithmetic mean is only 0.36 N·m and understates copper heating. Negative braking torque can also contribute through squared current. Squaring gives disproportionate weight to brief high-torque intervals. Equal durations at +1 and −1 N·m give zero mean torque but an RMS of 1 N·m. Reversing mechanical direction therefore does not cancel winding heating.

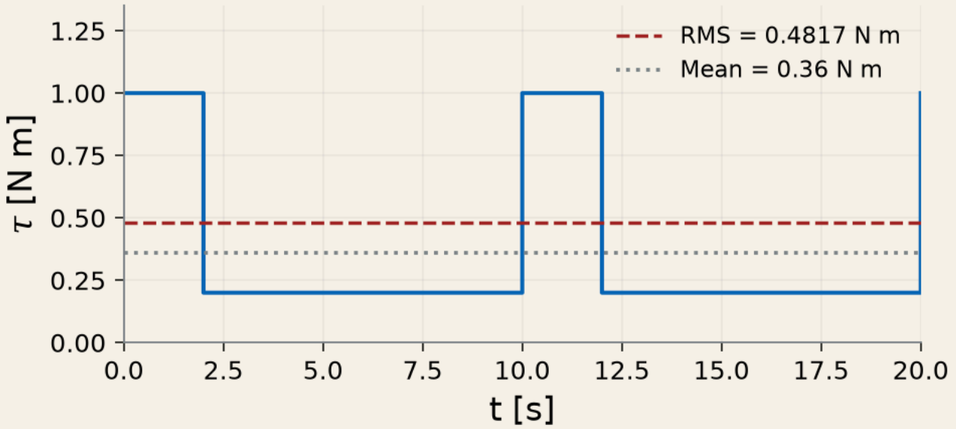
$$\tau_{RMS} = \sqrt{\frac{1^2(2) + 0.2^2(8)}{10}} = 0.4817 \text{ N m}$$

3 RMS alone is not sufficient

RMS summarizes average copper heating when cycles are short relative to thermal time constants and kt and R are approximately fixed. A long peak interval can still overheat the winding. Iron, bearing and drive losses, ambient temperature and mounting heat rejection also matter. Read continuous torque ratings together with their specified cooling and ambient conditions.



Blue shows duty-cycle torque and dashed red its RMS, which is not the arithmetic mean of the trace.



Assumptions and physical limits

The duty cycle is an educational example. Verify actual peak duration using the manufacturer’s thermal model or measurements; an RMS threshold alone does not guarantee protection.

S49 · maxon, thermal ratings; S63 · Asada, Robot Actuators; original RMS example.

Torque–speed and RMS derivations

From the steady electrical equation to power and heating

1 Eliminate current to obtain the torque line

At steady current Li=0, so i=(V−keω)/R. Substituting into τ=kt i gives a linear reduction in torque with speed. Power is τω; differentiation in ω gives its maximum. This is an ideal constant-voltage armature model. The second derivative is −2ktke/R, so positive parameters make the stationary point a maximum. Mechanical power vanishes at both ends: speed is zero at stall and torque at ideal no-load. These limits verify the physical meaning of the derivative result.

$$P(\omega) = \frac{k_t}{R}(V\omega - k_e\omega^2)$$

$$\frac{dP}{d\omega} = \frac{k_t}{R}(V - 2k_e\omega) = 0 \Rightarrow \omega_* = \frac{V}{2k_e}$$

2 Calculate the mathematical maximum

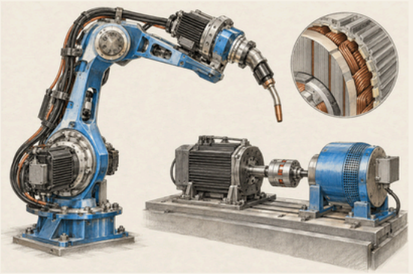
Take V=24 V, R=2 Ω and kt=ke=0.1 in SI units. Maximum power occurs at 120 rad/s, 0.6 N·m and 72 W. Current is 6 A and copper loss also 72 W; maximum output power does not guarantee efficiency or continuous operation.

$$P_{Cu} = I^2R = 6^2(2) = 72\ \text{W}$$

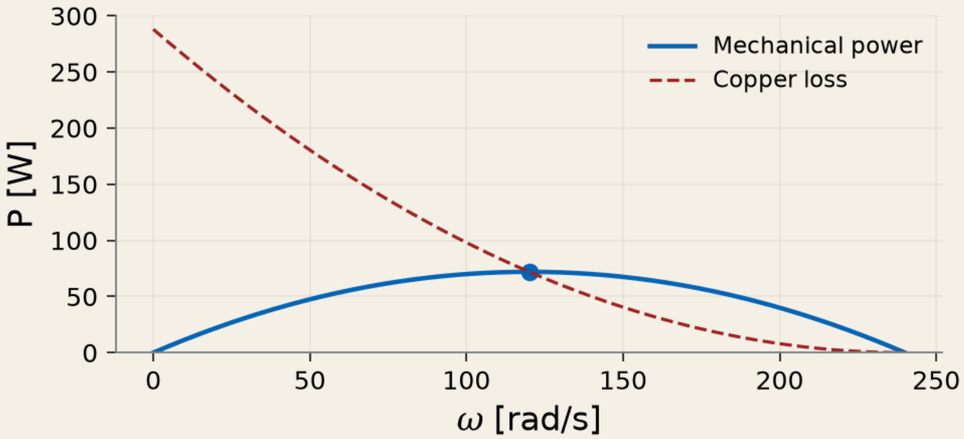
3 From heating to RMS

Average copper loss over a cycle is R/T times the integral of squared current. The constant current producing the same heat is RMS current. With fixed kt, apply the same definition to torque: 1 N·m for 2 s and 0.2 N·m for 8 s gives 0.4817 N·m RMS; the 0.36 mean is insufficient.

$$\tau_{RMS} = \sqrt{\frac{2(1)^2 + 8(0.2)^2}{10}} = 0.4817\ \text{N m}$$



At maximum power, mechanical output equals copper loss; this does not imply high efficiency.



Assumptions and physical limits

RMS equivalence assumes constant resistance and an appropriate thermal time scale. Check peak duration, speed limit and cooling separately.

Motor and gearbox selection

Reduction ratio decreases reflected load torque while changing motor speed and rotor acceleration torque.

1 Define time-varying output requirements

Compute output load torque from q(t), $\dot{q}(t)$, $\ddot{q}(t)$, payload and gravity. One maximum torque value is insufficient: specify its speed and duration. Each ratio candidate multiplies motor speed by N and approximately divides load torque by ηN. Here load torque excludes rotor inertia; avoid counting the same inertia twice. The rotor term multiplies by N rather than dividing by it: a fixed output acceleration requires N times the rotor acceleration. Larger reduction lowers the referred load torque but increases rotor acceleration torque. Selection must evaluate both effects throughout the task.

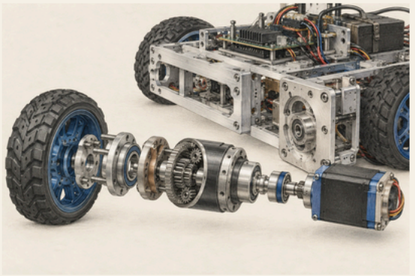
$$\omega_m = N\dot{q}, \qquad \tau_m \approx \frac{\tau_{load}}{\eta N} + J_m N \ddot{q}$$

2 Compare two candidate ratios

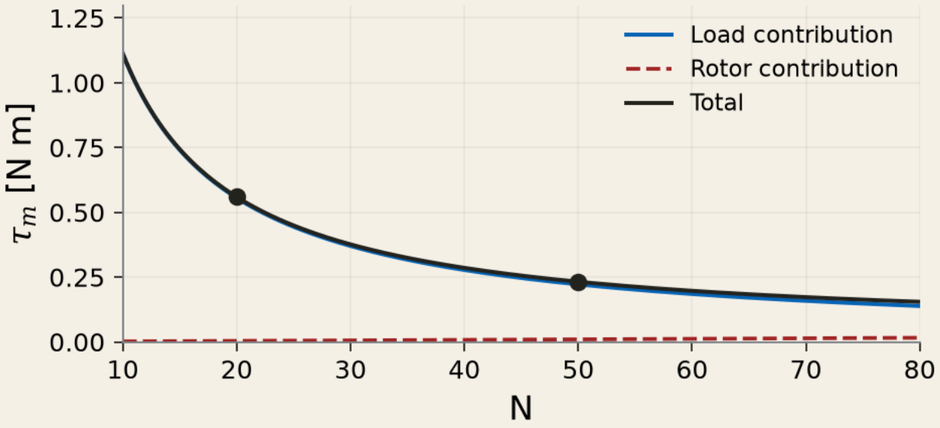
Let τload = 10 N·m, η = 0.90, Jm = 10^{-4} kg·m², $\ddot{q}$ = 2 rad/s² and $\dot{q}$ = 1 rad/s. N = 20 requires 0.5596 N·m at 20 rad/s; N = 50 requires 0.2322 N·m at 50 rad/s. Increasing ratio reduces the load contribution but raises the rotor contribution. Lower torque demand alone does not make the second candidate better.

3 Screen against every relevant limit

First place every torque–speed point inside the voltage/current envelope, then check RMS current, peak duration and temperature. Evaluate gearbox output-bearing loading, stiffness, backlash, lifetime and backdrivability separately. Efficiency can change with power-flow direction. Selection ends with validation under the actual task profile, not simply a catalog choice.



The graph separately computes load and rotor torque contributions and marks N = 20 and N = 50.



Assumptions and physical limits

This comparison assumes fixed forward-power efficiency. Model regeneration, friction and additional transmission inertias separately during detailed sizing.

Wheel-drive sizing

Adequate motor torque does not guarantee that wheel–ground traction can transmit the required force.

1 Sum forces along the slope

m is mass [kg], a forward acceleration [m/s²], θ slope and cr rolling-resistance coefficient. Sum acceleration demand, uphill gravity and rolling loss along the travel direction. Aerodynamic drag is neglected in this low-speed example. For 40 kg, 0.5 m/s², 5° and cr = 0.02, contributions are 20.00, 34.20 and 7.82 N, totaling about 62.02 N.

$$F_{req} = ma + mg\sin\theta + c_r mg\cos\theta$$

2 Convert force to wheel torque and power

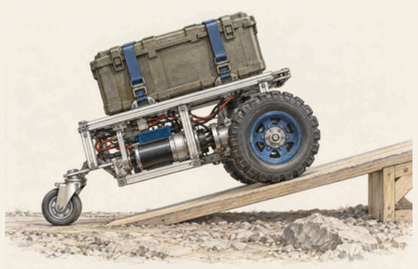
With two equally sharing driven wheels of radius rw = 0.10 m, torque is 3.10 N·m per wheel. At 1 m/s vehicle speed, wheel speed is 10 rad/s and total road mechanical power 62.02 W. N = 20 and η = 0.85 give approximately 0.1824 N·m and 200 rad/s per motor; rotor acceleration torque and drive loss have not yet been added. nd counts driven wheels sharing traction, not passive casters. Equal wheel torque also assumes symmetric sharing. Turning or unequal ground conditions change individual normal loads and traction capacities, requiring wheel-by-wheel checks.

$$\tau_w = \frac{F_{req} r_w}{n_d}, \quad \omega_w = \frac{v}{r_w}, \quad P = F_{req} v$$

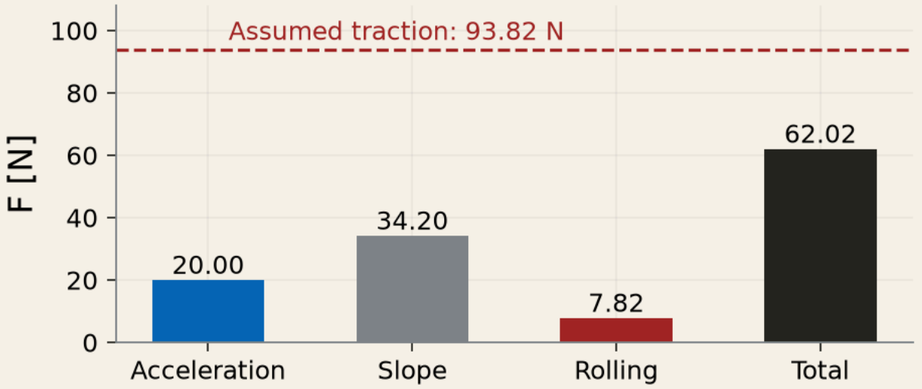
3 Complete the traction check

Transmittable force is limited by μNdrive, where Ndrive is total normal load on driven wheels, not necessarily mg. If that load is 60% of total normal force and μ = 0.40, traction capacity is about 93.82 N, exceeding the 62.02 N demand. The 60% share is an assumption; derive actual loading from slope, CoM height, acceleration and axle geometry.

$$F_{req} \le \mu N_{drive}$$



Force components are computed along the same slope; drive traction imposes an independent upper bound.



Assumptions and physical limits

Assumptions are equal wheel sharing, small rolling resistance and no longitudinal slip. Ramp transitions and caster load transfer need separate analysis; the illustration is not a measured prototype of the numerical model.

From command to motor current

PWM switches voltage; motor inductance turns that voltage into a time-varying current.

1 Duty ratio and average voltage

With switching period Ts and on-time ton, D = ton/Ts is dimensionless duty ratio. In a unidirectional buck-like drive, motor voltage is Vdc during on-time and approximately zero during freewheeling, giving mean D Vdc. A 24 V bus at D = 0.40 gives 9.6 V, not 9.6 A or necessarily 40% torque.

$$D = \frac{t_{on}}{T_s}, \qquad \bar{V} \approx DV_{dc}$$

2 The circuit equation determines current

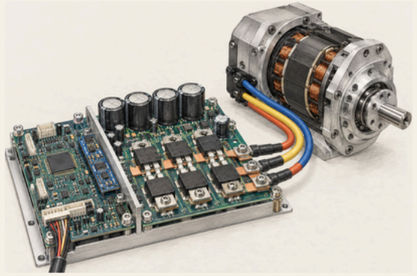
Motor inductance L [H] prevents an instantaneous current jump. Subtract resistive drop Ri and back-EMF eb from applied voltage to obtain inductor voltage. With R = 1 Ω, eb = 8 V and 9.6 V average voltage, steady average current is approximately 1.6 A. Changing speed changes eb, so identical duty ratios can produce different currents. Current can rise during the on interval and fall during the off interval while inductance maintains continuity. That is why the rectangular voltage and small current ripple have different shapes. Mean current governs ideal torque, while RMS current determines copper heating.

$$L\frac{di}{dt} = V_{applied} - Ri - e_b$$

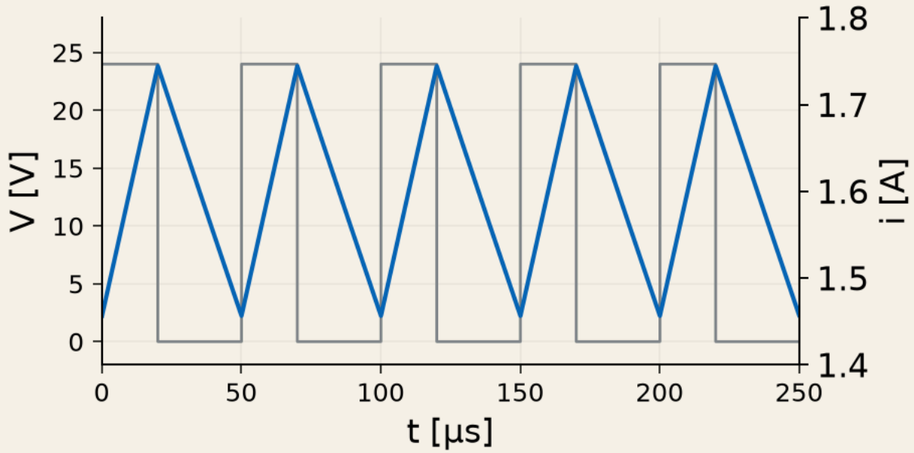
3 State the H-bridge switching mode

A four-switch H-bridge reverses motor-voltage polarity. Bipolar PWM uses +Vdc and −Vdc states, giving mean (2D−1)Vdc; D = 0.50 then gives zero average voltage. No single PWM-voltage formula applies to every drive without specifying conducting switches and freewheel paths.

$$\bar{V}_{bipolar} = (2D - 1)V_{dc}$$



0/Vdc PWM voltage and continuous motor current are plotted on the same time axis with separate units.



Assumptions and physical limits

The plot assumes ideal switching without device drops or dead-time effects. Averaging requires separation between electrical and switching time scales.

Switching and current paths

Turning a switch off does not instantaneously eliminate inductive load current.

1 Provide a path for stored magnetic energy

An inductor stores EL = Li²/2. For L = 1 mH and i = 2 A, stored energy is 2 mJ. When a high-side switch turns off, positive motor current may freewheel through a lower diode or synchronous MOSFET. Without a path, the inductor generates a large voltage to sustain current. Current and applied voltage need not have the same sign. When the switch opens, the source path is interrupted but the motor–diode loop remains closed. Diode orientation permits the existing current to continue. Treating an open switch as instantaneous zero current may fit a resistor but not an inductive motor load.

$$E_L = \frac{1}{2}Li^2 = 2\ \text{mJ}$$

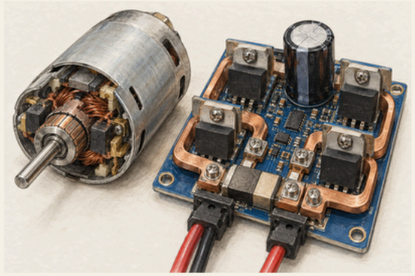
2 Shoot-through and dead time

If both MOSFETs in one half-bridge conduct simultaneously, they short the DC bus. Dead time separates one device turning off from the other turning on. Motor current does not vanish during that interval; it may commutate into a diode. Excessive dead time increases distortion and loss, so choose it from switching and gate-driver delays.

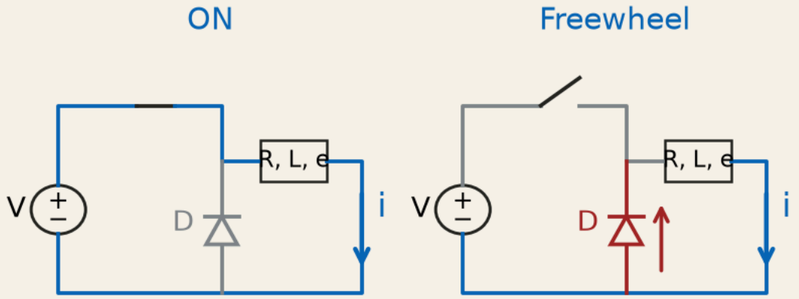
3 Conduction and switching losses

MOSFET conduction loss grows with squared RMS current and temperature-dependent RDS(on). During switching, voltage and current overlap briefly; a triangular overlap estimate gives the expression below. At 24 V, 2 A, 100 ns total transition time and 20 kHz, the estimate is 0.048 W. It excludes reverse recovery, gate charge and parasitic ringing.

$$P_{cond} \approx I_{RMS}^2 R_{DS(on)}, \quad P_{sw} \approx \frac{VI}{2}(t_r + t_f)f_s$$



The closed current path changes between ON and freewheel states; current remains continuous in both.



D: freewheel diode maintains current continuity

Assumptions and physical limits

Switches are ideal functional symbols. Real drives require coordinated validation of gate voltage, common-mode transitions, hardware overcurrent protection and layout.

BLDC/PMSM and servo drives

Electrical field angle and mechanical rotor angle are related by the number of pole pairs.

1 Coordinate rotor and stator fields

A permanent-magnet rotor interacts with the stator field generated by three phase currents. Three half-bridges switch phase voltages, while current control generates the required currents relative to rotor position. BLDC and PMSM labels can overlap in practice; specify the back-EMF waveform and commutation method in the model.

2 Count pole pairs, not poles

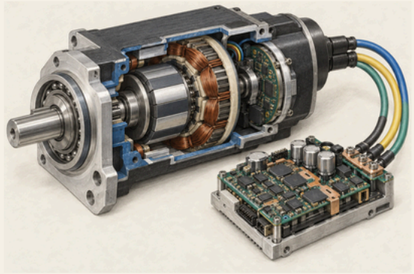
p is pole-pair count, θm mechanical angle, θe electrical angle and θoffset alignment offset. A p = 7 motor has 14 magnetic poles. Mechanical 1000 rpm is 16.667 revolutions per second, so electrical frequency is 116.67 Hz. Encoder zero must be aligned with the rotor magnetic axis to obtain the correct commutation angle. Electrical angle counts repetition of the rotor field relative to the stator windings. One mechanical revolution passes p pole pairs, producing p electrical cycles. Omitting that factor gives incorrect commutation even when mechanical position is measured accurately.

$$\theta_e = p\theta_m + \theta_{offset}, \qquad f_e = p\frac{n_m}{60}$$

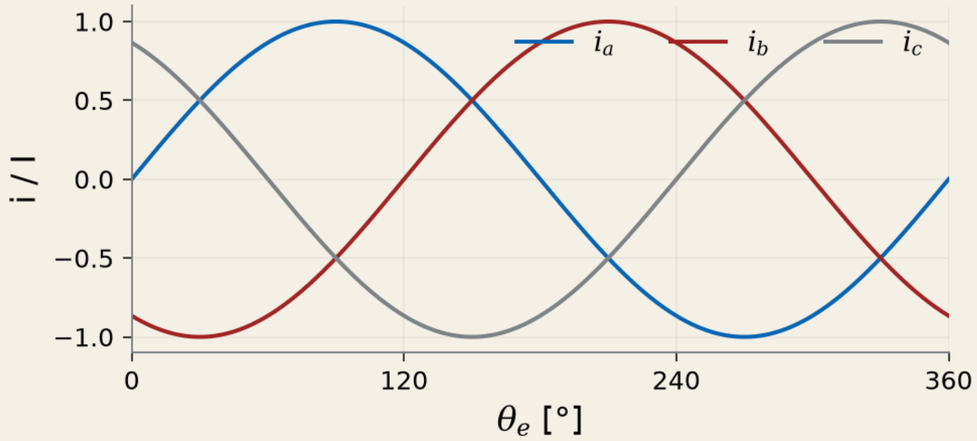
3 Six-step versus sinusoidal drive

Six-step commutation changes phase pairs by electrical sector; Hall sensors can identify sectors. Sinusoidal control/FOC regulates the current vector using continuous angle measurement or estimation. In a balanced three-wire motor, ia+ib+ic is approximately zero, so two measurements may reconstruct the third. Sampling timing and switching noise complicate that reconstruction.

$$i_a + i_b + i_c \approx 0$$



Three sinusoidal phase currents are separated by 120 electrical degrees; the graph is an ideal balanced-current example.



Assumptions and physical limits

Hall sector information is not high-resolution absolute angle measurement. Sensorless estimation can be limited at low speed; specify the method and operating range.

S46 · ST, PMSM/BLDC control and FOC documentation; S51 three-phase inverter; original pole-pair example.

Servo architecture in a robot joint

Position error becomes torque demand, then motor current; measurement location is part of the architecture.

1 The outer loop defines the task

Position error is desired qd minus measured q. Proportional action responds to position error, derivative action to velocity error, and feedforward τff can compensate expected gravity or acceleration loading. Gain units matter: Kp [N·m/rad], Kd [N·m·s/rad]. The commanded torque below is referenced to the gearbox output joint. The proportional term behaves like a restoring spring and the derivative term like damping against velocity error. No physical spring is added; the behaviour is produced through measurement and the current loop. Delay limits how closely this analogy holds.

$$\tau^*_{out} = K_p(q_d - q) + K_d(\dot{q}_d - \dot{q}) + \tau_{ff}$$

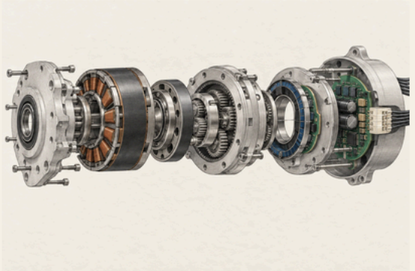
2 The inner loop regulates physical current

Convert output torque demand to motor current using ratio N and efficiency η; kt is the motor-shaft constant. A 2 N·m demand with N = 20, η = 0.85 and kt = 0.08 requires about 1.47 A. The current loop compares measured and demanded current and adjusts inverter voltage. Voltage saturation may prevent that current at some speeds.

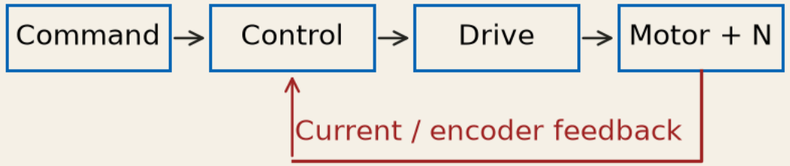
$$i^* \approx \frac{\tau^*_{out}}{\eta N k_t} = 1.47 \text{ A}$$

3 Check where the encoder is mounted

A motor encoder measures θm; ideal rigid transmission gives q = θm/N. Backlash or compliance breaks that exact load-side relation. An output encoder measures actual joint position, but dual-sensor feedback must account for flexible modes. A cascade may use position to command velocity and velocity to command current; architectures need not all use identical blocks.



Each feedback signal is tied to a physical sensor location; motor and load angles are separate variables in a compliant transmission.



Functional architecture; not a circuit schematic

Assumptions and physical limits

These equations explain architecture, not validated gains for a real robot. Evaluate current/torque limits, measurement delay and mechanical resonance together.

Field-oriented control

FOC regulates the three-phase current vector in rotor-aligned coordinates to simplify torque control.

1 First move to two stationary axes

Balanced currents satisfy ia+ib+ic = 0, leaving two independent components. An amplitude-invariant Clarke transform maps them into stationary stator αβ axes. State the normalization explicitly: this slide preserves phase peak amplitude, so a dq current is not directly the phase RMS current.

$$i_\alpha = i_a, \qquad i_\beta = \frac{i_a + 2i_b}{\sqrt{3}}$$

2 The rotor-aligned d-q axes

Align d with rotor magnet flux and q 90° ahead in the positive rotation direction. θe is electrical rotor angle. The Park transform changes coordinates of the same current vector rather than physical current. Synchronously rotating sinusoidal currents become approximately constant in this frame and can be tracked by PI regulators. An observer rotating with a rotating arrow sees an approximately stationary arrow. Likewise, d–q control regulates nearly constant components instead of continuously varying sinusoids. An incorrect rotor angle misaligns the transformation and mixes the intended current components.

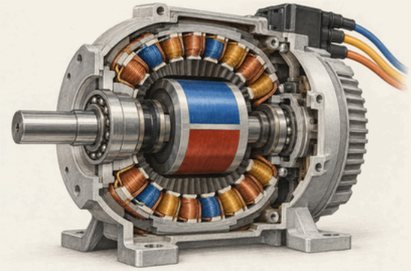
$$i_d = i_\alpha \cos\theta_e + i_\beta \sin\theta_e$$

$$i_q = -i_\alpha \sin\theta_e + i_\beta \cos\theta_e$$

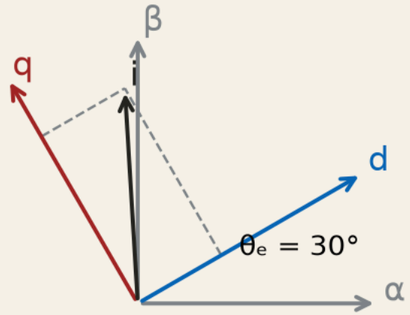
3 Two contributions to electromagnetic torque

p is pole-pair count, ψf magnet flux linkage [Wb], and Ld/Lq axis inductances [H]. The first term is magnet torque; the second is saliency-based reluctance torque. For an ideal surface-magnet motor Ld ≈ Lq. With id = 0, p = 4, ψf = 0.02 Wb and iq = 3 A give 0.36 N·m. High-speed voltage limits may require field weakening.

$$\tau_e = \tfrac{3}{2} p \left[\psi_f i_q + (L_d - L_q) i_d i_q\right]$$



The αβ and dq axes describe the same current vector; changing coordinates does not create a new physical force.



Assumptions and physical limits

Convention: amplitude-invariant transform, peak current and electrical angle. id = 0 is not optimal for every motor and speed; saliency and voltage/current limits change the strategy.

Steps of the Clarke–Park transform

From three phases to a fixed plane and then rotor axes

1 State the Clarke scaling

Assume ia+ib+ic=0 in a three-wire system. The amplitude-invariant Clarke transform projects axes spaced 120° apart and scales by 2/3. This gives iα=ia and iβ=(ia+2ib)/√3. Zero-sequence current is absent under this assumption.

$$i_\alpha = \tfrac{2}{3}(i_a - i_b/2 - i_c/2), \quad i_\beta = \tfrac{\sqrt{3}}{3}(i_b - i_c)$$

2 Park expresses the same vector in a rotating basis

Let d point along rotor electrical angle θe, with q +90° ahead. Dot products of the current vector with these unit axes give id and iq. Reverse the transform using −θe; the physical vector stays unchanged while coordinates change.

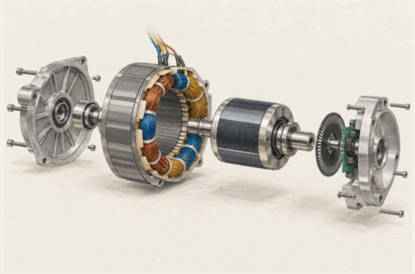
$$i_d = i_\alpha \cos\theta_e + i_\beta \sin\theta_e$$

$$i_q = -i_\alpha \sin\theta_e + i_\beta \cos\theta_e$$

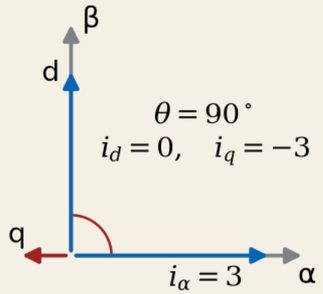
3 A 90° test and power scaling

For ia=3 A and ib=ic=−1.5 A, iα=3 A and iβ=0. At θe=90°, id=0 and iq=−3 A. An amplitude-invariant transform does not preserve the three-phase sum norm: instantaneous power includes a 3/2 factor, consistent with the torque convention. In the 90° example, current still points right in stator coordinates; only the new d axis points upward. Negative q current therefore does not mean the physical vector was reversed. This sign test separates coordinate representation from an actual current command.

$$p = v_a i_a + v_b i_b + v_c i_c = \tfrac{3}{2}(v_d i_d + v_q i_q)$$



In the 90° example, the current vector remains along +α but has a negative q component.



Assumptions and physical limits

Electrical angle is mechanical angle times pole-pair count. A power-invariant Clarke scaling requires corresponding coefficient changes.

Current, velocity and position loops

An outer loop must account for what the inner loop can deliver and how long it takes.

1 Why make the inner loop faster?

The current loop regulates electrical dynamics, velocity control regulates mechanical acceleration, and position control tracks trajectory error. If the outer loop changes demands too quickly, inner-loop tracking lag adds phase delay. Bandwidth separation is useful, but a fixed “ten times faster” rule is not a stability proof.

2 PI action and saturation

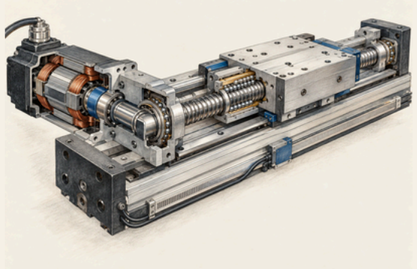
Let PI output be u = Kp e+xI with integral state ẋI = Ki e. If voltage saturates while the integral keeps accumulating, windup occurs. Stored integral can hold the command high even after the error reverses. Higher gain does not automatically solve this; feed the actuator limit back into the controller. Integrating error can generate the steady command needed against a persistent load. Once the drive saturates, further accumulation cannot create more actual torque. Windup is the divergence between the controller’s internal demand and the physically available input.

$$C_{PI}(s) = K_p + \frac{K_i}{s}, \qquad u_{sat} = \mathrm{clip}(u, -U_{max}, U_{max})$$

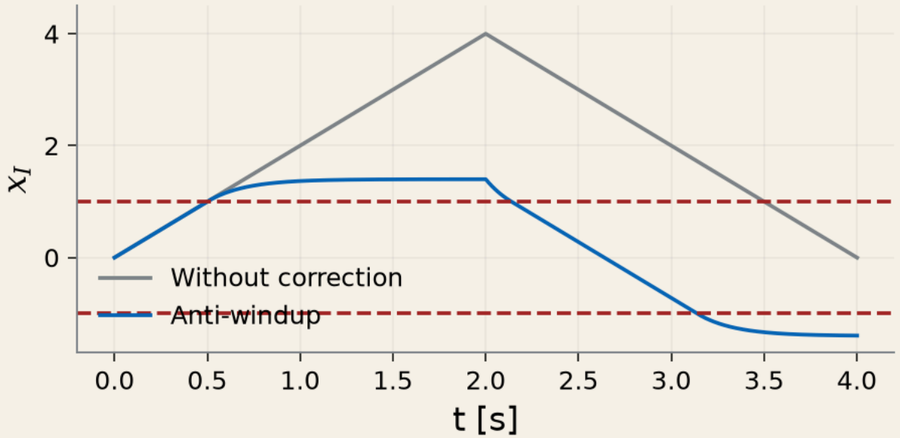
3 Back-calculation anti-windup

Add the difference between achievable usat and requested u to the integrator derivative. At upper saturation, usat−u is negative, so positive Kaw reduces accumulation. To isolate the effect, the example uses Kp = 0, Ki = 2, Kaw = 5 and limits ±1. The plot compares integrator states under the same error, not validated motor-control tuning.

$$\dot{x}_I = K_i e + K_{aw}(u_{sat} - u)$$



Integrator states are compared under the same error profile; negative feedback of the saturation difference limits accumulation.



Assumptions and physical limits

The example error is +1 for 2 s then −1 for 2 s. In a real loop error comes from the plant; also model sampling, filtering and computation delay.

S46 · ST UM1052, current/speed loop architecture; original back-calculation illustrative simulation.

Actuator feedback sensors

Resolution, accuracy and knowing position after restart are three different requirements.

1 Incremental encoder and quadrature

A and B channels are shifted by one quarter cycle. Edge order determines direction, and counted edges determine displacement. With 1024 lines per revolution and x4 decoding of both edges on both channels, there are 4096 counts/revolution. Angular increment is 360°/4096 = 0.08789°. Do not multiply by four again if the manufacturer's PPR/CPR definition already includes decoding.

$$N_{counts} = 4N_{lines}, \qquad \Delta\theta = \frac{2\pi}{N_{counts}}$$

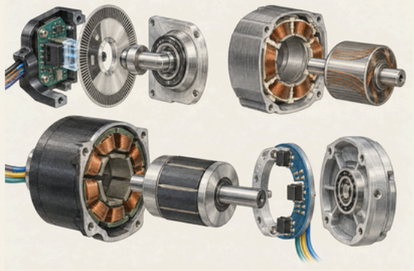
2 Absolute measurement and referencing

An incremental encoder counts changes after startup and needs homing, an index or another reference. A single-turn absolute encoder directly encodes angle within one revolution; multi-turn information may require additional hardware or retention. A resolver measures sine/cosine amplitudes relative to excitation. Hall sensors can identify commutation sectors but do not provide equivalent angular resolution.

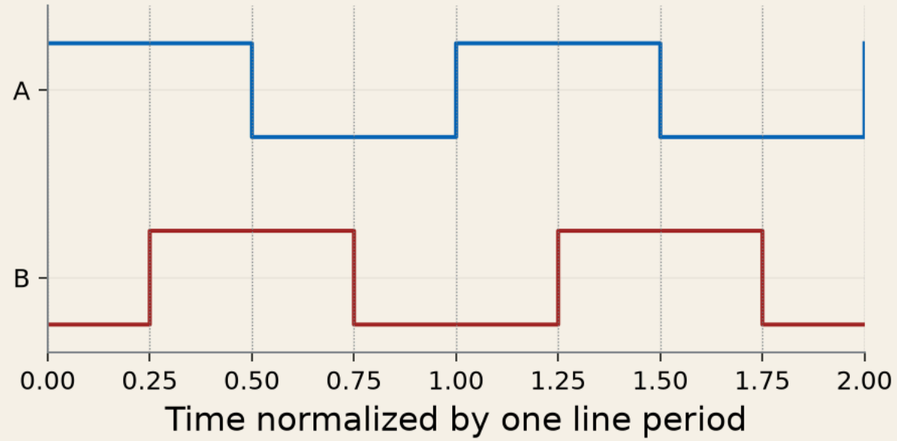
3 Time scale in velocity estimation

Estimate speed from count change over a fixed Δt. With 4096 counts/revolution and a 1 ms window, one count corresponds to about 1.534 rad/s. At low speed, longer windows or edge-period measurement can help, at the cost of delay. Mechanical eccentricity, interpolation and mounting errors limit accuracy independently of resolution. Zero counts in a short window do not prove that the motor stopped; motion may not yet cross a count boundary. A longer window resolves smaller speeds but averages over older motion. This creates the resolution–latency trade-off.

$$\hat{\omega} = \frac{2\pi\Delta n}{N_{counts}\Delta t}$$



Four A/B edges are counted per line period; direction follows which channel transitions first.



Assumptions and physical limits

Keep counts, lines and revolutions distinct. A motor-shaft encoder relates to output angle through ratio and transmission errors; extra bits do not remove backlash.

Robot measurement chain

A correct ADC code cannot repair a wrong sensor scale or a saturated analog stage.

1 Convert the physical quantity to voltage

Let x be the measured quantity and Vs the sensor output. A linear model is Vs = Sx+V0, where S is sensitivity and V0 zero offset. Amplifier gain G and offset adapt this range to the ADC. Check sensor range, amplifier input/common-mode limits and ADC full scale separately.

$$V_s = Sx + V_0, \qquad V_{ADC} = GV_s + V_{offset}$$

2 Calibration with explicit units

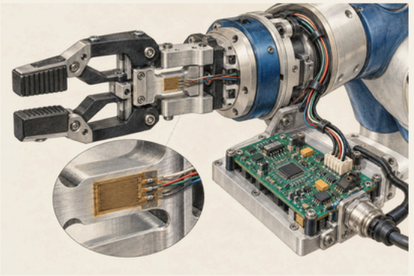
A 0–10 A sensor producing 0.2–2.8 V has sensitivity (2.8−0.2)/10 = 0.26 V/A. For a 1.5 V measurement, first subtract the 0.2 V offset, then divide by 0.26 to obtain 5 A. Dividing 1.5 by 0.26 alone mistakes offset for current. Two-point calibration corrects linear gain/offset, not all nonlinearity or temperature drift. Calibration inverts the measurement model: subtract the fixed voltage offset, then divide by sensitivity to recover amperes. Follow the same unit-consistent chain through amplifier gain and ADC code-to-voltage conversion rather than mixing intermediate representations.

$$I = \frac{V_s - 0.2\ \mathrm{V}}{0.26\ \mathrm{V/A}} = 5\ \mathrm{A}$$

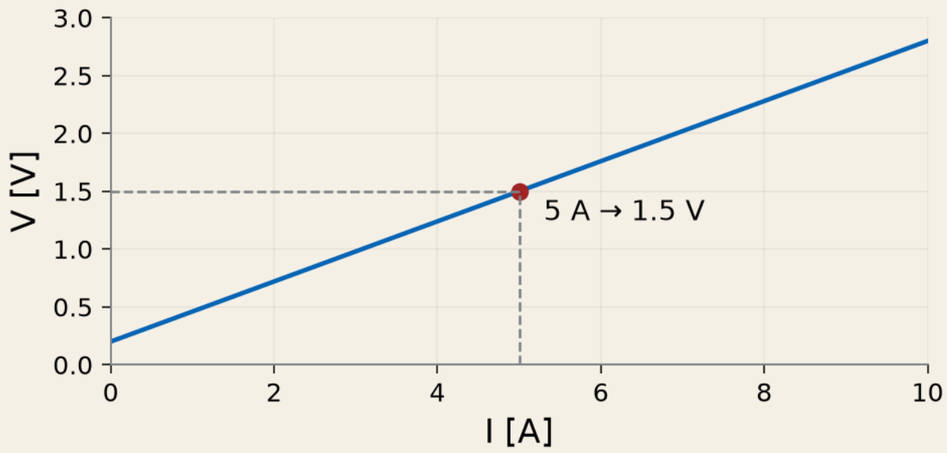
3 Filtering, noise and saturation

An analog filter before the ADC attenuates unwanted high frequencies; a single RC cutoff is 1/(2πRC). Filtering reduces noise but adds delay and attenuation. An out-of-range ADC input clips, and later digital filtering cannot restore the lost information. The slowest or most range-limited stage may dominate the measurement chain.

$$f_c = \frac{1}{2\pi RC}$$



The calibration line maps current to voltage; the red point shows 1.5 V corresponding to 5 A.



Assumptions and physical limits

Calibration, filtering and sampling address different error mechanisms. None replaces the others. Ground-reference error between sensor and ADC can shift the measurement.

Resolution, sampling and latency

More bits do not compensate for incorrect sampling or measurement latency.

1 Compute the ideal ADC step

An ideal N-bit ADC has 2^N code intervals. For full-scale span VFS [V], ideal step q = VFS/2^N; 3.3 V at 12 bits gives 0.8057 mV. If sufficiently excited ideal quantization error is uniform over [−q/2,q/2], its standard deviation is q/√12. That assumption may fail for constant or deterministic low-amplitude signals. Adding one bit halves the ideal code step, not sensor offset or reference-voltage error. More codes allow smaller differences to be represented. Agreement with the physical value still requires a separate measurement-error budget.

$$q = \frac{V_{FS}}{2^N}, \qquad \sigma_q = \frac{q}{\sqrt{12}}$$

2 Aliasing makes different frequencies indistinguishable

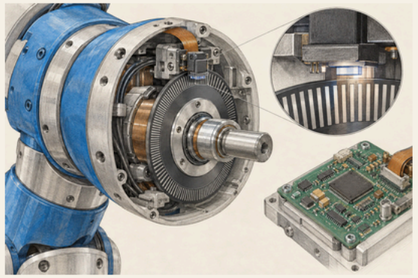
At 1 kHz sampling, a 900 Hz cosine has the same sampled values as a 100 Hz cosine. At sample instants, 900k/1000=k−100k/1000; remove the integer turns and use the even symmetry of cosine. A band-limited signal must remain below fs/2, with extra margin for a real anti-alias filter transition band. Post-sampling filtering cannot separate components already aliased together.

$$\cos(2\pi 900k/1000) = \cos(2\pi 100k/1000)$$

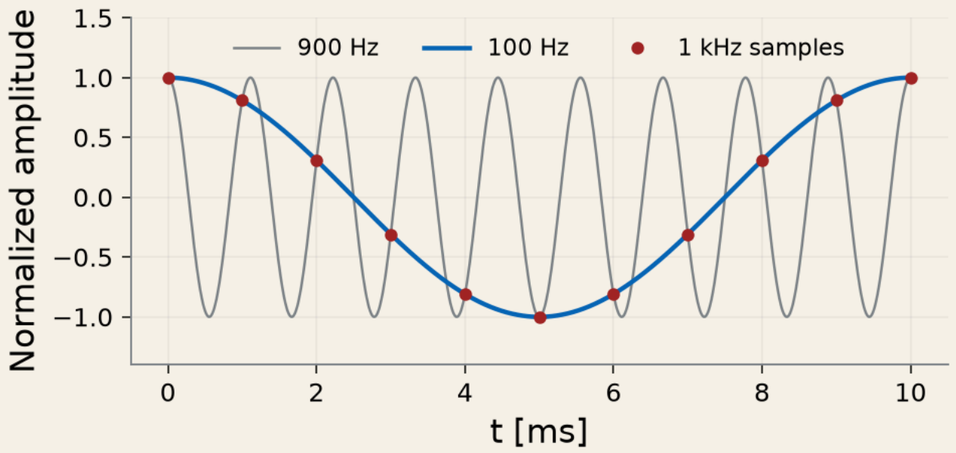
3 Measurement age is part of the information

Latency is the time between an event and using its data; jitter is variation in that delay. A timestamp at acquisition is more informative than one assigned on arrival. A robot moving at 1 m/s travels 2 cm during 20 ms latency. Higher ADC resolution does not remove this geometric discrepancy.

$$\Delta x \approx v\Delta t = 0.02 \text{ m}$$



The 900 Hz and 100 Hz curves differ, but 1 ms sample points coincide. This is a computed aliasing example, not measured data.



Assumptions and physical limits

LSB interval and a 2^N-1 endpoint calibration scale are different definitions. Resolution is not accuracy or effective number of bits; include noise and reference error separately.

ADC and sampling equations

Code spacing, quantization error and frequency ambiguity

1 Divide the voltage range into codes

N bits provide 2^N codes. An ideal uniform ADC has step $q=VFS/2^N$. For 3.3 V and 12 bits, q=0.8057 mV. Endpoint calibration may instead use 2^N-1, a different code-to-voltage convention; do not mix them without explanation.

$$q = 3.3/4096 = 0.805664 \text{ mV}$$

2 Integrate the error variance

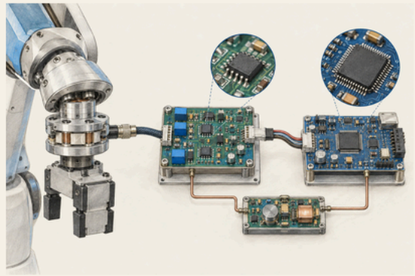
Assume ideal rounding error is uniform from −q/2 to +q/2. Its mean is zero and density 1/q, so integrating the second moment gives $q^2/12$. Standard deviation is $q/\sqrt{12}=0.2326$ mV. This is not total sensor or ADC accuracy. The density height is 1/q so its total probability area equals one. Symmetry makes mean error zero, but squared error is positive on both sides. Zero mean error therefore does not imply zero variance or zero measurement uncertainty.

$$\sigma_q^2 = \frac{1}{q}\int_{-q/2}^{q/2} e^2\,de = \frac{q^2}{12}$$

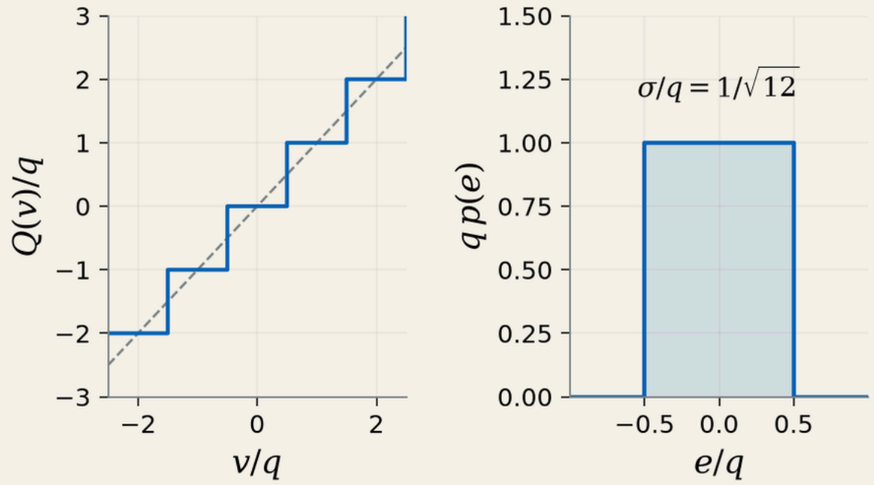
3 See alias equivalence at the sample index

At t=n/fs, adding integer kfs to frequency adds phase 2πkn, leaving samples unchanged. At fs=1000 Hz, cos(2π900n/1000)=cos(2π100n/1000). The corresponding sine example has reversed sign. Digital processing cannot restore the lost distinction; limit analog bandwidth.

$$\cos(2\pi(f+kf_s)n/f_s) = \cos(2\pi fn/f_s)$$



The staircase quantizer and error density are two views of the same ideal model.



Assumptions and physical limits

The independent uniform-error model can fail for small or constant signals. Real ADCs also have offset, gain, nonlinearity and reference noise.

Valvano Shape the World Ch13–14; original variance integral and alias phase check.

Robot-joint electronics

Power switching and precision sensing meet on one board; return paths and measurement references are central.

1 Connect functions through actual signals

MCU PWM outputs feed gate drivers, which drive MOSFET gates. Power MOSFETs switch motor phases; these current paths differ from logic-level PWM pins. Current feedback returns through the shunt–amplifier–ADC chain. Encoder inputs provide rotor position, and communications carry commands from the upper-level controller.

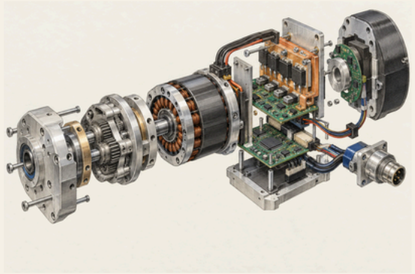
2 Shunt example: small signal, real heating

At I = 10 A and Rsh = 5 mΩ, shunt drop is 50 mV and loss 0.5 W. Gain G = 20 produces a 1 V signal. An input-referred 1 mV offset corresponds to 0.2 A current error before amplification. Kelvin sensing separates measurement terminals from additional copper voltage drop in the power path. A smaller shunt dissipates less heat but also gives a smaller signal, increasing the relative importance of amplifier offset and noise. Minimizing resistance purely for efficiency can degrade current feedback. Selection must balance power loss and measurement quality.

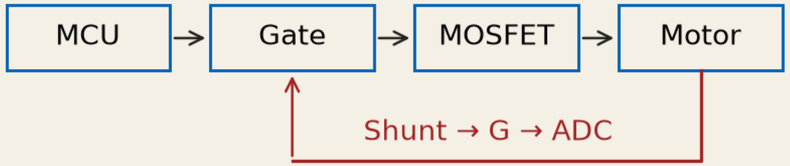
$$V_{sh} = IR_{sh}, \quad P_{sh} = I^2R_{sh}, \quad V_{ADC} = GV_{sh}$$

3 References, isolation and protection

Variable ground drop between ADC and shunt references creates current error. An isolation barrier separates the required voltage domains; not every sensor line must be isolated. Hardware overcurrent shutdown can be complemented by software logging and controlled recovery. Define which fault each protection path is intended to cover.



Power, control and sensing paths are distinguished within one joint board; current feedback is physically tied to the motor phase.



Functional architecture; not a circuit schematic

Assumptions and physical limits

A functional schematic is not a production-ready PCB. The TI reference separates tested power/sensing functions from incompletely tested diagnostic/shutdown options; these are not certification claims.

Robot power architecture

Distribute energy while preserving control and measurement

1 One battery, different power paths

After protection, the battery feeds the DC bus. The inverter supplies motor phases from this bus; DC/DC converters create regulated rails for compute and sensors. Define voltage range, continuous and peak current, and fault behavior for each branch. Power and data cables serve different functions. Do not add currents measured on different voltage rails as though they were all battery currents. A 2 A load at 5 V differs from 2 A at 24 V. Convert each through power and efficiency to the common battery side first.

$$P_{battery} = P_{drive,\,in} + \sum_{j} P_{rail,\,j}/\eta_j$$

2 Place conversion losses correctly

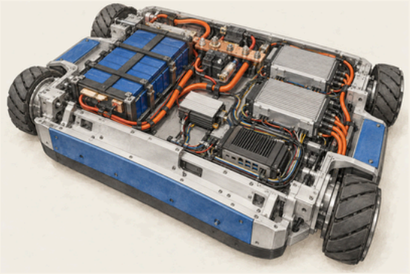
A load consuming 24 W on a 12 V rail does not draw only 24 W from the battery. A converter with efficiency 0.90 needs 26.67 W; 2.67 W becomes heat. Efficiency is output power divided by input power. Efficiencies of cascaded converters multiply.

$$P_{in} = 24/0.90 = 26.67\ \text{W}, \quad P_{loss} = 2.67\ \text{W}$$

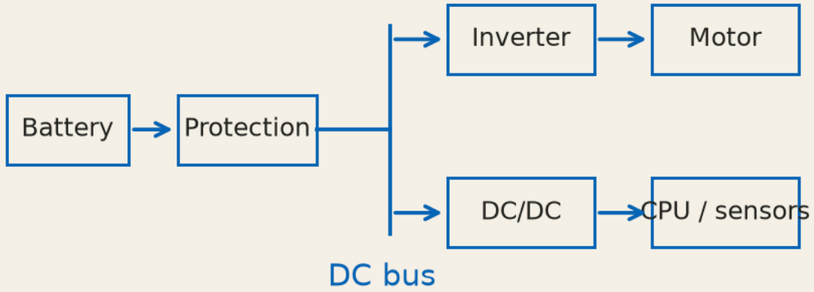
3 Regeneration requires an energy sink

During braking the motor can generate electrical energy. A full battery, limited charge current or an open protection switch may prevent its absorption. Model energy flow, bus capacitance and any braking resistor together; rising bus voltage is not merely a software issue.

$$\frac{d}{dt}\left(\tfrac{1}{2}CV_{bus}^2\right) = P_{regen} - P_{absorbed}$$



Drive and compute use separate power branches; the shared battery remains a common failure source.



Assumptions and physical limits

The diagram is a functional power architecture. Select protection thresholds, isolation and discharge paths using actual component limits.

Battery pack and BMS

Measurement, estimation, balancing and protection are distinct

1 Measure cells, manage the pack

The BMS monitors cell voltages, pack current and selected temperatures. Series cells carry the same current; if one cell reaches its limit, operation must stop even when total pack voltage looks normal. Capacity ageing and present charge level are different quantities. SOC describes the fraction of currently available charge remaining; capacity SOH compares that available capacity with the new battery. An aged battery can show 100% SOC while providing less runtime than a new one.

$$\mathrm{SOC} = Q_{remaining}/Q_{available}, \quad \mathrm{SOH}_Q = Q_{available}/Q_{new}$$

2 Integrating current informs charge estimation

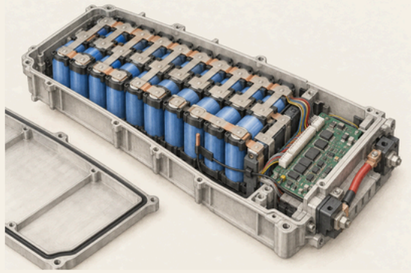
Choose positive current for discharge. When capacity is in ampere-hours, divide the time integral in seconds by 3600. Initial SOC error and current-sensor bias do not disappear automatically. Voltage-based correction depends on chemistry, temperature, rest and hysteresis.

$$\mathrm{SOC}(t) = \mathrm{SOC}(0) - \frac{1}{3600 Q_{Ah}} \int_0^t I(\xi)\, d\xi$$

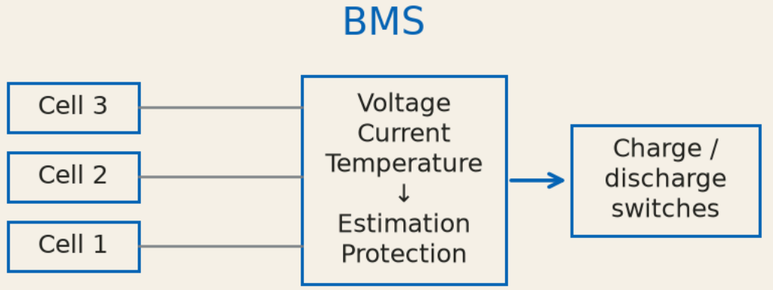
3 Balancing does not replace protection

Passive balancing dissipates excess charge as heat; active balancing transfers energy between cells. A 100 Ω resistor across a 4 V cell carries 40 mA and dissipates 0.16 W. Protection limits charge or discharge for overvoltage, undervoltage, current and temperature conditions.

$$I_{bal} = V/R = 0.04 \text{ A}, \quad P_{bal} = V^2/R = 0.16 \text{ W}$$



Cell terminals are monitored, pack current is shared, and temperature needs multiple physical measurement locations.



Assumptions and physical limits

SOC is an estimate based on models and measurements, not a directly sensed quantity. SOH here is defined only through capacity.

Pack voltage and energy capacity

Series connection raises voltage; parallel connection raises capacity

1 Size from cell to pack

For identical cells define Ns series groups, each containing Np parallel cells. Series voltages add while Ah capacity is unchanged. In parallel, voltage is unchanged and capacities add. Under ideal current sharing, cell current is pack current divided by Np. In series, the same charge passes through every cell, so Ah capacities do not add. Energy increases because each unit of charge spans a higher total voltage. In parallel, charge is shared among branches, increasing capacity instead.

$$V_{pack} = N_s V_{cell}, \quad Q_{pack} = N_p Q_{cell}, \quad I_{cell} = I_{pack}/N_p$$

2 A 7S4P pack example

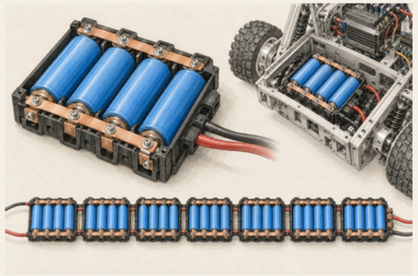
Choose a cell rated 3.6 V nominal, 4.2 V fully charged and 2.5 Ah. Seven series groups give 25.2 V nominal and 29.4 V maximum. Four parallel cells give 10 Ah. There are 28 cells in total; nominal energy is 252 Wh. These values describe the chosen example chemistry.

$$E_{nom} \approx V_{nom} Q_{Ah} = 25.2 \times 10 = 252 \text{ Wh}$$

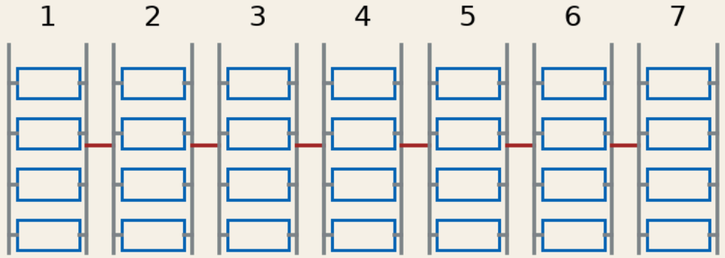
3 Nominal voltage is not a component maximum

Check inverter and DC/DC input ranges against 29.4 V fully charged voltage and possible bus rises. Usable energy changes with low-voltage cutoff, temperature, ageing and load current. Unequal cell resistance prevents equal current sharing in parallel.

$$E_{Wh} = \frac{1}{3600} \int V(t) I(t)\, dt$$



The exact topology shows four parallel cells in each of seven series groups.



Assumptions and physical limits

4.2 V is not universal across battery chemistries. Use the selected cell data sheet for its voltage range and protection thresholds.

Charge, energy and SOC in a battery

Current integration gives charge; energy also needs voltage

1 From coulombs to ampere-hours

Current is charge flow per time: 1 A = 1 C/s. Integrating over seconds gives coulombs; divide by 3600 because 1 Ah = 3600 C. A 2 A current for one hour consumes 7200 C, or 2 Ah. Choose positive current for discharge.

$$Q_{Ah} = \frac{1}{3600}\int I(t)\,dt$$

2 Energy as an area

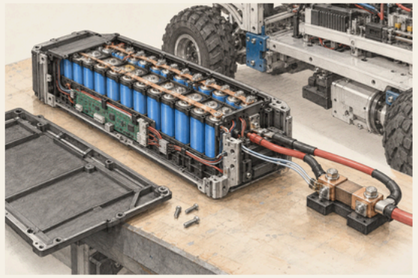
Because voltage can vary, actual energy is the integral of VI. Using charge as the variable makes energy the area under the voltage–charge curve. Constant voltage yields the Vnom×QAh approximation. If voltage falls linearly from 28 to 22 V over 10 Ah, mean voltage is 25 V and energy 250 Wh. Ah measures electric charge; Wh measures energy. Two packs delivering the same 10 Ah at different voltages do not store equal energy. Including V in the integral accounts for that difference; capacity alone is insufficient for a runtime comparison.

$$E_{Wh} = \frac{1}{3600}\int VI\,dt = \int V\,dQ_{Ah}$$

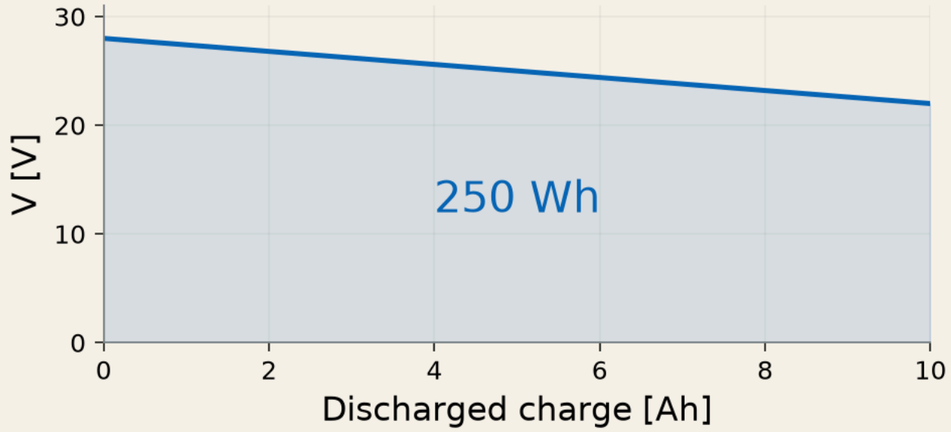
3 Current bias causes SOC drift

A constant +10 mA measurement bias counts an extra 0.1 Ah discharge over 10 hours. For 5 Ah capacity, estimated SOC becomes 0.1/5=0.02, or two percentage points too low. Faster sampling does not remove constant bias; calibration and model-based correction are needed.

$$\Delta\mathrm{SOC} = -\frac{I_{bias}t_h}{Q_{Ah}} = -\frac{0.01(10)}{5} = -0.02$$



Voltage–charge area is energy; bias–time area is accumulated charge error.



Assumptions and physical limits

Capacity depends on temperature, ageing and use. SOC integration needs an initial value; charge-direction coulomb efficiency may need separate modeling.

Power budget and runtime

Separate average energy consumption from peak power demand

1 Weight power by duty duration

Let Pi denote battery-side power of each operating mode and di its time fraction; fractions sum to one. Driving uses 150 W for 40%, sensing 80 W for 30%, and idle 53.33 W for 30%. Average power is 100 W; these modes are mutually exclusive here. The fractions describe mutually exclusive modes. If sensing also operates during driving, include it in that mode’s total power. Counting it both within the mode and as a separate load double-counts energy. Define the task timeline before averaging power.

$$P_{avg} = \sum_i d_i P_i \approx 60 + 24 + 16 = 100 \text{ W}$$

2 Energy divided by power gives time

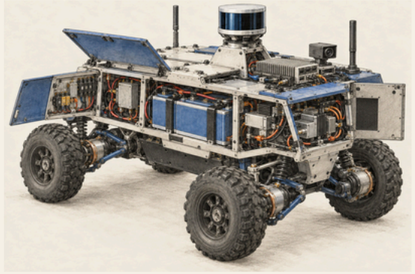
If 80% of a nominal 252 Wh pack is usable, energy is 201.6 Wh. At 100 W average battery power, estimated runtime is 2.016 hours. Do not apply converter efficiency again if powers already refer to the battery. Dividing Ah by watts does not yield hours.

$$E_{usable} = 0.80(252) = 201.6 \text{ Wh}, \quad t = 201.6/100 = 2.016 \text{ h}$$

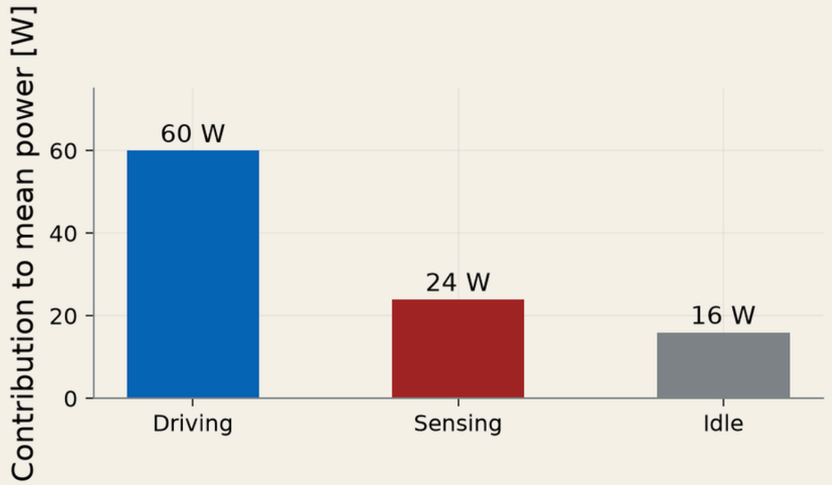
3 Peak power is a separate design condition

A brief acceleration may consume little energy yet exceed the BMS current limit. Check simultaneous peak currents, durations and voltage sag. Reserving energy for return or a safe stop is an engineering decision that determines the usable SOC window.

$$E_{mission} = \sum_i P_i t_i, \quad P_{peak} \neq P_{avg}$$



Bars show each mode’s contribution to the average budget, not its instantaneous power.



Assumptions and physical limits

The estimate assumes fixed mission statistics. Validate real runtime against payload, terrain, temperature and ageing measurements.

Peak current and voltage sag

Adequate energy does not guarantee adequate instantaneous supply

1 A simple Thevenin model

Let Voc be open-circuit voltage and R the combined internal and cable resistance. As discharge current I rises, terminal voltage falls by IR. The resistance dissipates I²R. This model describes a slowly varying operating point; cell polarization and cable inductance are omitted.

$$V_{load} = V_{oc} - IR, \quad P_{load} = V_{load} I, \quad P_{loss} = I^2 R$$

2 Where is the stated voltage measured?

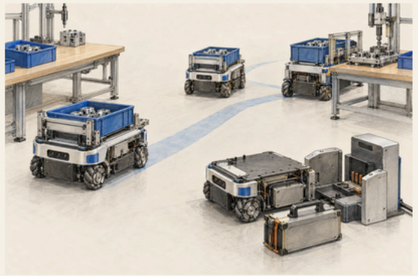
For 22 V measured at the load and 400 W, I = 400/22 = 18.18 A. With R = 0.1 Ω, drop is 1.818 V and loss is 33.06 W; required Voc is 23.818 V. If 22 V were open-circuit voltage, this division would not give the correct current.

$$I = 18.18 \text{ A}, \quad V_{oc} = 22 + 1.818 = 23.818 \text{ V}$$

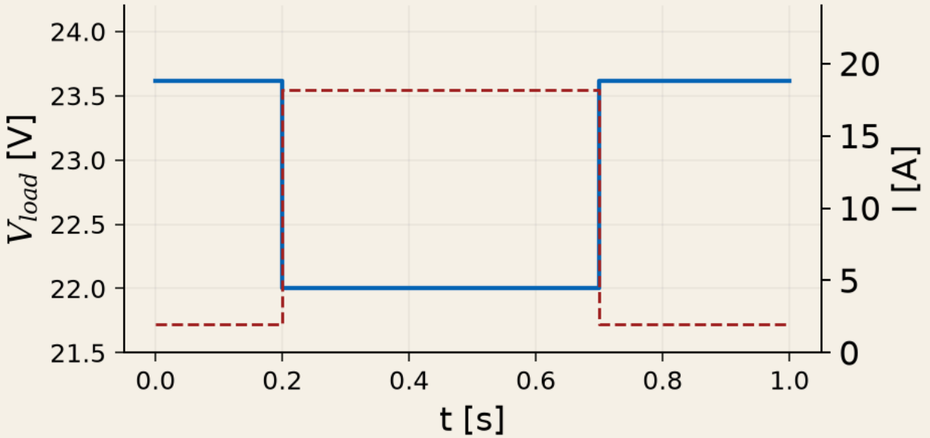
3 Brownout can propagate through a shared bus

Motor acceleration can pull the shared bus below the compute converter’s minimum input. More capacitance can reduce a brief transient but cannot repair a persistently inadequate source. Its energy calculation requires permitted initial and final capacitor voltages. A constant-power converter draws more current as voltage falls, potentially increasing series voltage drop further. A resistive load instead draws less current when voltage decreases. Understanding brownout therefore requires identifying the load’s input behaviour.

$$\Delta E_C = \tfrac{1}{2} C (V_1^2 - V_2^2), \quad \Delta t \approx \Delta E_C / P_{deficit}$$



A load-current step produces a simultaneous reduction in shared-bus voltage.



Assumptions and physical limits

The plotted voltage drop is analytical, not a measured robot oscilloscope trace. Actual brownout thresholds depend on the device data sheet.

Power and brownout from internal resistance

Current is an implicit unknown for a constant-power load

1 Substitute voltage sag into power

Let Voc be open-circuit voltage and R total series resistance. For constant power P, I=P/Vload, but Vload itself depends on I. Substitute Vload=Voc−RI into P=Vload I to obtain a quadratic. P/Voc is only an approximation when sag is small.

$$P = V_{oc}I - RI^2 \quad \Rightarrow \quad RI^2 - V_{oc}I + P = 0$$

2 Select the low-current branch

The minus-sign root gives I→0 as power becomes small. For Voc=24 V, R=0.1 Ω and P=400 W, I=18.0196 A and Vload=22.198 V. Estimating 400/24 gives 16.667 A and understates loss.

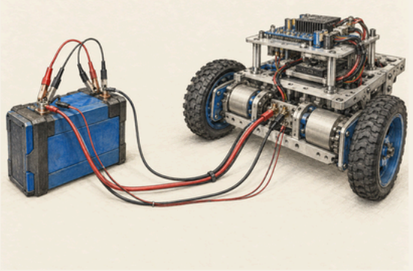
$$I = \frac{V_{oc} - \sqrt{V_{oc}^2 - 4RP}}{2R} = 18.0196 \text{ A}$$

$$P_{loss} = I^2 R = 32.471 \text{ W}$$

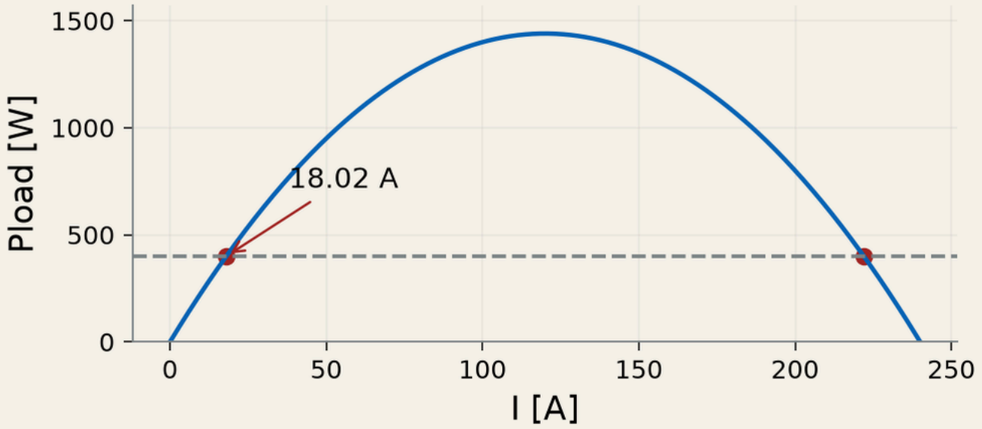
3 Physical meaning of the discriminant

A real root requires Voc²−4RP≥0, giving a model maximum power Voc²/(4R). Here it is 1440 W, not a battery operating recommendation: current, temperature and undervoltage limits may bind much earlier. At the parabola maximum, the two algebraic roots merge; a higher power line has no intersection. The constant-source/resistance model then has no steady solution at that power. Predicting shutdown behaviour additionally requires converter and protection dynamics.

$$P \le \frac{V_{oc}^2}{4R}$$



Intersections of load power with the Thevenin power parabola show the two algebraic roots.



Assumptions and physical limits

The high-current root is not automatically a feasible operating point. This constant-resistance model does not establish dynamic stability or electrochemistry.

TI BMS protection context; original Thevenin constant-power derivation and root analysis.

Protection and DC-bus precharge

Interrupting current and charging capacitance are different tasks

1 Fuse, contactor and resistor

A fuse interrupts fault current according to its breaking capability and time-current curve. A contactor provides commanded connection; connecting an empty DC-link capacitor directly can create a large inrush. With the main contactor open, an auxiliary precharge branch charges it through a resistor. The precharge resistor limits initial current; as the capacitor charges, resistor voltage and current decrease. Closing the main contactor too early bypasses that limitation. Verify the source-to-bus voltage difference rather than relying only on a fixed waiting time.

$$I(0) = V/R, \quad \tau_{RC} = RC$$

2 A 24 V bus example

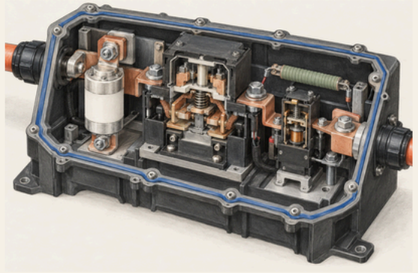
With an initially empty 2 mF capacitor and a 100 Ω resistor, initial current is 0.24 A and the time constant is 0.2 s. For a constant source and unloaded bus, five time constants elapse after 1 s; capacitor voltage reaches 23.84 V, or 99.33% of supply.

$$V_C(t) = V(1 - e^{-t/(RC)}), \quad I(t) = \frac{V}{R} e^{-t/(RC)}$$

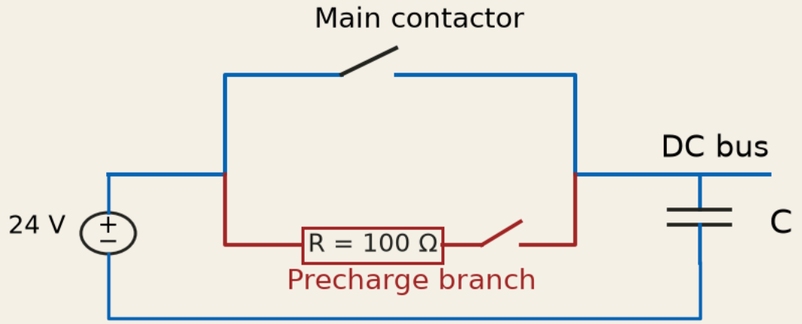
3 Energy and closing sequence

The capacitor stores 0.576 J; the resistor also dissipates 0.576 J during ideal RC charging. Check its pulse-energy rating as well as continuous watts. After voltage difference is sufficiently small, close the main contactor, open the auxiliary branch and enable the load.

$$E_C = \frac{1}{2} C V^2 = 0.576 \text{ J}$$



The precharge branch and main contactor connect in parallel between the same two bus nodes.



Assumptions and physical limits

This precharge concerns the DC link; it is distinct from limited-current charging of deeply discharged cells. Verify voltage before sequencing.

RC circuit laws; TI TIDA-00792 §2.2.7 distinguishes cell precharge; original DC-link example.

RC solution for precharge

Initial condition, target voltage and resistor energy

1 From circuit to differential equation

For a source V, resistor R and capacitor C in series, Kirchhoff gives V=Ri+VC. Using i=C V̇C gives RC V̇C+VC=V. With initial voltage VC(0)=V0, the solution approaches final voltage V exponentially.

$$RC\dot{V}_C + V_C = V, \quad V_C(t) = V + (V_0 - V)e^{-t/(RC)}$$

2 From target percentage to waiting time

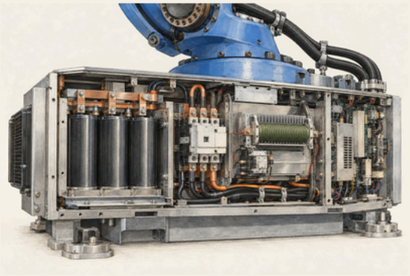
For an initially empty capacitor and target VC=αV, solve e^(−t/RC)=1−α and take logarithms. With R=100 Ω and C=2 mF, RC=0.2 s. Reaching 95% takes 0.5991 s and 99% takes 0.9210 s; exactly 100% takes infinite time mathematically.

$$t_{target} = -RC\ln(1-\alpha)$$

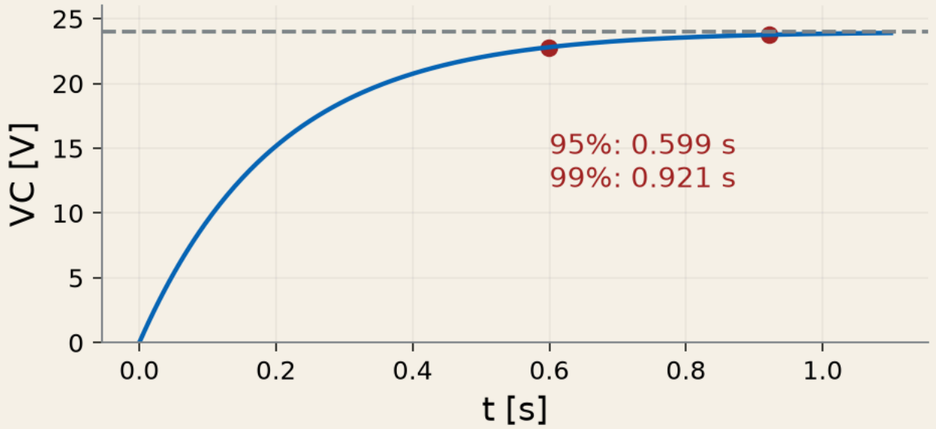
3 Energy dissipated in the resistor

Integrate I²R from zero to infinity using current (V/R)e^(−t/RC). The result is CV²/2, equal to stored capacitor energy. At 24 V and 2 mF, each is 0.576 J and the source supplies 1.152 J. Increasing R does not reduce ideal total loss; it increases charging time. The half-energy loss applies to this specific constant-voltage RC charge from an empty capacitor. Small resistance dissipates energy at greater power over a shorter interval; large resistance does so more slowly. Total energy and peak power are therefore different design quantities.

$$E_R = \int_0^\infty \frac{V^2}{R} e^{-2t/(RC)} dt = \frac{CV^2}{2}$$



Voltage percentage is mapped directly to the time constant; 95% and 99% points are calculated.



Assumptions and physical limits

Assume a constant source, empty initial capacitor and unloaded bus. A connected load changes the target and energy result; verify by voltage measurement.

PCB, wiring and EMC

Interference has a physical source and coupling path

1 Fast current change generates voltage

Parasitic inductance is not zero during MOSFET switching. If current changes by 10 A in 100 ns through 20 nH, the induced excursion is 2 V. The issue is not only large current but its rate of change. Draw forward and return paths together. The inductance is often a property of the cable or PCB current loop, not a separate drawn inductor. Faster switching can reduce transition loss while increasing di/dt and voltage overshoot. Layout and switching speed must therefore be designed together.

$$V_L = L\frac{di}{dt} = 20 \times 10^{-9}\frac{10}{100 \times 10^{-9}} = 2\ \text{V}$$

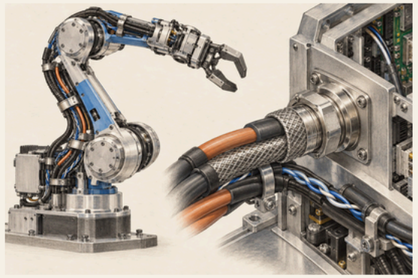
2 Capacitive coupling and common mode

A switching node has parasitic capacitance to chassis or adjacent conductors. Changing voltage drives current through that capacitance. For example, 100 pF and 100 V/µs can produce 10 mA of common-mode current. Returning through a sensor reference turns it into measurement error.

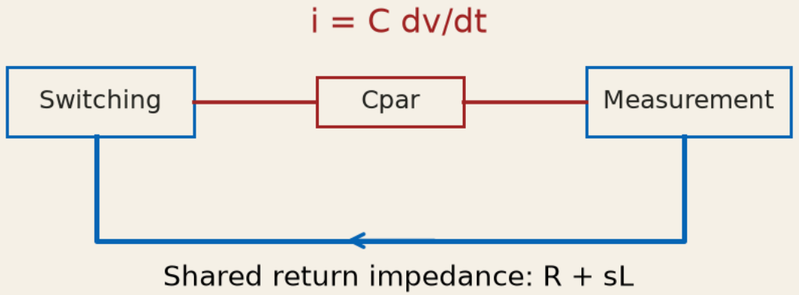
$$I_C = C\frac{dv}{dt} = 0.01\ \text{A}$$

3 From layout to verification

Closely paired forward and return paths reduce loop area; a suitable reference layer supports high-frequency return. Choose shields, connectors and chassis bonds for the coupling mechanism. Arbitrary ground splits can lengthen return paths. Test emissions and immunity separately.



Inductive and capacitive coupling are shown separately; the return path is part of the circuit.



Assumptions and physical limits

The drawing is a causal coupling model, not proof of product EMC compliance. ESD protection also depends on its physical discharge path.

Drive layout and cable routing

Separate critical switching loops from sensitive measurement connections

1 Physically shorten the power loop

The DC-link capacitor, high/low MOSFETs and return connection form a fast switching loop. A distant capacitor can increase connection inductance. Keep gate-drive and source-return traces compact too; proximity on the schematic is not enough.

$$\Delta V \approx L_{loop}\Delta I/\Delta t$$

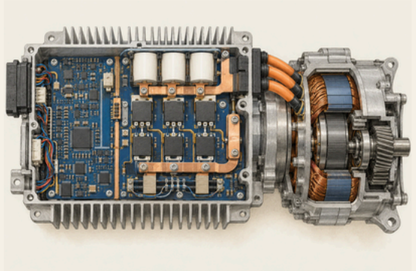
2 Why is Kelvin sensing a separate connection?

Load current flows through the shunt; separate sense traces connect directly to its terminals. This excludes extra drops along power connections. At 10 A, a 1 mΩ copper connection adds 10 mV error, which is 20% of the 50 mV signal from a 5 mΩ shunt. A high-impedance measurement input draws little current through the Kelvin sense traces, making their voltage drop small. The amplifier therefore observes the resistor element rather than including high-current connection drops. This is the physical benefit of four-terminal sensing.

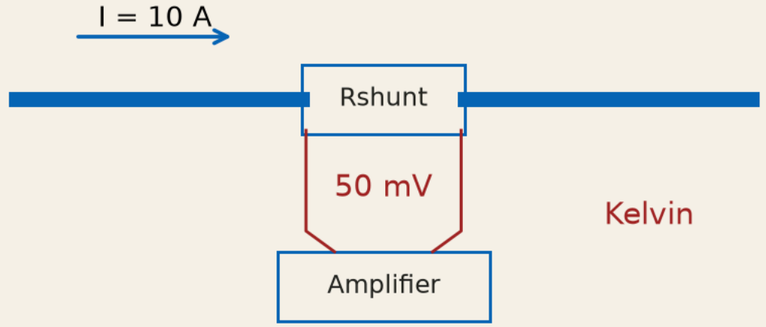
$$V_{error} = IR_{trace} = 10\ \text{mV}, \quad P_{trace} = I^2R_{trace} = 0.1\ \text{W}$$

3 Layers and cables

Keep sensitive analog traces away from fast switching nodes. Plan an appropriate continuous return reference for signal layers, including layer transitions. Motor and encoder cables should not share avoidable coupling paths; resolve shield termination for the system topology.



Load current uses the heavy copper path; a separate Kelvin pair carries the sensed voltage.



Assumptions and physical limits

This is a layout-principle drawing, not a production PCB. Size clearance and creepage separately for voltage, material and environment.

Thermal design of robot electronics

Translate electrical loss into temperature and time limits

1 A resistance network for heat

Loss P is in watts; thermal resistance Rθ in kelvin per watt relates temperature difference to heat flow. At steady state, 10 W loss, a total 4 K/W path and 35 °C ambient give 75 °C. This belongs to the modeled node, not automatically the surface or semiconductor junction.

$$T_{ss} = T_a + PR_\theta = 35 + 10(4) = 75\ ^\circ\mathrm{C}$$

2 Thermal capacitance sets the transient

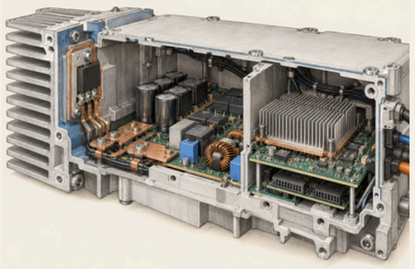
Thermal capacitance Cθ is in J/K. Input power minus heat flow to ambient equals the rate of stored energy. With Cθ = 20 J/K, the time constant is 80 s. Starting at ambient, one time constant reaches 63.2% of the final temperature rise. Increasing thermal capacitance slows temperature rise but does not reduce the steady temperature at fixed loss. Lower steady temperature requires less loss or lower thermal resistance to ambient. A larger metal mass and a better heat path are therefore different solutions.

$$C_\theta \dot{T} = P - \frac{T - T_a}{R_\theta}, \quad \tau_\theta = R_\theta C_\theta = 80\ \mathrm{s}$$

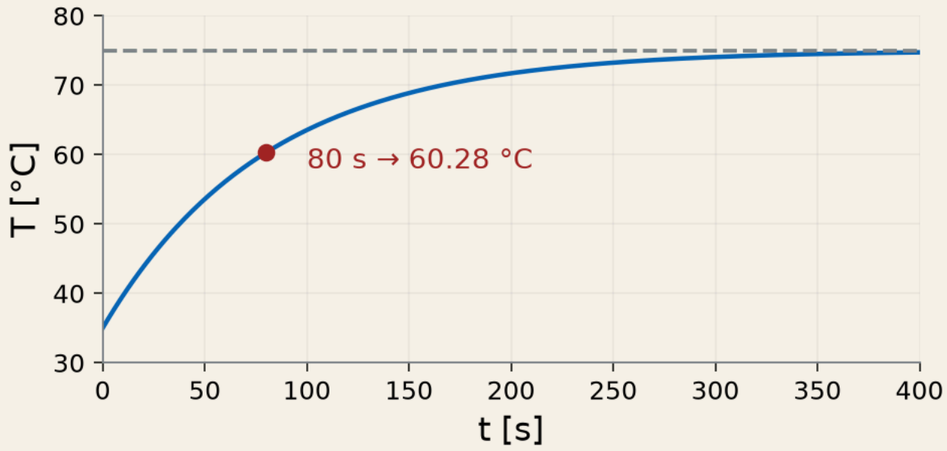
3 Design the actual heat path

Die, package, interface, housing and ambient form series/parallel heat paths. Contact pressure, copper area and airflow change the result. Data-sheet RθJA cannot be transferred unchanged to another enclosure; estimating junction temperature from a case measurement needs a suitable model.

$$T(t) = 35 + 40(1 - e^{-t/80}) \quad [^\circ\mathrm{C}]$$



The curve is a calculated single-node response, not a thermal-camera image or measured data.



Assumptions and physical limits

The single-node model assumes constant loss and ambient. Motor winding, MOSFET junction and processor may need separate nodes and limits.

Deriving the thermal RC model

Energy conservation determines the temperature response

1 Change in stored heat

Let one body have thermal capacitance Cθ. A small temperature rise dT stores Cθ dT energy. Heat flow to ambient is (T−Ta)/Rθ. Subtract this flow from loss power to obtain the temperature-rate equation. Cθ is in J/K and Rθ in K/W.

$$C_\theta \dot{T} = P - \frac{T - T_a}{R_\theta}$$

2 Steady value and initial-condition solution

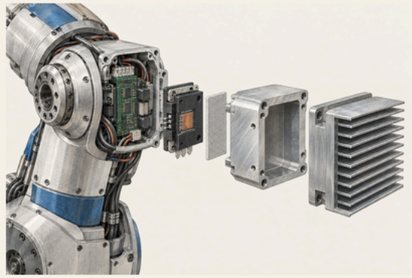
Setting Ṫ=0 gives Tss=Ta+PRθ. The deviation T−Tss follows a homogeneous first-order equation. Thus temperature approaches the steady value from T0 with time constant RθCθ. Electrical RC is a mathematical analogy, not an identity of physical quantities. Defining deviation T−Tss removes the constant forcing term, leaving an exponential decay toward zero deviation. Checking T(0)=T0 and the long-time limit Tss verifies that the integration constant has been chosen correctly.

$$T(t) = T_{ss} + (T_0 - T_{ss})e^{-t/\tau_\theta}, \quad \tau_\theta = R_\theta C_\theta$$

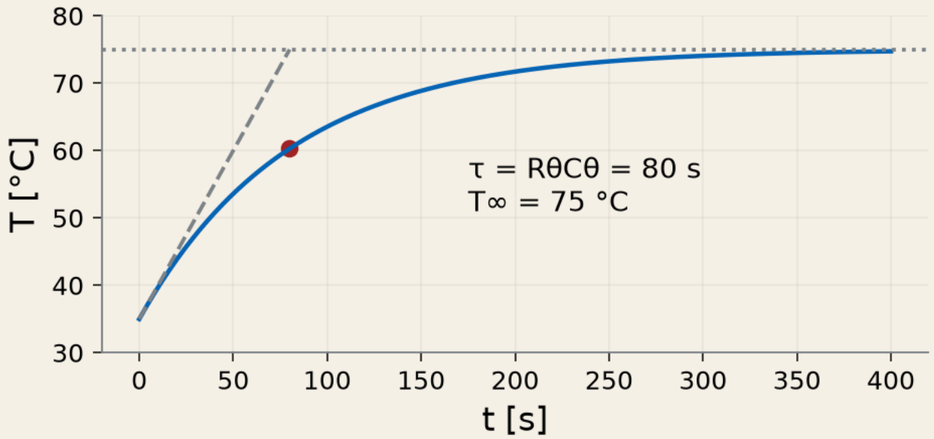
3 Numerical and dimensional check

Take P=10 W, Rθ=4 K/W, Cθ=20 J/K and Ta=T0=35 °C. Steady temperature is 75 °C and time constant 80 s. At 80 s, T=35+40(1−e⁻¹)=60.285 °C. K/W times J/K equals J/W=s, confirming dimensional consistency.

$$T(80) = 60.285\ ^\circ\mathrm{C}, \quad T(\infty) = 75\ ^\circ\mathrm{C}$$



The initial tangent and final temperature identify the time constant and steady value.



Assumptions and physical limits

This is a single-node model. Junction–case–ambient may require several resistances and capacitances; convection can make Rθ temperature-dependent.

maxon thermal time constants; ADI thermal-path context; original lumped energy-balance derivation.

Embedded computing layers

Choose hardware by task, timing constraint and failure behavior

1 Compute throughput is not the only criterion

An MCU supports tightly timed loops close to timers, ADC and PWM. An MPU/SBC provides an operating system, networking and large memory; a GPU suits parallel vision and an FPGA specialized data paths and precise timing. Startup, power, data transfer and maintainability also shape the architecture.

2 An initial capacity estimate

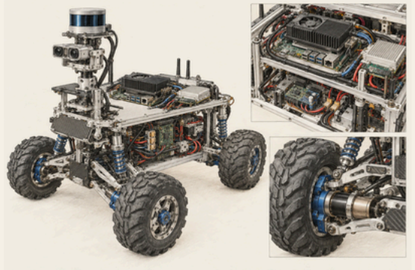
Let Ci be execution time and Ti the task period. A task needing 0.2 ms every 1 ms consumes 20% CPU time. This is only a capacity estimate: interrupts, shared-resource blocking, cache and operating-system overhead affect completion time. U is a fraction of processor time consumed by work. It does not describe exactly when jobs become ready or how long they block one another. Low average utilization therefore cannot establish that a short deadline will be met.

$$U = \sum_i C_i/T_i, \quad U_1 = 0.2/1 = 0.20$$

3 A contract between layers

The high-level computer sends target velocity or position; the low-level controller checks freshness and limits. A message carries units, frame, timestamp, sequence number and validity duration. A delayed GPU computation must not leave motor control using an old command indefinitely.

$$t_{age} = t_{now} - t_{command} \leq T_{valid}$$



Fast physical feedback and high-volume perception operate on different time scales.

Layer	Example task	Main criterion
MCU	Current / encoder	Short, bounded delay
SBC	Planning / network	OS, memory, boot
GPU	Vision inference	Data movement / power
FPGA	Custom I/O path	Clock / pipeline

Assumptions and physical limits

The allocation is an architecture example. Neither FPGA nor GPU is mandatory; the same function can have different hardware implementations.

Inside the robot MCU

Peripherals, CPU and memory working together

1 Hardware events establish timing

A timer counter produces PWM edges; a compare event triggers ADC acquisition at an appropriate phase. DMA transfers samples to memory; an interrupt or task starts computation. GPIO carries digital levels, while an encoder timer counts edges. The CPU need not generate every edge in software.

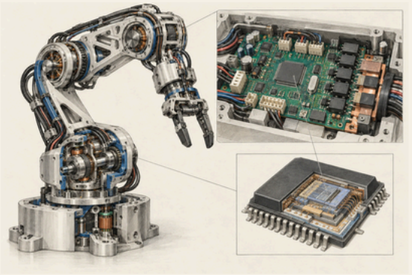
2 Calculate the PWM period

In edge-aligned up-counting, the counter visits 0 through ARR, requiring ARR+1 ticks. The prescaler divides the timer clock by PSC+1. An 80 MHz timer with PSC = 0 and ARR = 3999 gives 20 kHz and a 50 µs period. The same formula cannot be applied unchanged to center-aligned counting. ARR=3999 gives 4000 counter states because zero is included. Likewise, PSC=0 divides by one rather than stopping the clock. These +1 details can cause systematic frequency errors, so inspect the counter range before applying the formula.

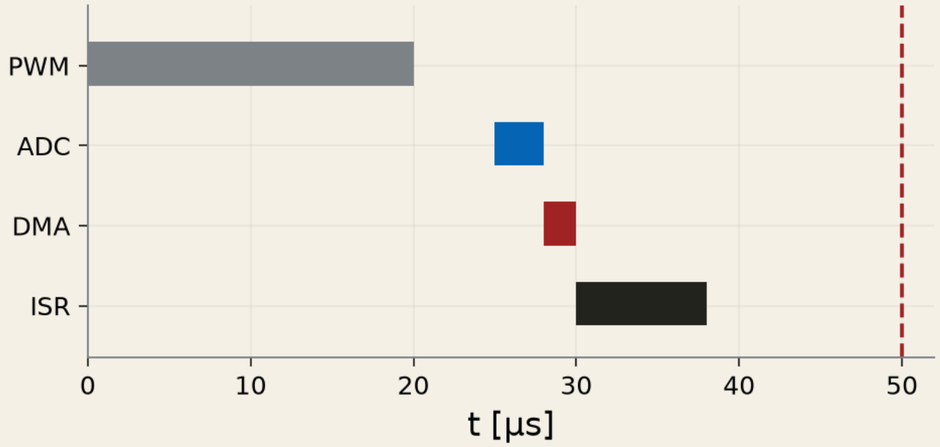
$$f_{PWM} = \frac{f_{timer}}{(PSC+1)(ARR+1)} = 20\ \text{kHz}$$

3 Data ownership and the ISR boundary

Peripheral registers are accessed through memory addresses; drivers separate these details from the application. The CPU must not read a buffer while DMA is updating it: double buffering and completion events can help. Keep ISRs short; move lengthy computation or blocking waits to a suitable task.



Events within the 50 µs period form an illustrative timing budget, not a device measurement.



Assumptions and physical limits

Timer clock need not equal CPU clock. Verify clock tree, cache coherency, ADC settling and trigger details for the chosen MCU.

Robot communication interfaces

Select bit rate, access method and physical layer together

1 Board-level and robot-level links

UART provides asynchronous serial bits; voltage levels and physical transceiver are separate choices. SPI is clocked and usually short-range; devices may need individual select lines. I²C uses addressed open-drain data/clock lines; rise time depends on pull-ups and bus capacitance.

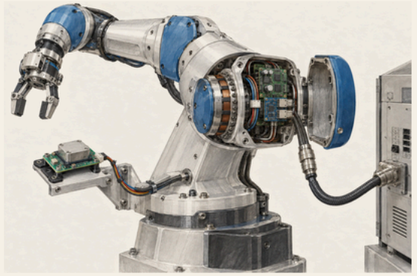
2 Distributed motion control

CAN provides identifier-based priority and non-destructive arbitration; lower-priority waiting needs analysis. CAN FD can use a different data-phase rate and a longer payload. EtherCAT supports cyclic transfer and distributed clocks for multi-axis systems; cycle time depends on topology. Bandwidth describes data carried per unit time; determinism concerns how tightly delay can be bounded. A faster link is not automatically more predictable when priority and queue behaviour are unknown. Synchronized motor axes require clock coordination as well as data rate.

3 A UART wire-time example

An 8N1 frame has one start, eight data and one stop bit: 10 bits per byte. A 32-byte packet at 115200 baud on binary UART takes 320/115200 = 2.778 ms. This excludes software queues, responses, retries and processing.

$$t_{wire} = \frac{32(1+8+1)}{115200} = 2.778 \text{ ms}$$



The table is a selection map; no single protocol is best for every robot data flow.

Interface	Physical signals	Access / timing
UART	TX/RX	Asynchronous
SPI	CLK / data / CS	Clock + selection
I²C	SDA / SCL	Addressed bus
CAN FD	CAN H/L	Priority arbitration
EtherCAT	Ethernet PHY	Cycle + clocks

Assumptions and physical limits

Controller, transceiver and connector are different components. Terminate CAN at bus ends and design EtherCAT to its physical-layer requirements.

Communication and timing budget

Count every delay from sensing to actuation

1 Components of end-to-end latency

A sensor sample is acquired, queued, transmitted, processed and applied by an actuator. Define the start and end events for every term. Measuring only packet transmission does not measure the entire control delay. Direct addition applies to sequential dependent steps. If computation cannot begin before its last required sample arrives, that sample's queue delay lies on the critical path. Arbitrarily summing all parallel activities can misrepresent the actual end-to-end delay.

$$T_{total} = T_{sense} + T_{queue} + T_{wire} + T_{compute} + T_{act}$$

2 A one-millisecond loop

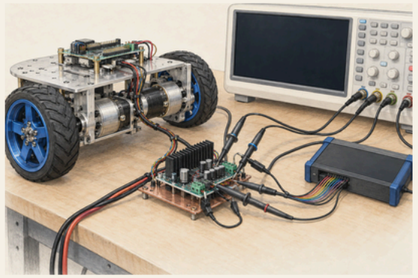
Assume upper bounds of 0.15, 0.20, 0.25, 0.20 and 0.10 ms. Total latency is 0.90 ms, leaving 0.10 ms to a 1 ms deadline. If these are averages, this is not a worst-case guarantee; queue and arbitration bounds are still needed.

$$T_{total} = 0.90 \text{ ms}, \quad M = D - T_{total} = 0.10 \text{ ms}$$

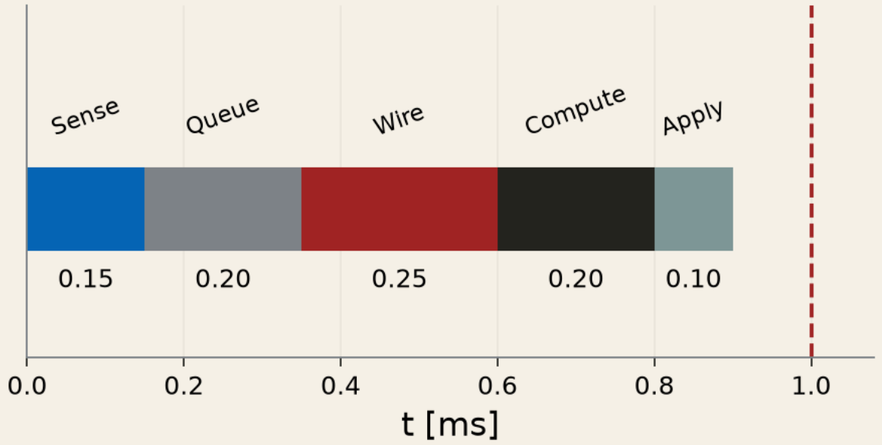
3 Clock synchronization and jitter

Subtracting timestamps from different devices does not reveal latency unless clocks share a reference. Account for clock offset, drift and synchronization error. Jitter is deviation of a specified event from its ideal timing; low average delay does not imply low jitter.

$$\varepsilon_k = t_k - (t_0 + kT), \quad J_{max} = \max_k |\varepsilon_k|$$



Acquisition, queue, wire, computation and application share one time axis.



Assumptions and physical limits

The chart sums the stated bounds and is not a measurement. When stages overlap, use critical-path analysis for the actual end-to-end bound.

Wire time and end-to-end latency

Bit-level overhead becomes visible at the task level

1 Actual length of a UART byte

8N1 carries eight data bits plus start and stop bits. A 32-byte packet takes 320 symbols; for binary UART, baud and bit/s are numerically equal. At 115200 baud this takes 2.778 ms. DMA reduces CPU transfer work, not physical wire time.

$$t_{UART} = \frac{N_{bytes}(1 + N_{data} + N_{parity} + N_{stop})}{baud}$$

2 The one-millisecond limit

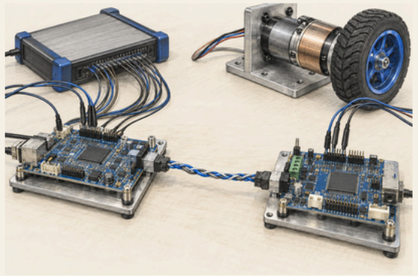
Sending that packet every 1 ms would require 277.8% wire occupancy, which is impossible. At 1 Mbaud it falls to 0.32 ms, but queueing, response and computation remain. This calculation is for UART; CAN has a different frame-bit count.

$$U_{wire} = t_{wire}/T = 2.778/1 = 2.778 > 1$$

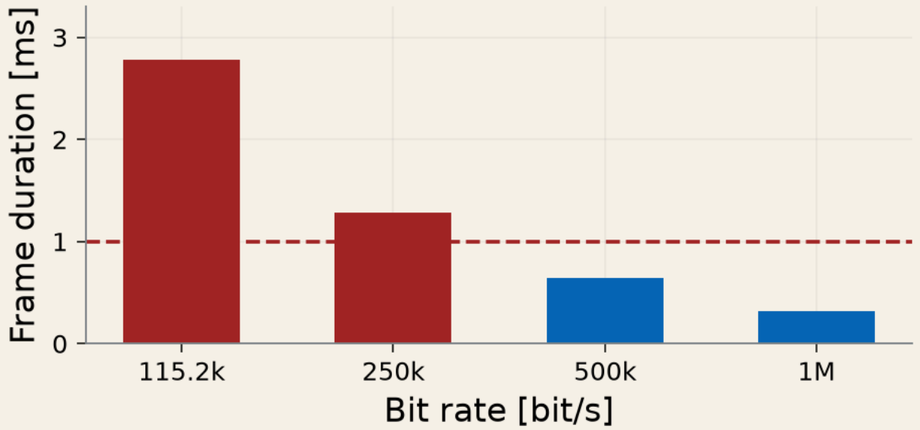
3 Clock domains in timestamping

With clock offset Δ between sender and receiver, apparent latency equals true latency plus Δ. Synchronize clocks or estimate offset first. Distinguish data age from wire time: rapidly transmitting an old sample does not make it fresh. If the sender clock is 5 ms behind the receiver, a true 1 ms transfer can appear as 6 ms. Correcting a fixed offset may not suffice when clock rates differ and drift accumulates. Latency measurement requires more than subtracting two timestamps.

$$t_{rx}^{(B)} - t_{tx}^{(A)} = T_{latency} + \Delta_{BA}$$



One byte frame and the entire packet duration follow the same bit budget.



Assumptions and physical limits

Add bit stuffing, arbitration and retries according to the protocol. Average occupancy is not a worst-case waiting-time bound.

Robot firmware architecture

Explicit states for boot, operation and update

1 Responsibilities of the layers

Drivers manage hardware registers; a HAL offers a common access interface. The application state machine responds to sensor, command and fault events. A bootloader runs before the application; flash partitioning, validation and update transitions are separate responsibilities.

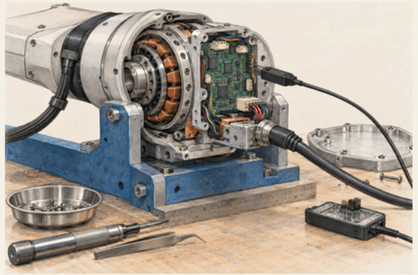
2 Guard state transitions

Transition from BOOT to SELF-TEST and then READY. RUN additionally requires valid commands, sensors and enable conditions. Keep motor outputs inactive during boot. FAULT must not return to RUN until its cause and required reset conditions are checked. A guard is a transition condition distinct from the event: an enable message must not enter RUN if sensing is invalid. An action specifies what happens on transition, such as clearing stale commands. This turns a state diagram into implementable behaviour.

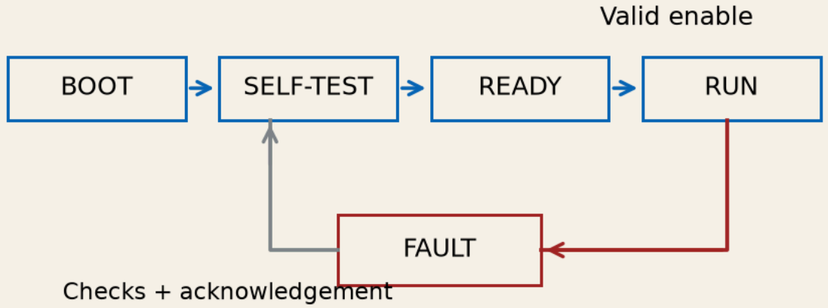
$$(s, event, guard) \longmapsto (s_{next}, action)$$

3 Update recovery after power loss

A candidate image can occupy a separate slot, undergo integrity/signature checks and boot as a trial. If it is not confirmed, a supported revert scheme selects the previous image. Anti-rollback security and recovery from a failed trial are different policies; parameter compatibility also matters.



The state graph ties motor enable to READY and valid conditions.



Assumptions and physical limits

A signature authenticates origin and integrity, not bug-free behavior. Not every bootloader mode supports reverting; test the selected flash layout.

Fault-tolerant firmware

A running processor is not necessarily a working controller

1 Choose evidence for each failure

A watchdog helps detect stalled software, a heartbeat missing communication from another node, and a timeout stale data. Range and plausibility checks can detect physically impossible sensor values. CRC can detect corrupted messages, but does not guarantee detection of plausible yet wrong measurements.

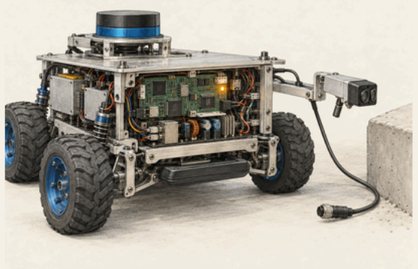
2 Command age matters to control

At 100 Hz the expected command interval is 10 ms. An illustrative 50 ms limit permits roughly five periods after the last valid command. The control task checks age every cycle; decision and application delay add to the timeout. The last valid message must pass integrity, sequence and content checks; not every received packet should refresh the timer. Otherwise, corrupted or repeatedly replayed stale packets can make the system appear healthy. Communication activity and usable fresh information are different.

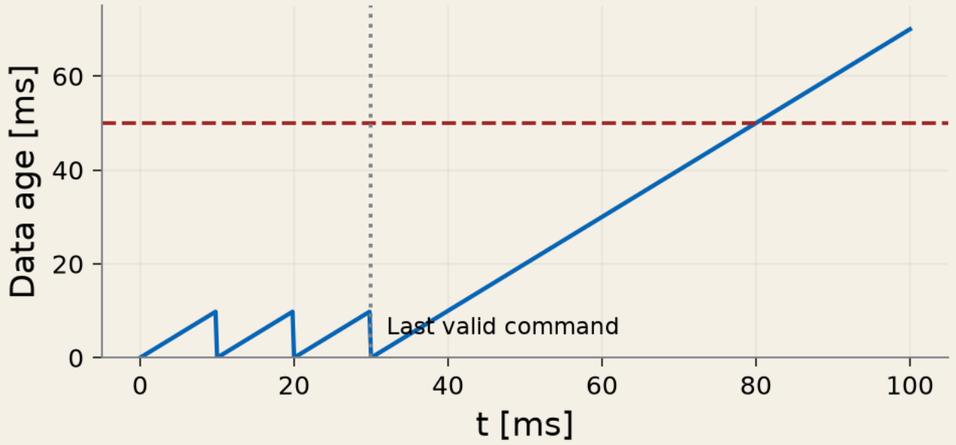
$$t_{age} = t_{now} - t_{last\ valid}, \quad t_{age} > 50\ \text{ms} \Rightarrow fault$$

3 Feed a watchdog from evidence of progress

Interrupts may continue while the control task is stuck. Blindly feeding the watchdog in every timer ISR hides this failure. A supervisor checks task progress counters, sample freshness and required checkpoints. Fault codes, first occurrence time and measurements support diagnosis.



After commands stop, data age rises; timeout sets the start of the physical response.



Assumptions and physical limits

50 ms is not a universal safe value; choose it from stopping distance and robot hazards. One detector cannot cover every failure.

Safe recovery after a fault

Detect, stop appropriately, verify and restart

1 Fault response depends on robot physics

A horizontal mobile base may decelerate under control; removing torque suddenly from a vertical joint can drop a load. Distinguish transient and latched faults. Preserve the fault record and check that the cause is gone and motion conditions are acceptable before restarting.

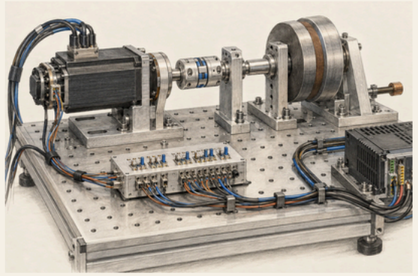
2 Detection time becomes distance

Let initial speed be v, detection/response delay td and achievable constant deceleration a. Travel during delay is vtd; braking adds a distance proportional to speed squared. At 1 m/s, 0.1 s and 1 m/s², total distance is 0.6 m. Areas under velocity versus time have units of distance. Reducing detection delay shrinks only the rectangle; insufficient braking leaves a large triangular area. Recovery therefore depends on both software reaction time and achievable physical deceleration.

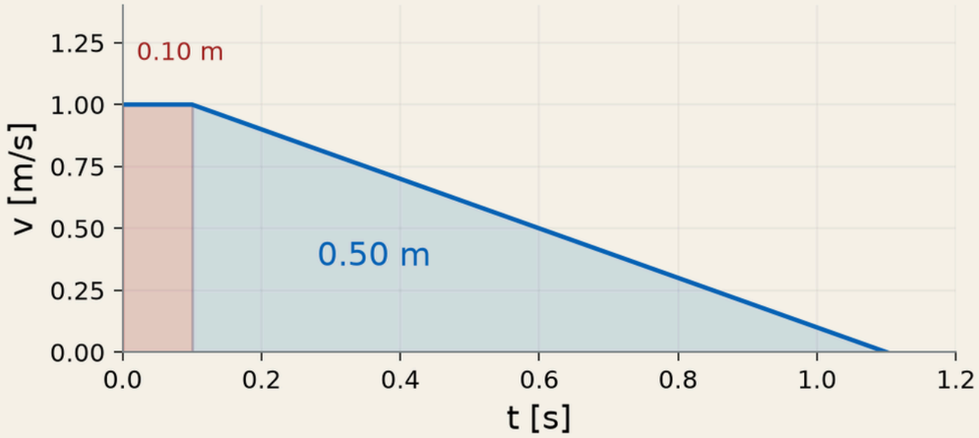
$$d_{stop} = vt_d + \frac{v^2}{2a} = 0.1 + 0.5 = 0.6 \text{ m}$$

3 Reset is not motion permission

Reset restarts a processor; it does not fix a mechanical fault or make an old command safe. Returning to READY requires sensor checks, fault acknowledgement and homing when needed. Enter RUN through a new enable event; do not automatically replay the old velocity command at boot.



Delay and braking distances are separate areas under the velocity–time curve.



Assumptions and physical limits

The distance model assumes level ground and achievable constant deceleration. Actual traction, brake capability, jerk and measurement uncertainty require separate treatment.

Schneider safe motion functions (STO/SS1 distinction); original fault-state and stopping-distance example.

RTOS and real-time robotics

Producing the correct result on time is also a requirement

1 Task, period and deadline

A task can be ready, running or waiting for an event. The scheduler assigns the CPU to an eligible ready task. Period is the interval between releases; deadline is the required completion bound. A hard real-time deadline miss is unacceptable; soft real-time misses may degrade quality. C is actual processor execution time; R includes all elapsed waiting from release to completion. A task with little code can still finish late due to higher-priority work. Compare the deadline with the appropriately bounded response time, not execution time alone.

$$U_i = C_i/T_i, \quad R_i = t_{finish} - t_{release}, \quad R_i \le D_i$$

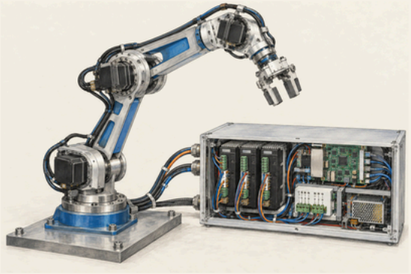
2 Queue, semaphore and mutex

A queue organizes data and ownership transfer; lifetime must be clear for copies or pointers. A semaphore signals an event or resource count. A mutex provides owned mutual exclusion for a short critical section. Blocking on a mutex inside an ISR is inappropriate; hand an event to a task.

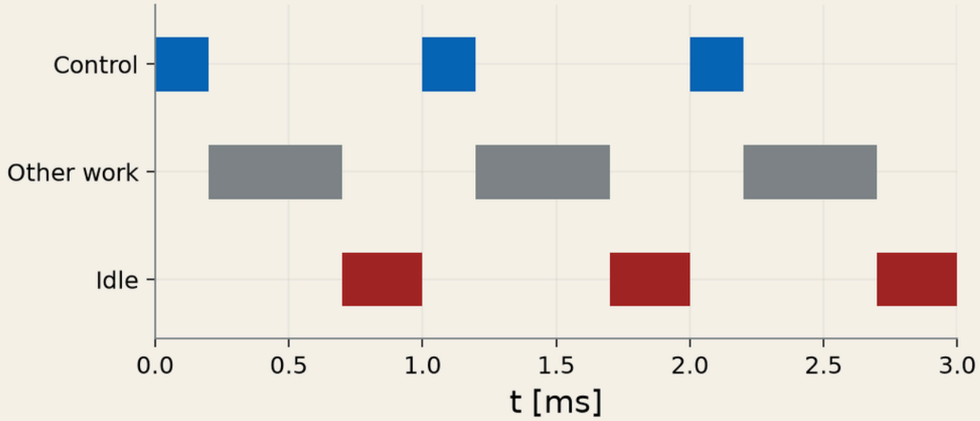
3 CPU utilization is not sufficient evidence

Execution of 0.2 ms every 1 ms gives 20% utilization. But waiting 0.9 ms for a shared resource can consume the budget before work completes. WCET bounds the longest possible execution; an observed mean or single-test maximum is different. An RTOS name alone guarantees nothing.

$$M_i = D_i - R_i$$



While one task waits, the CPU can execute another; ready does not mean running.



Assumptions and physical limits

This uses a single-core task model. Include interrupt cost, context switching, blocking and release jitter in timing analysis.

Priority inversion and deadlines

A shared resource can delay a higher-priority task

1 The three-task problem

While low-priority L holds a mutex, H requests it and blocks. If medium-priority M preempts L, H indirectly waits for M too. Priority inheritance temporarily raises L to H’s priority so it can finish the critical section. It neither removes locking time nor guarantees every deadline. When H waits for L, interference by M effectively reverses the intended priority order. Inheritance lets L finish the required critical section ahead of M. It does not shorten that section; long or nested locks still require blocking analysis.

2 A fixed point for response time

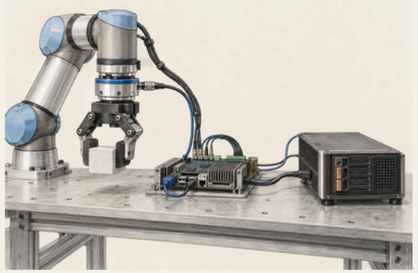
Let Ci be the task’s worst-case execution and Bi lower-priority blocking bound. Each higher-priority task h releases at most ceiling(Ri/Th) jobs in the candidate interval. Add this interference and recompute Ri until it stabilizes or exceeds the deadline.

$$R_i^{(k+1)} = C_i + B_i + \sum_{h \in hp(i)} \left\lceil \frac{R_i^{(k)}}{T_h} \right\rceil C_h$$

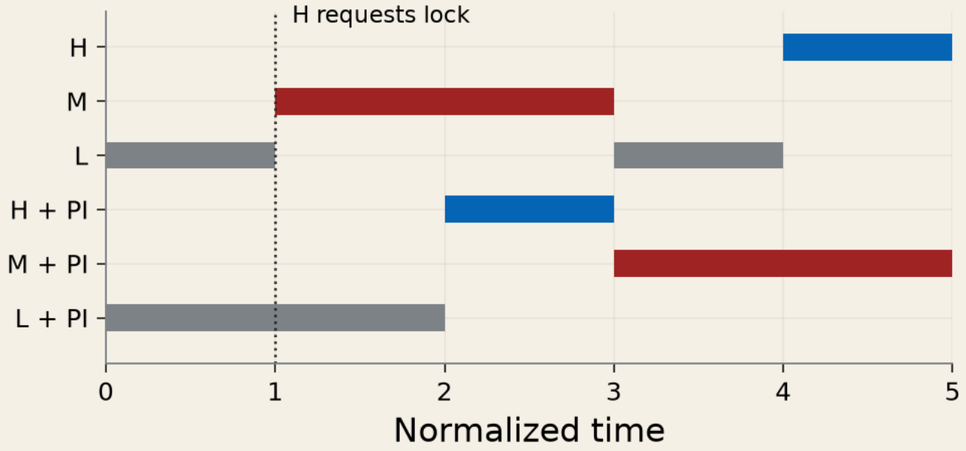
3 Numerical check

Take Ci = 1 ms, Bi = 0.2 ms, and one higher task with Ch = 0.2 ms and Th = 1 ms. Start at 1.2 ms; the next value is 1.6 ms and remains unchanged. With Di = 5 ms, the model gives a 3.4 ms margin.

$$1.2 \longrightarrow 1.2 + 2(0.2) = 1.6 \longrightarrow 1.6 \text{ ms}$$



The timelines compare the same locking problem without and with inheritance.



Assumptions and physical limits

Assume one core, preemptive fixed priorities, bounded blocking and zero release jitter. Do not directly generalize to multicore or arbitrary lock chains.

FreeRTOS Kernel Book §8.3.2–8.3.3, PDF210–212; fixed-priority response-time analysis; original example.

Calculating real-time response time

Bound the number of preemptions with a ceiling function

1 How many jobs arrive in the candidate window?

Let Th be a higher-priority task’s minimum release interval. If it can release simultaneously with the job under analysis, account for at most ceiling(R/Th) higher-priority jobs in a window R. Own execution Ci and lower-priority blocking Bi form the base cost. The ceiling counts whole job arrivals: a 1.2-period window can contain one at its start and another one period later. Interference can extend the window enough to admit more jobs. Fixed-point iteration resolves that mutual dependence.

$$R^{(0)} = C_i + B_i, \quad R^{(k+1)} = C_i + B_i + \sum_h \lceil R^{(k)}/T_h \rceil C_h$$

2 First and second iterations

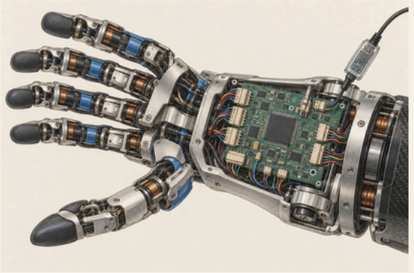
With Ci=1 ms and Bi=0.2 ms, start at R0=1.2 ms. The higher task has Ch=0.2 ms and Th=1 ms. The 1.2 ms window counts two higher jobs: R1=1.2+2×0.2=1.6 ms. The 1.6 ms window also counts two, so R2 remains 1.6 ms.

$$\lceil 1.2/1 \rceil = 2, \quad \lceil 1.6/1 \rceil = 2, \quad R = 1.6 \text{ ms}$$

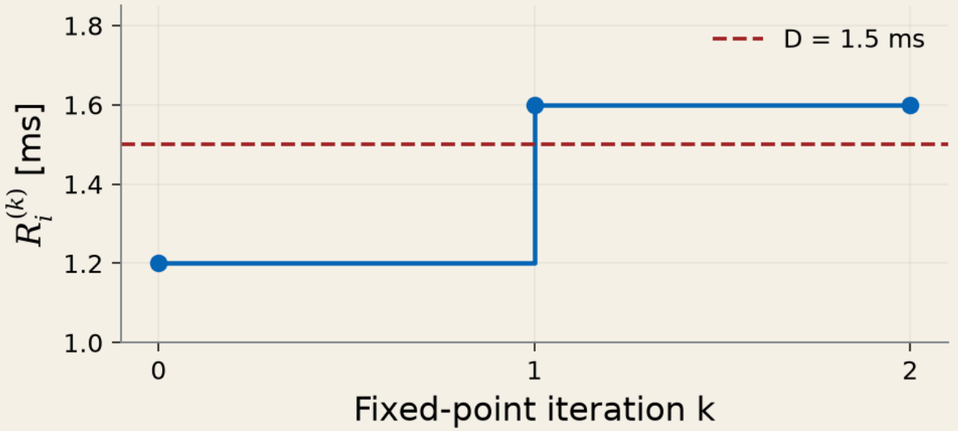
3 Compare against the deadline

For Di=5 ms, R≤Di with 3.4 ms margin. With Di=1.5 ms the same task set would fail this bound. Extend the model for release jitter, interrupts and scheduler cost when non-negligible; use an execution bound rather than measured mean Ci.

$$D_i - R_i = 5 - 1.6 = 3.4 \text{ ms}$$



The candidate window expands from 1.2 to 1.6 ms while the counted higher-priority jobs remain two.



Assumptions and physical limits

Assume one core, preemptive fixed priorities, independent periodic/sporadic tasks and bounded blocking. This is not a multicore guarantee.

Embedded Linux in robots

Assess a rich application environment together with timing needs

1 From user space to hardware

Perception and planning processes run in user space; drivers expose kernel interfaces to hardware. Device tree describes hardware addresses, interrupts and connections; it does not create hardware. Cross-compilation builds for the target processor. Service management organizes boot and restart.

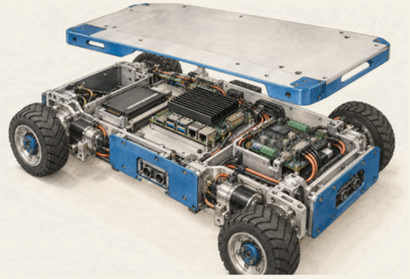
2 What does PREEMPT_RT change?

PREEMPT_RT makes more kernel contexts preemptible, threads many interrupts and uses priority-inheritance-aware locks. This can reduce waiting for a ready high-priority task. Application locks, drivers, I/O and hardware delays remain part of the system. Preemptibility allows running kernel work to yield appropriately when higher-priority work becomes ready. It does not make every interrupt or driver delay-free. Real-time behaviour must be evaluated across operating-system configuration, application and hardware together.

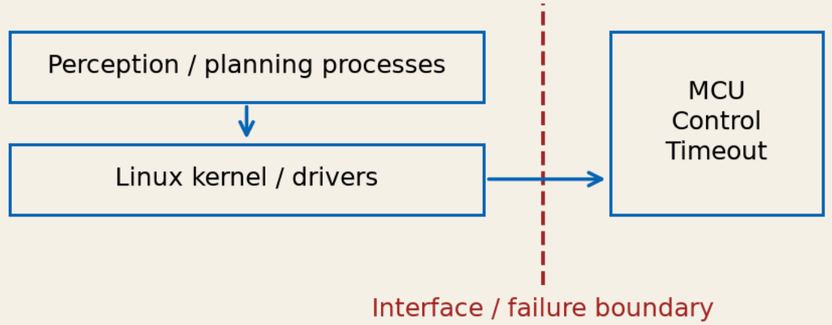
3 Define a measurement protocol

For a 1 kHz task, compare ideal release t0+kT with actual start. Report CPU, disk and network load, priority, affinity, memory locking and test duration. A histogram is not a worst-case proof. No invented latency distribution is shown here because no measurement was performed.

$$\ell_k = t_{start,k} - (t_0 + kT), \quad \ell_{max,observed} = \max_k \ell_k$$



The diagram shows latency and failure boundaries between application, kernel and MCU.



Assumptions and physical limits

A low-level drive controller can bound high-level delay through a timeout. This architectural boundary complements Linux configuration.

Robot safety architecture

Define stopping functions from hazards and energy

1 Stop functions are not equivalent

An emergency stop is a human response to danger; a protective stop may be triggered by a safeguarding condition. STO prevents the motor from receiving energy that can generate torque. It does not alone imply controlled braking, electrical isolation or a discharged DC bus.

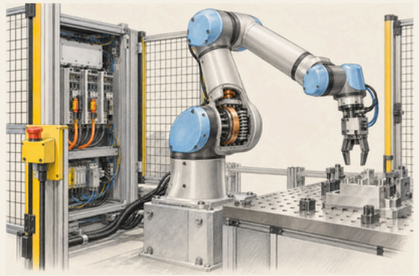
2 Motion and gravitational energy

A moving base’s kinetic energy grows with speed squared. A vertical load retains potential energy after torque is removed. A 10 kg load falling 0.5 m releases 49.05 J. Design brakes, mechanical holding and stop sequencing around this physical energy.

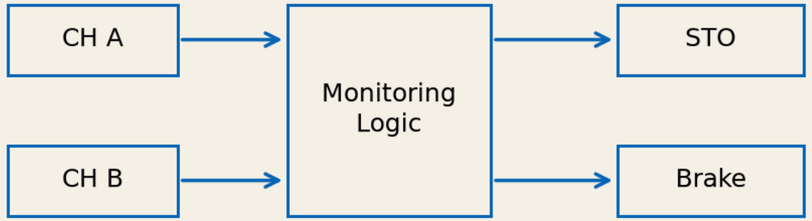
$$E_k = \frac{1}{2}mv^2, \quad E_p = mgh = 49.05\ \mathrm{J}$$

3 Independent paths and diagnostics

Dual-channel inputs, monitored switching and brake feedback can help detect specific failures. Shared power, software and wiring faults can undermine independence. Reset, deliberate restart and restoration of safeguarding conditions are separate events. Two channels are not independent if one common failure disables both. Diagnostics detect faults, but the physical risk-reducing response must also occur. Evaluate the complete chain of input, decision, energy interruption and, where required, mechanical holding.



For a vertical load, inhibiting torque and holding the load are different physical functions.



Functional example; not a certified circuit

Assumptions and physical limits

This teaching diagram is neither certification nor a normative design. Required performance follows the robot risk assessment and applicable standards.

Stopping distance and safe energy

How does squared speed change stopping design?

1 Travel during the delay

Suppose detecting a fault and initiating braking takes td while a robot travels at speed v. If speed stays constant during the delay, the velocity–time area is a rectangle: ddelay=vtd. Count communication, decision and actuator delays together in td.

$$d_{delay} = vt_d$$

2 Braking distance at constant deceleration

For positive deceleration magnitude a, braking time is v/a. The triangular velocity–time area is v(v/a)/2; equating kinetic energy to braking work gives the same result. Total distance is the sum of rectangle and triangle. With constant deceleration, stopping time is v/a and average braking speed is v/2, giving distance v²/(2a). Mass disappears only because achievable deceleration a is prescribed. If braking force is fixed instead, mass still affects the result.

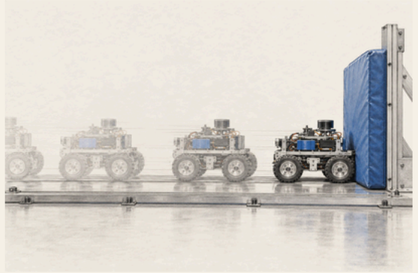
$$d_{brake} = \frac{v^2}{2a}, \quad \frac{1}{2}mv^2 = (ma)d_{brake}$$

$$d_{stop} = vt_d + \frac{v^2}{2a}$$

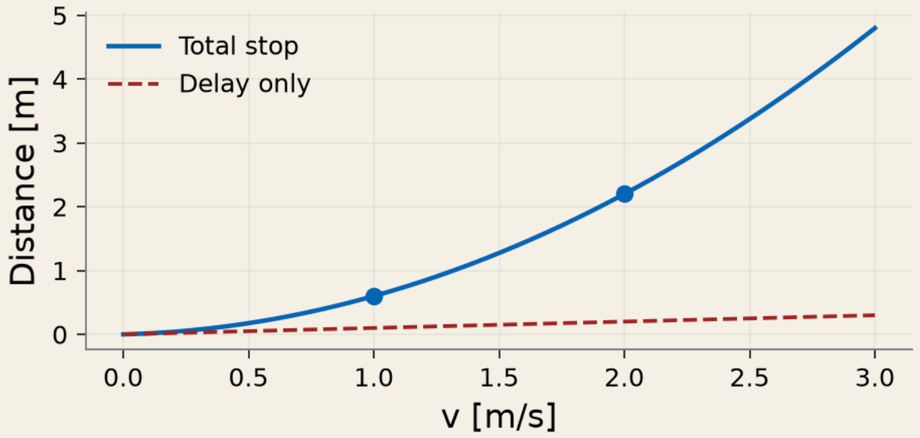
3 When speed doubles

With td=0.1 s and a=1 m/s², v=1 m/s gives 0.1+0.5=0.6 m. At v=2 m/s the distance becomes 0.2+2=2.2 m. Braking distance quadruples while delay distance doubles. A vertical load also has mgh energy; an STO command alone does not hold it.

$$d_{stop}(1) = 0.6 \text{ m}, \quad d_{stop}(2) = 2.2 \text{ m}$$



Stopping distance grows nonlinearly with speed; the dashed line isolates the delay contribution.



Assumptions and physical limits

a must be achievable under actual conditions. Slope, traction, jerk and measurement uncertainty lie outside this ideal calculation; it is not certification.

Verification from model to robot

Each testing stage exposes different classes of error

1 MIL, SIL and PIL

MIL runs controller and plant models together. SIL compares compiled controller code against simulation, exposing sign, unit and numerical implementation errors. PIL runs code on the target processor to examine data types and processor effects. The PIL connection need not run in real time.

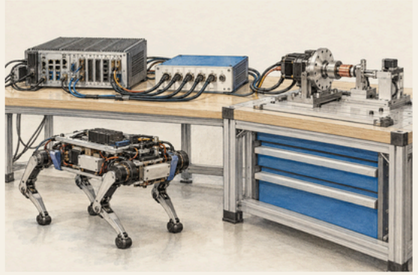
2 HIL uses real interfaces

HIL connects real controller hardware to a real-time plant simulator. It can exercise ADC, PWM, communication and timing; it does not automatically cover every mechanical effect of the robot. The simulator must meet its own time step and emulate sensor/actuator interfaces adequately. In HIL, the controller and electrical interfaces are real, but simulated-load fidelity still depends on the plant model. Passing with an incorrect tyre or motor model does not validate that physics. State explicitly which classes of error each stage tests.

3 One requirement, traceable evidence

Example requirement: assert a fault within 50 ms after valid commands cease. Record initial conditions, injection time, expected result and acceptance tolerance. Bench and robot tests then compare the model against physical measurements; successful simulation alone is not field validation.

$$\Delta t_{detect} = t_{fault} - t_{last\ valid} \leq 50\ \text{ms}$$



The table distinguishes where controller and plant execute at each stage.

	Controller	Plant
MIL	Model	Model
SIL	Compiled code	Simulation
PIL	Target processor	Simulation
HIL	Real controller	Real-time model

Assumptions and physical limits

50 ms is an example test requirement. Define product tolerances including control period and measurement uncertainty.

MathWorks, Hardware-in-the-Loop: MIL/SIL/PIL/HIL comparison; original requirements-based test example.

Fault injection on a robot

Make fault response observable and repeatable

1 Select one controlled fault

First choose sensor freeze, packet delay, command loss or supply sag in simulation or HIL. Injecting many faults at once obscures causality. For robot testing, bound the motion area, energy and stopping conditions through the test plan.

2 Record four response events

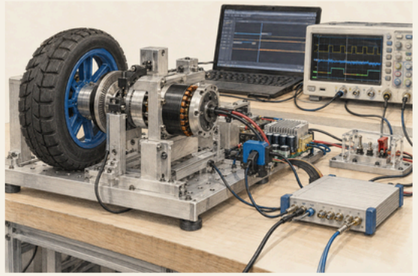
Record fault onset t0, detection t1, stop application t2 and achievement of the safe condition t3 separately. One “fault occurred” log does not capture the response. A shared or synchronized clock separates detection, software/communication and physical stopping time. t1−t0 is detection delay, t2−t1 command application delay, and t3−t2 physical stopping time. A fast software fault log may establish only the first part. Observe the final safe condition through velocity, brake or energy measurements as appropriate.

$$T_{response} = (t_1 - t_0) + (t_2 - t_1) + (t_3 - t_2)$$

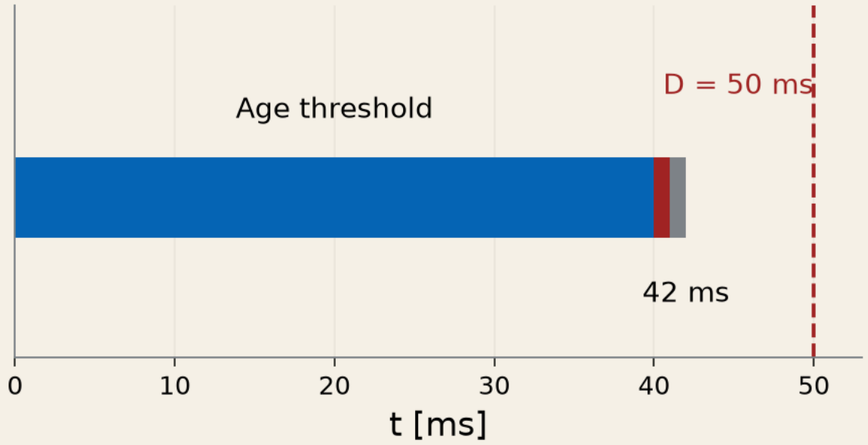
3 Write acceptance criteria beforehand

In the example, interrupt 100 Hz commands, use a 40 ms age threshold and a 1 ms supervisor period. If additional execution delay is bounded by 1 ms, detection is bounded by 42 ms, leaving 8 ms to a 50 ms requirement. This is a test design, not an experimental result.

$$T_{detect} \leq 40 + 1 + 1 = 42 \text{ ms}$$



The timeline shows an example acceptance budget, not measured data.



Assumptions and physical limits

Record software version, parameters, device clocks and repetition count. A frozen sensor and a completely interrupted link are different tests.

Integrated robot engineering

From requirements to models, hardware and measured behavior

1 A platform requires coupled decisions

On a mobile robot, payload and slope set wheel torque, which drives motor and gearbox selection. Motor current sets drive requirements, losses affect thermal design, and duty cycle sets battery size. Command delay changes control and stopping distance. These are coupled decisions, not independent catalogs.

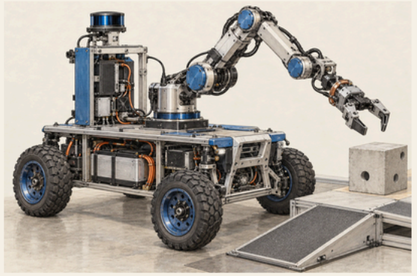
2 The same method, different physics

Tires and speed range dominate a vehicle model; inertia and gravity an arm; contact feasibility a humanoid; thrust direction and rotor limits a drone. For each, state, inputs, constraints, power and timing budgets must be explicit. State which effects a simplified model omits.

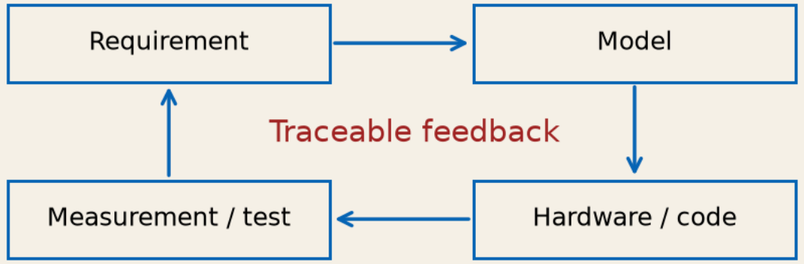
$$\dot{x} = f(x, u), \quad u \in \mathcal{U}, \quad g(x, u) \leq 0$$

3 Closing a design requires evidence

Map each requirement to a calculation, drawing, software component and acceptance test. If measurement disagrees with the model, check assumptions, parameters, signs and time reference before merely tuning gains. Verification findings feed back into models and requirements, closing the engineering loop. For example, increasing battery size improves energy capacity but changes mass, wheel torque and braking requirements. Validation must close these dependencies rather than checking only the replaced component. The design record should identify which requirements and calculations each decision changes.



Requirement, model, component and test form one traceable engineering chain.



Assumptions and physical limits

The outcome is not memorizing component names, but connecting a physical robot’s model, energy, computation and verification.

References 1/3

Sections used by topic and original calculations

Lynch & Park · Modern Robotics

Ch. 2–6, 8, 12–13: geometry, dynamics, contact and wheeled robots.

Tedrake · Underactuated Robotics

Multibody Dynamics, Quadrotors, Humanoids and Contact Dynamics.

MIT 2.72 · Mechanical Design

Selected lectures: loads, stiffness, fatigue, bearings, transmission and vibration.

Bathe · MIT 2.092

Lecture 1: finite-element modeling and verification framework.

CommonRoad · Vehicle Models

Kinematic/dynamic single-track models; assumptions and model hierarchy.

Lecture_2.1_vehicle_dynamics.pdf

48-page lecture supplied by the user; diagrams independently redrawn.

Drake · QuadrotorPlant

Newton–Euler and rotor moments; local source snapshot recorded with SHA256.

Harmonic Drive / Nabtesco

Strain-wave catalog p. 3; operating principle of the RV mechanism.

University of Florida · EML2322L

Tolerances: tolerance types, manufacturing and assembly allowances.

References 2/3

Sections used by topic and original calculations

Asada · MIT 2.12, Robot Actuators
Chapter 2: motor equivalent circuit, power, gearing and reflected inertia.

maxon · Motor Data Sheets Explained
Torque–speed characteristics, continuous/peak operation and thermal parameters.

ST · UM1052, §4.3–4.5
PMSM control, Clarke/Park transforms and current/speed loops.

TI · TIDA-010956 / TIDUF82B
Three-phase inverter, gate drive and current sensing; PDF pp. 2, 4–7.

Valvano & Yerraballi · Shape the World
Chapters 13–14: sampling, ADC, resolution and measurement chain.

TI · TIDA-00792 / TIDUCN1
BMS, cell monitoring, estimation and protection; PDF pp. 1–8.

Analog Devices · EMI Series, Part 4
PCB stack-up, return-current paths, routing and placement.

Analog Devices · EMI Series, Part 5
Shielding, common-mode interaction and thermal-design context.

ST · AN4776, Rev. 4
Timer, PSC/ARR, capture/compare, PWM and DMA; February 2026.

References 3/3

Sections used by topic and original calculations

Lee & Seshia · Embedded Systems

Chapters 1–2: cyber-physical systems, time, concurrency and architecture.

Linux Kernel · PREEMPT_RT

Theory of operation: preemption, threaded IRQs and priority inheritance.

FreeRTOS / Zephyr

Tasks and mutexes; FreeRTOS §8.3.2–8.3.3; Zephyr scheduler/priority inheritance.

Schneider / TI TIDA-01599

Distinguish STO, SS1 and braking; product examples do not imply certification.

Zuhily · York, PhD thesis (2009)

§2.3.2, Eqs. 2.11–2.12: fixed-priority response-time analysis with blocking.

MathWorks · Hardware-in-the-Loop

MIL / SIL / PIL / HIL comparison, fault scenarios and verification.

CiA / EtherCAT Technology Group

CAN FD technical brochure; EtherCAT Distributed Clocks, printed pp. 16–17.

Görseller ve hesaplar / Visuals & calculations

Scientific diagrams and numerical examples are original. Robot illustrations are AI-assisted, not manufacturing drawings.

MCUboot · Design

Image slots and test/confirm/revert flow; original application architecture.